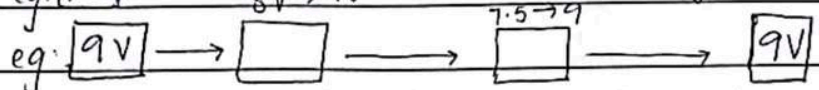


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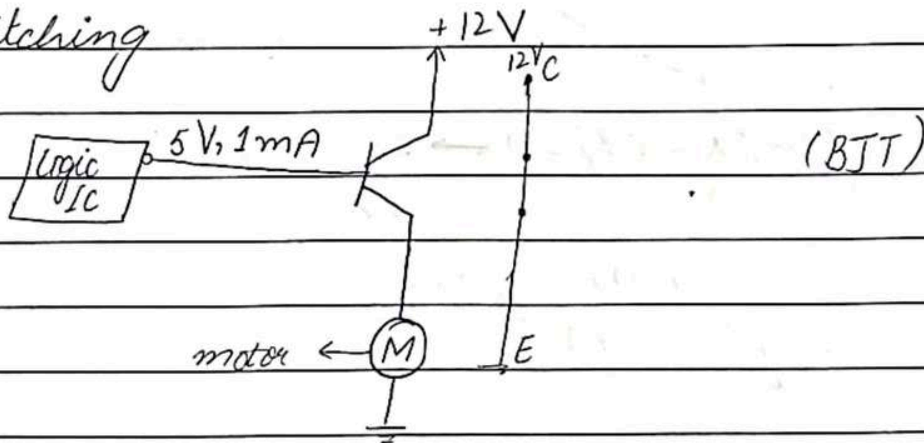
ELECTRIC CIRCUIT DESIGN

CHAPTER #1

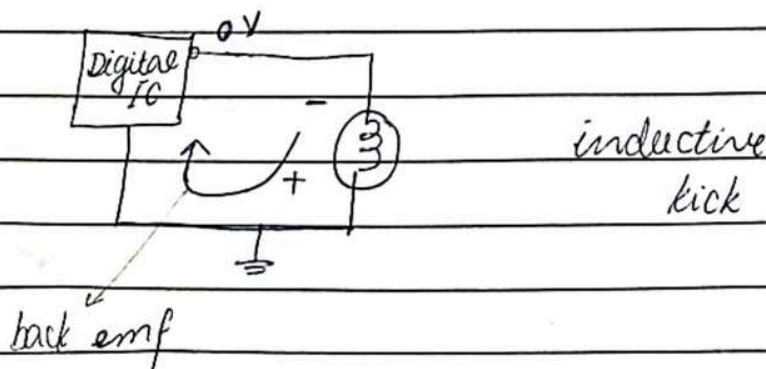
- 1) Amplification $(2m \rightarrow 2)$ (smaller I/P $\rightarrow$ larger O/P)
- 2) Switching eg. 
- 3) Buffering eg.  (I/P = O/P)
- 4) Isolating
- 5) Filtering
- 6) Math operation

larger I/P $\rightarrow$ smaller O/P (attenuation)

- 2) Switching



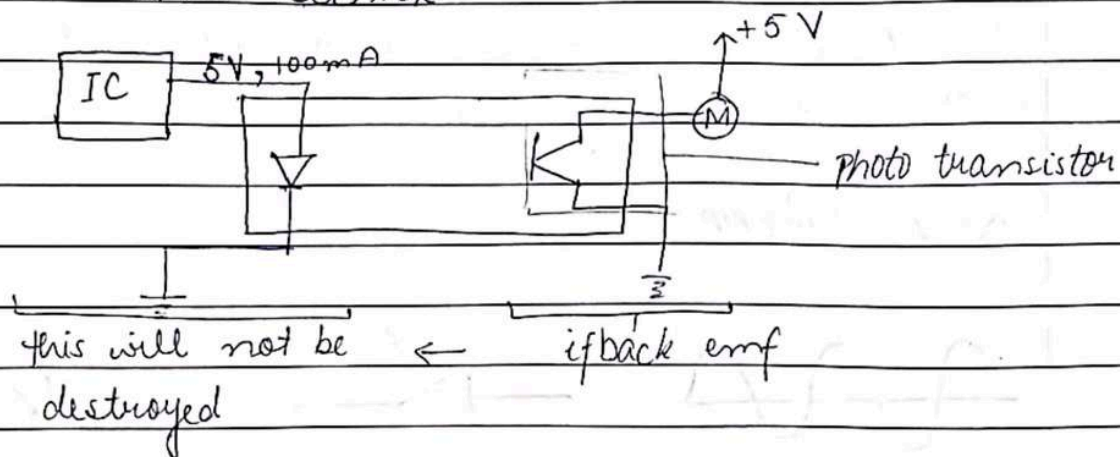
- 4) ISOLATION:



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OPTO ISOLATOR



1.4.2

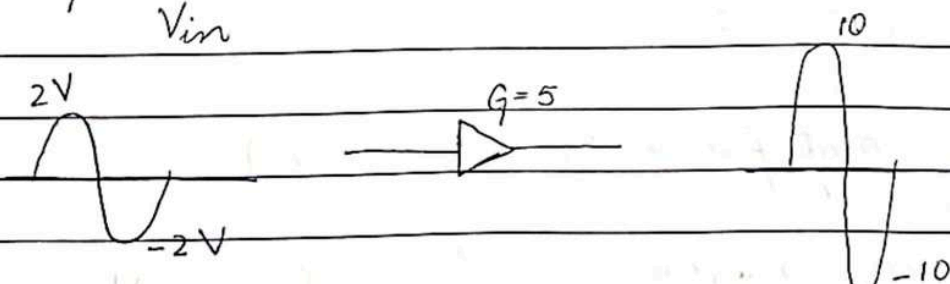
SIGNAL AMPLIFICATION

- Most practical sensor generator O/P of mV, MV, mA, ~~MA~~
- Since the signals are very weak, they need to be amplified.

* $V_{in} \rightarrow [G] \rightarrow V_{out}$

$$V_{out} = V_{in} \times G$$

$$G = \frac{V_{out}}{V_{in}}$$

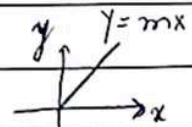
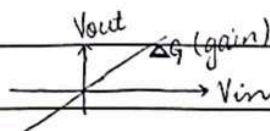


$$2 \sin(\omega t) \times 5 = 10 \sin(\omega t)$$

(amplitude magnification)

AMPLIFIERS:

- Linear (eg. $V_{out}/V_{in} = 5$
 $\Rightarrow V_{out} = 5V_{in}$)
- Non-linear

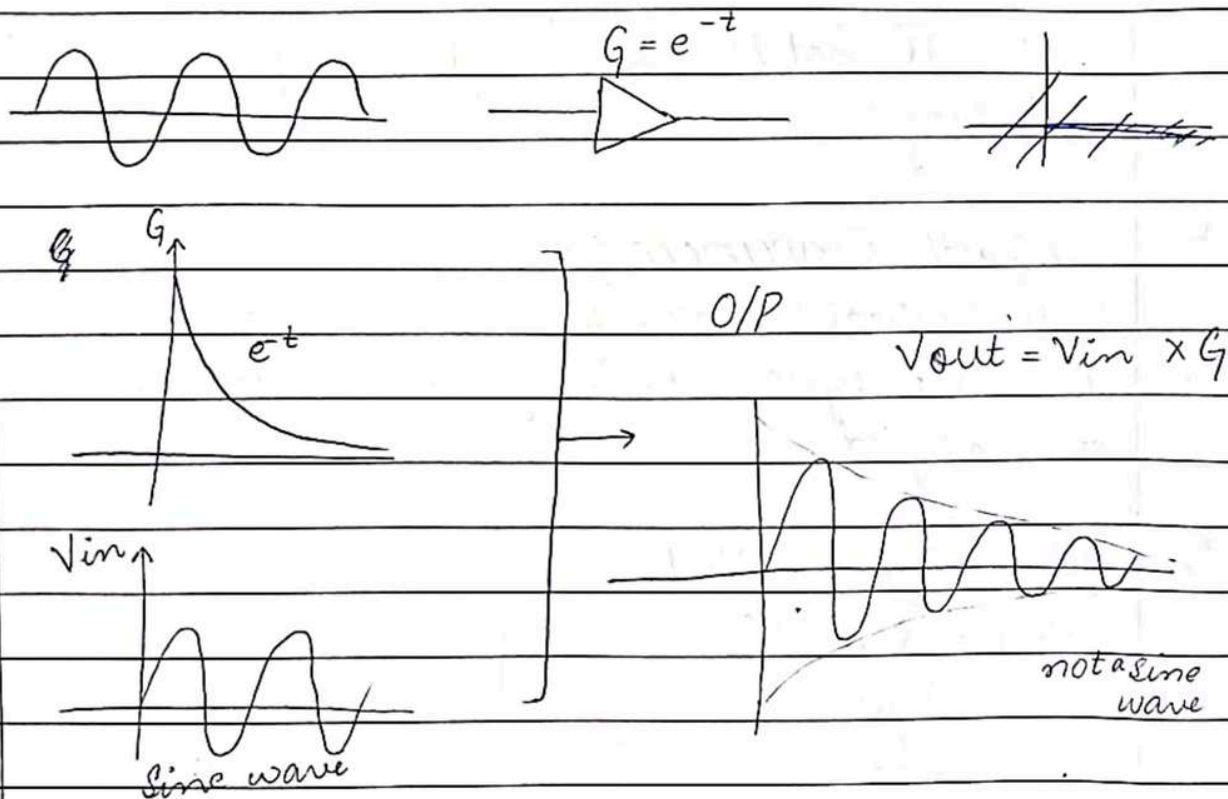


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1) **LINEAR:** V_{out} vs V_{in} transfer characteristic is a straight line — hence linear amplifier or gain (G) is constant.

2) **NON - LINEAR:**

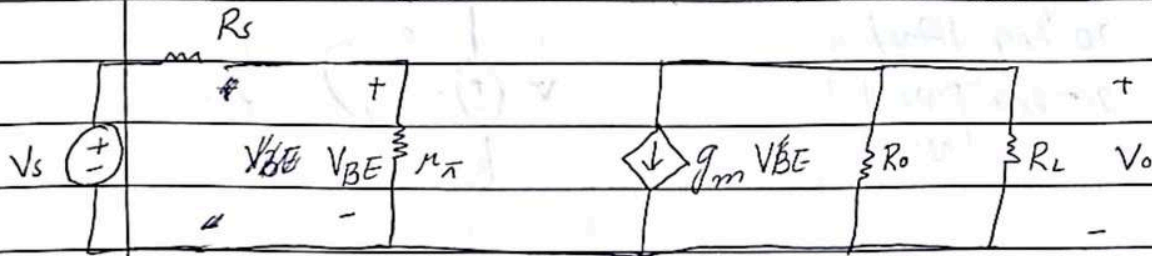


3) Voltage amplifier = $\frac{V_{out}}{V_{in}} = A_v (V/V)$

4) Current amplifier = $\frac{I_{out}}{I_{in}} = A_i (A/A)$

* 5) Power amplifier = $\frac{P_{out}}{P_{in}} = \frac{V_{out} \times I_{out}}{V_{in} \times I_{in}}$
 $= \left(\frac{V_{out}}{V_{in}} \right) \left(\frac{I_{out}}{I_{in}} \right) = A_v \times A_i$
 $= A_p (W/W)$

BJT MODEL (Trans-conductance amplifier)



$$R_s = 5k\Omega$$

$$r_{\pi} = 2.5k\Omega$$

$$g_m = 40mA/V$$

$$R_o = 100k\Omega$$

$$R_L = 5k\Omega$$

$$V_o = ?$$

$$V_s$$

$$V_o =$$

2/8

AMPLIFICATION GAINS (in dB's)

$$A_v = \frac{V_o}{V_i} \quad (V/V) \quad (\text{voltage gain})$$

$$A_v (dB) = 20 \log |A_v|$$

$$= 20 \log |V_o/V_i| \text{ dB}$$

$$A_i = I_o / I_i \quad (A/A) \quad (\text{current gain})$$

$$A_i (dB) = 20 \log |A_i|$$

$$= 20 \log |I_o/I_i| \text{ dB}$$

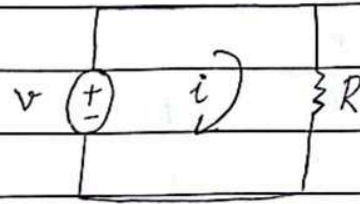
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POWER GAIN:

$$20 \log |A_v|$$

$$20 \log \left| \frac{v_o}{v_i} \right|$$



$$= 10 \log \left| \frac{v_o}{v_i} \right|^2$$

$$= 10 \log \left(\frac{v_o^2}{v_i^2} \right)$$

$$= 10 \log \left| \frac{v_o^2/R}{v_i^2/R} \right| = 10 \log \left| \frac{P_o}{P_i} \right| \text{ dB}$$

$$A_p = P_o/P_i \quad (\text{W/W})$$

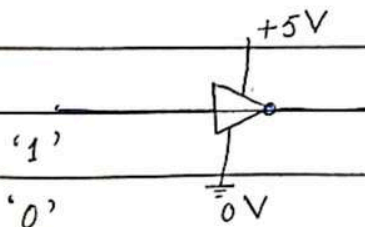
$$A_p (\text{dB}) = 10 \log \left| \frac{P_o}{P_i} \right|$$

$$A_v = \frac{v_o}{v_i} < 0 \quad (\text{It means it is inverted})$$

$$A_v (\text{dB}) = 20 \log \left| \frac{v_o}{v_i} \right| < 0 \quad (\text{It has been attenuated})$$

(lessen the value)

AMPLIFIER POWER SUPPLIES & SATURATION:



$$'0' \rightarrow 0V, '1' \rightarrow 5V$$

NOTE: logic '1' = +Vcc and logic '0' = ground (0V)

Max. v_o of your Ic circuit = +Vcc

Min v_o of your Ic circuit = -Vcc

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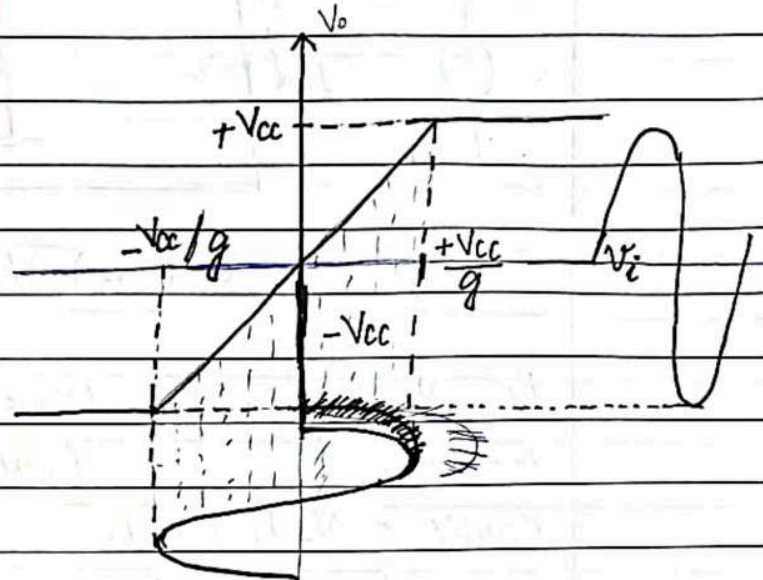
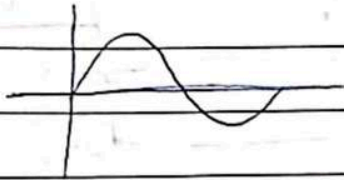
IDEAL TRANSFER CHARACTERISTICS:

$$G = \frac{V_o}{V_i} = 5$$

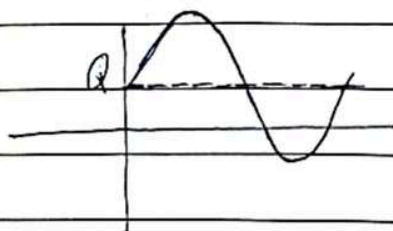
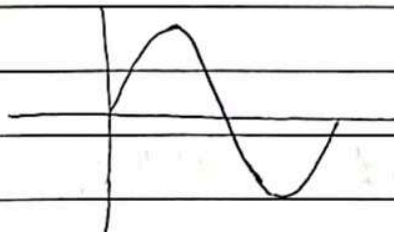
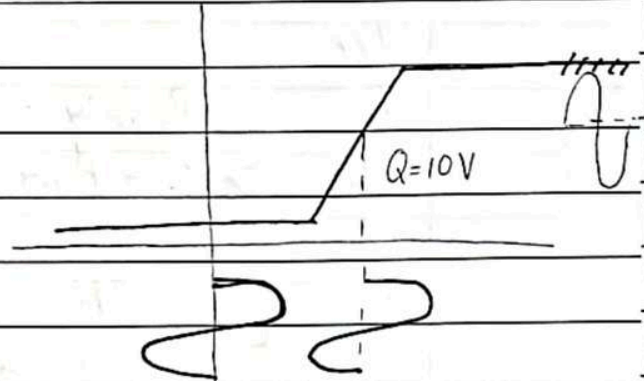
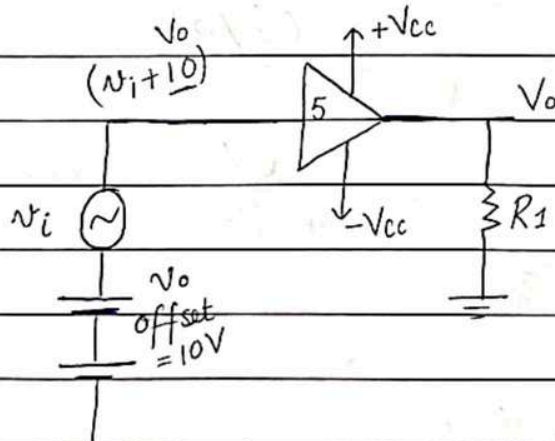
$$\begin{aligned} V_o &= 5V_i \\ y &= mx \end{aligned}$$

$$\pm V_{CC} = \pm 15V$$

$$V_i = 3 \sin(\omega t)$$

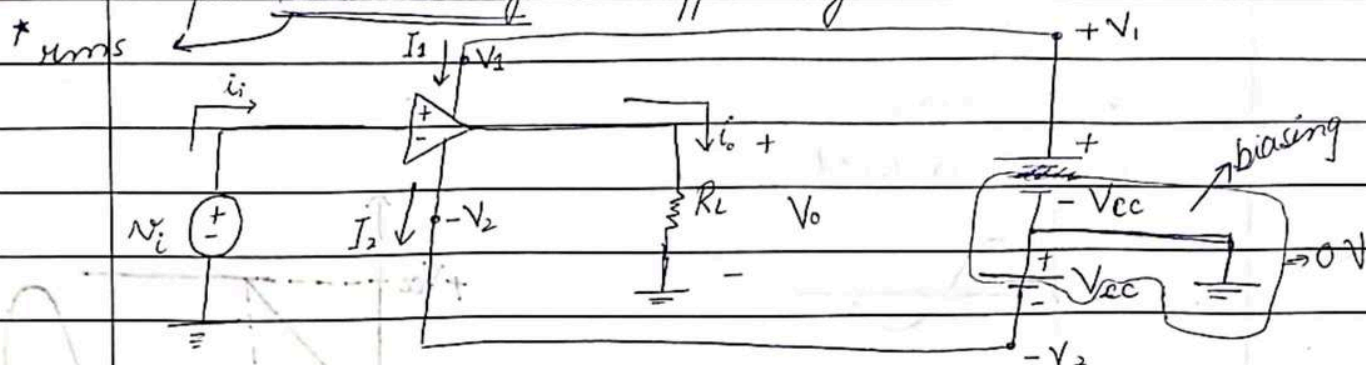


PRACTICAL TRANSFER CHARACTERISTICS:



- * capacitor blocks DC
- * capacitor allows AC

Consider an amplifier operating at $\pm 10V$ power supplies. It is fed sine voltage efficiency



$$a \cdot (-v_2) = v_2$$

$$P_i = v_i i_i \quad (\text{input})$$

$$P_o = V_o \cdot i_o \quad (\text{output})$$

$$P_{\text{supply}} = V_1 I_1 + V_2 I_2 \quad (i)$$

$$V_1 = V_2 = V_{CC}$$

$$I_1 = I_2 = I_{CC}$$

for DC $\Rightarrow (i) P_{\text{supply}} = 2 V_{\text{oc}} I_{\text{sc}} \quad (I/P)$

$$\begin{array}{c} \text{I/P} \\ P_i + P_{\text{supply}} \end{array} = \begin{array}{c} \text{O/P} \\ P_o + P_{\text{loss}} \end{array} \quad \left(\begin{array}{c} \text{Law of conservation} \\ \text{of energy} \end{array} \right)$$

$$P_{\text{loss}} = P_i + P_{\text{supply}} - P_o$$

$$\text{efficiency} = \eta = \frac{P_o}{P_i + P_{\text{supply}}} \times 100\%$$

$$V_o = 9V = 9/\sqrt{2} \text{ rms}, \quad i_i = 0.1 \text{ peak mA} = 0.1/\sqrt{2}, \quad R_L = 1k\Omega$$

$$V_i = 1V = 1/\sqrt{2} \text{ rms (peak voltage)}, \quad I_1 = I_2 = 9.5 \text{ mA}$$

Date: _____

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$$a) A_v = \frac{V_{o, rms}}{V_{i, rms}} = \frac{9/\sqrt{2}}{1/\sqrt{2}} = 9 V/V$$

$$A_v \text{ to db} = 20 \log(9) = 19.08 \text{ dB}$$

$$i_{i, rms} = \frac{0.1 \text{ m}}{\sqrt{2}} = 0.0707 \text{ mA}$$

$$i_{o, rms} = \frac{9/\sqrt{2}}{1 \text{ k}\Omega \rightarrow R_L} = 6.3 \text{ mA}$$

$$v = iR \Rightarrow i = v/R$$

$$b) A_i = \frac{I_o}{I_i} = \frac{6.3 \text{ mA}}{0.0707 \text{ mA}} = 90 A/A$$

$$\Rightarrow 20 \log(90) = 39.08 \text{ dB}$$

$$\begin{aligned} \text{(avg. power)} P_o &= V_{o, rms} \times I_{o, rms} \\ \text{1/p} &= 9/\sqrt{2} \times 6.3 \text{ mA} \\ &= 40.5 \text{ mW} \end{aligned}$$

$$\begin{aligned} \text{(avg. power)} P_i &= V_{i, rms} \times I_{i, rms} \\ \text{1/p} &= 1/\sqrt{2} \times 0.0707 \text{ mA} \\ &= 0.05 \text{ mW} \end{aligned}$$

$$c) A_P = P_o/P_i = 40.5 \text{ mW}/0.05 \text{ mW} = 810 \text{ W/W} \text{ or } A_P = A_v \times A_i$$

$$\Rightarrow 10 \log(810) = 29.08 \text{ dB}$$

$$\begin{aligned} d) P_{supply} &= 2V_{cc} I_{cc} \\ &= 2(10)(9.5 \text{ mA}) \\ &= 190 \text{ mW} \end{aligned}$$

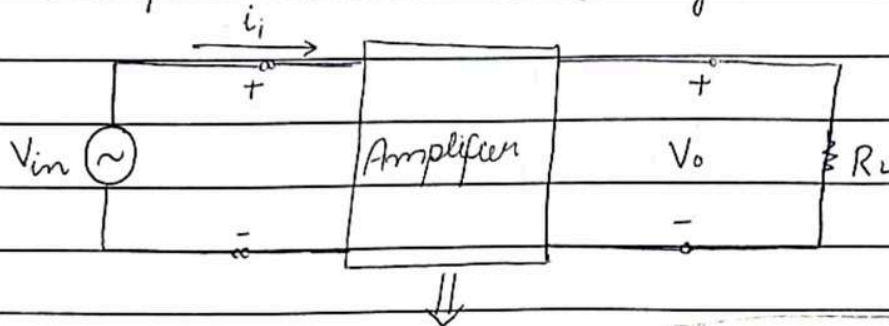
$$\begin{aligned} e) P_{loss} &= P_i + P_{supply} - P_o \\ &= 0.05 \text{ mW} + 190 \text{ mW} - 40.5 \text{ mW} \end{aligned}$$

$$P_{loss} = 149.55 \text{ mW}$$

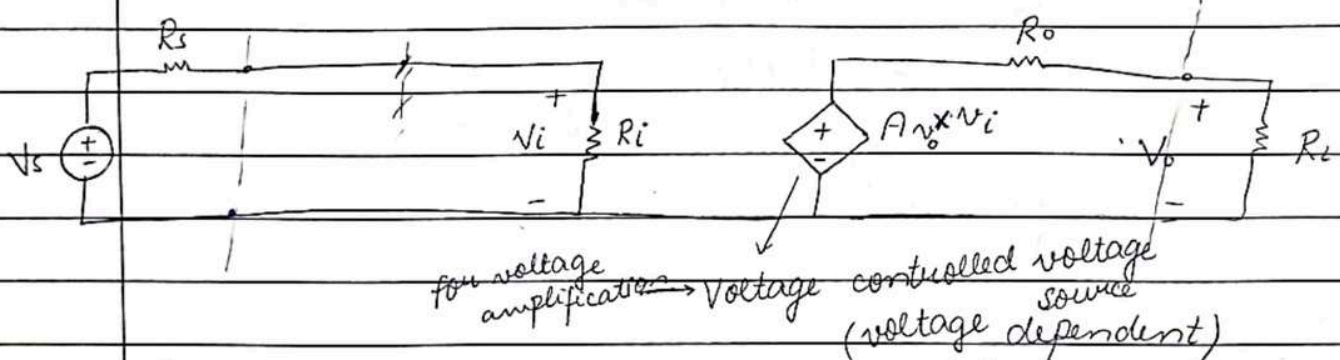
$$f) \eta = \frac{40.5 \text{ mW}}{190 \text{ mW} + 0.05 \text{ mW}} \times 100\% = 21.3\%$$

CIRCUIT MODEL FOR AMPLIFIERS

1) VOLTAGE AMPLIFIER: (voltage divides)



$$V_o = \frac{R_L}{R_L + R_o} \times A_{v_o} V_i$$



R_i accounts for the fact that amplifier draws a finite input current from the input voltage source.

* A load resistor ~~is connected~~ ^{across} with voltage source $\rightarrow$ a voltage divider

* Every source has an internal resistance.

• For Voltage source (V_s) $\approx V_i$

$\Rightarrow R_s \ll R_i$ (R_i should be much larger)

Ideally: $R_i \rightarrow \infty$

Ideally: $R_o \rightarrow 0$ (s.c) (for $A_{v_o} \times V_i \approx V_o$)

R_i (larger)

R_o (smaller)

Use VDR:

$$V_i = \left(\frac{R_i}{R_i + R_s} \right) \times V_s \quad (i)$$

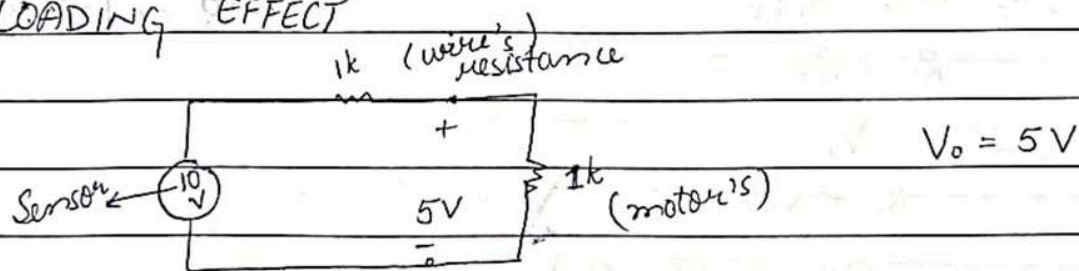
$$V_o = \left(\frac{R_L}{R_o + R_L} \right) \times A_{v_o} \times V_i \quad (ii)$$

 $\Rightarrow$ (ii) in (i)

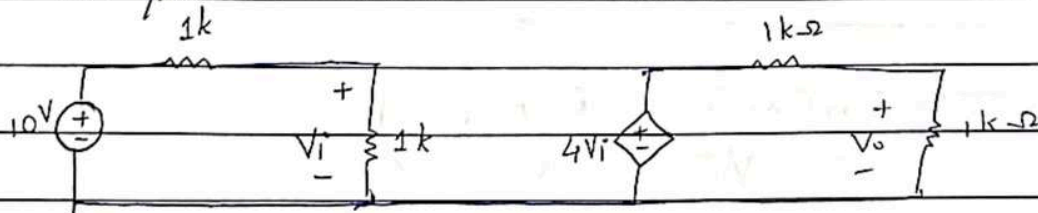
$$V_o = A_{v_o} \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{R_i}{R_i + R_s} \right) V_s$$

$$G = \frac{V_o}{V_s} = A_{v_o} \left(\frac{R_L}{R_L + R_o} \right) \left(\frac{R_i}{R_i + R_s} \right)$$

LOADING EFFECT



UNITY GAIN / BUFFER AMPLIFIER



$$\frac{V_o}{V_s} = A_{v_o} \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{R_i}{R_s + R_i} \right)$$

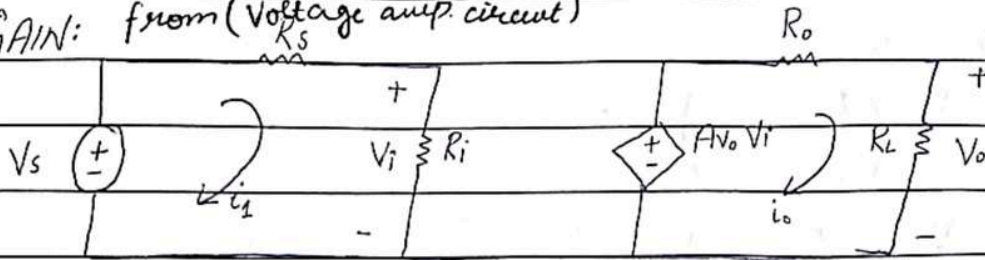
$$= 4 \left(\frac{1k}{2k} \right) \left(\frac{1k}{2k} \right) = 4(0.5)(0.5)$$

$$\frac{V_o}{V_s} = 1 \Rightarrow V_o = V_s = 10V \quad (\text{No signal droppage})$$

Date: 29/8

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CURRENT GAIN: from (Voltage amp. circuit)



$$G = \frac{V_o}{V_s} = A_v \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{R_i}{R_i + R_s} \right)$$

Q. $\frac{V_o}{V_i} = A_v \left(\frac{R_L}{R_o + R_L} \right)$

CURRENT GAIN:

$$\frac{i_o}{i_i} = ?$$

$$i_o = \frac{A_v V_i}{R_o + R_L}$$

$$V = IR \Rightarrow I = V/R$$

$$i_i = \frac{V_s}{R_i + R_s}$$

$$\frac{i_o}{i_i} = \frac{A_v V_i / (R_o + R_L)}{V_s / (R_i + R_s)}$$

$$= A_v \times \frac{V_i}{V_s} \left(\frac{R_i + R_s}{R_o + R_L} \right)$$

$$V_i = \left(\frac{R_i}{R_i + R_s} \right) \times V_s$$

$$\frac{V_i}{V_s} = \frac{R_i}{R_i + R_s}$$

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$$\frac{i_o}{i_i} = A_{V_o} \left(\frac{R_i}{R_i + R_s} \right) \left(\frac{R_i + R_s}{R_o + R_L} \right)$$

$$\frac{i_o}{i_i} = A_{V_o} \left(\frac{R_i}{R_o + R_L} \right)$$

OR

$$G = \frac{V_o}{V_s} = A_{V_o} \left(\frac{R_i}{R_i + R_s} \right) \left(\frac{R_L}{R_o + R_L} \right) \quad (V/V)$$

$$\frac{i_o}{i_i} = \frac{V_o/R_L}{V_s/(R_s + R_i)} = \left(\frac{V_o}{V_s} \right) \left(\frac{R_s + R_i}{R_L} \right) \quad (A/A)$$

$$= A_{V_o} \left(\frac{R_i}{R_i + R_s} \right) \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{R_s + R_i}{R_L} \right)$$

$$= A_{V_o} \left(\frac{R_i}{R_o + R_L} \right)$$

$$\frac{V_o}{i_i} = \frac{V_o}{V_s/(R_i + R_s)} = \frac{V_o}{V_s} \times (R_i + R_s) \quad \begin{matrix} \text{(Trans-resistance)} \\ \text{gain} \end{matrix}$$

(V/A-unit)

$$\frac{i_o}{V_s} = \frac{A_{V_o} V_i / (R_o + R_L)}{R_i / V_s} = \frac{A_{V_o} V_i}{V_s (R_o + R_L)} \times \frac{V_s}{R_i}$$

$$i_o = V_o / R_L$$

$$\frac{i_o}{V_s} = \frac{V_o}{V_s} \times \frac{1}{R_L} \quad \begin{matrix} (A/V) \\ \downarrow \\ \text{unit} \end{matrix} \quad \begin{matrix} \text{(Trans-conductance)} \\ \text{gain} \end{matrix}$$

Date: _____

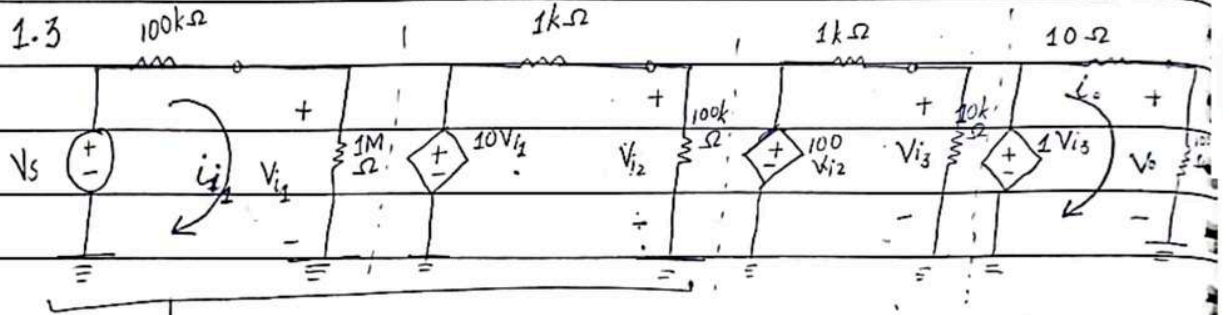
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2) CASCADE AMPLIFIERS:

$$V_i \longrightarrow [H_1] \longrightarrow V_o \Rightarrow V_o = H_1 V_i$$

$$V_i \longrightarrow [H_1] \longrightarrow [H_2] \longrightarrow [H_3] \longrightarrow V_o \Rightarrow V_o = (H_1 H_2 H_3) V_i$$

EXAMPLE # 1.3



(relatively high) stage 1
I/p resistance

(larger I/p) (moderate)

stage 2 (gain) (moderate gain)
(high gain factor & low I/p resistance)

(smaller output)

stage 3
(unity gain)

$$VDR: \quad \frac{V_o}{V_{i3}} = \frac{100}{100 + 10} = \frac{100}{110} = 0.909 \text{ V/V}$$

$$\frac{V_{i3}}{V_{i2}} = \frac{10k}{1k + 10k} \times 100 = 90.909 \text{ V/V}$$

$$\frac{V_{i2}}{V_{i1}} = \frac{100k}{100k + 1k} \times 10 = 9.9090 \text{ V/V}$$

$$\frac{V_{i1}}{V_s} = \frac{1M}{1M + 100k} = 0.9090 \text{ V/V}$$

$$\frac{V_o}{V_s} = \frac{V_o}{V_{i3}} \times \frac{V_{i3}}{V_{i2}} \times \frac{V_{i2}}{V_{i1}} \times \frac{V_{i1}}{V_s}$$

$$= 0.909 \times 90.909 \times 9.900 \times 0.9090$$

$$G = \boxed{\frac{V_o}{V_s} = 743.6 \text{ V/V}}$$

$\Rightarrow$ in dB

$$= 20 \log_{10} (743.6)$$

$$= 57.4 \text{ dB}$$

$$A_v = \frac{V_o}{V_{i1}} = \frac{V_o}{V_{i3}} \times \frac{V_{i3}}{V_{i2}} \times \frac{V_{i2}}{V_{i1}}$$

$$= 818.0 \text{ V/V}$$

in dB

$$= 58.255 \text{ dB}$$

$V = IR$

$$A_{ii} = \frac{i_o}{i_{i1}} = \frac{V_o / 100 + 10 R_L}{V_s / R_{i1} + R_s} \quad \text{or} \quad \frac{V_o / R_L}{V_{i1} / R_{i1}}$$

$$= \frac{V_o}{V_{i1}} \times \frac{R_{i1}}{R_L}$$

$$= (818) (1\text{M}/100) =$$

$$= 818 \times 10^4 \text{ A/A}$$

$$= 8.18 \times 10^6 \text{ A/A}$$

in dB

$$= 138.25 \text{ dB}$$

Date: 20-07-2020

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Power gain : (w.r.t input power)

$$A_p = A_v \times A_i = \left(\frac{V_o}{V_{i_1}} \right) \left(\frac{i_o}{i_{i_1}} \right)$$

$$= (818) (8.18 \times 10^6)$$

$$= 66.9 \times 10^8 \text{ W/W}$$

in dB:

$$= 10 \log (66.9 \times 10^8)$$

$$= 98.25 \text{ dB}$$

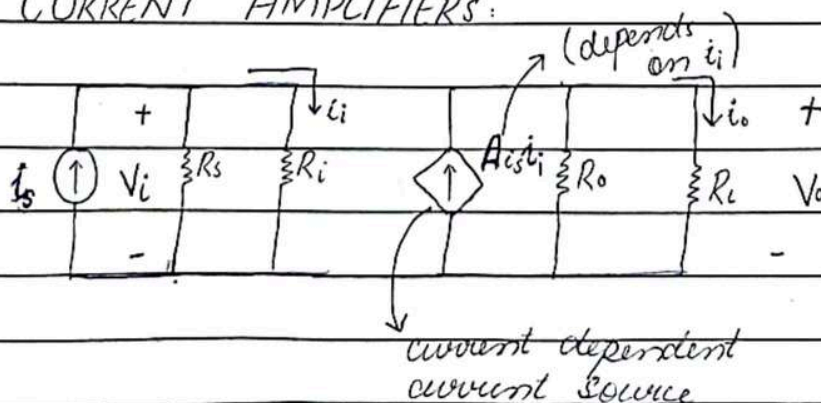
Power gain w.r.t source power

$$A_p = \frac{V_o}{V_s} \times \frac{i_o}{i_s} \quad \text{" } i_{i_1} = i_s$$

✓ FOUR AMPLIFIER MODELS

- 1) Voltage amplifiers
- 2) Current amplifiers
- 3) Trans-resistance amplifiers V_o/I_s
- 4) Trans-conductance amplifiers I_s/V_o

2) CURRENT AMPLIFIERS:



Date: _____

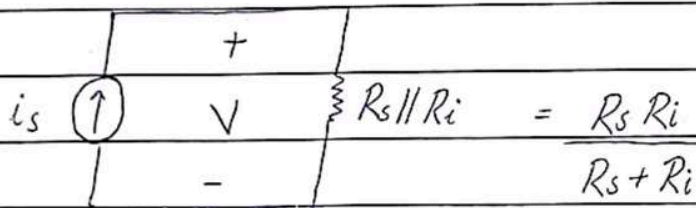
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CDR:

$$i_i = \frac{R_s}{R_s + R_i} \times i_s$$

$$i_o = \frac{R_o}{R_o + R_L} \times A_{is} i_i$$

$$A_i = \frac{i_o}{i_i} = A_{is} \left(\frac{R_o}{R_o + R_L} \right) \left(\frac{R_s}{R_s + R_i} \right)$$

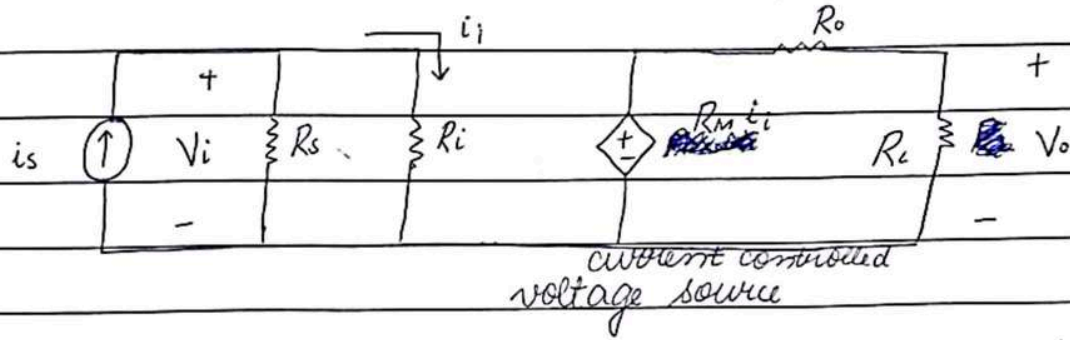


$$V = \frac{R_s R_i}{R_s + R_i} \times i_s \quad (V = IR)$$

$$i_o = \frac{V}{R_i} = \frac{1}{R_i} \times \frac{R_s R_i}{R_s + R_i} \times i_s$$

$$i_o = \frac{R_s}{R_s + R_i} \times i_s$$

$$\begin{aligned} \frac{i_o}{i_s} &= \frac{V_o}{V_i} = \frac{i_o \times R_L}{i_s \times (R_i \parallel R_s)} \\ &= \frac{i_o}{i_s} \times \frac{R_L \cancel{R_i}}{(R_i R_s) / (R_i + R_s)} \\ &= \frac{i_o}{i_s} \times \frac{R_L (R_i + R_s)}{R_i R_s} \end{aligned}$$

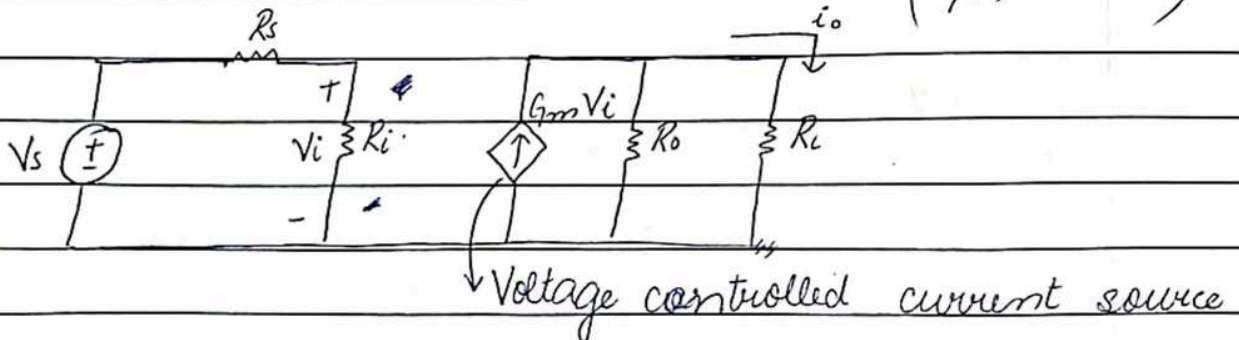
3) TRANS-RESISTANCE AMPLIFIER: ($R_m = V_o/I_s$)

$$\frac{V_o}{i_s} = \left(\frac{R_L}{R_o + R_L} \right) \times R_m i_i$$

$$i_i = \left(\frac{R_s}{R_i + R_s} \right) \times i_s$$

$$V_o = R_m \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{R_s}{R_i + R_s} \right) i_s$$

$$\frac{V_o}{i_s} = R_m \left(\frac{R_L}{R_o + R_L} \right) \left(\frac{R_s}{R_i + R_s} \right)$$

TRANS-CONDUCTANCE AMPLIFIER: ($G_m = I_o/V_s$)

Date: _____

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$$V_i = \frac{R_i}{R_s + R_i} \times V_s$$

$$i_o = \frac{R_o}{R_o + R_L} \times g_m V_i$$

$$\frac{i_o}{V_i} = g_m \left(\frac{R_o}{R_L + R_o} \right) \left(\frac{R_i}{R_s + R_i} \right)$$

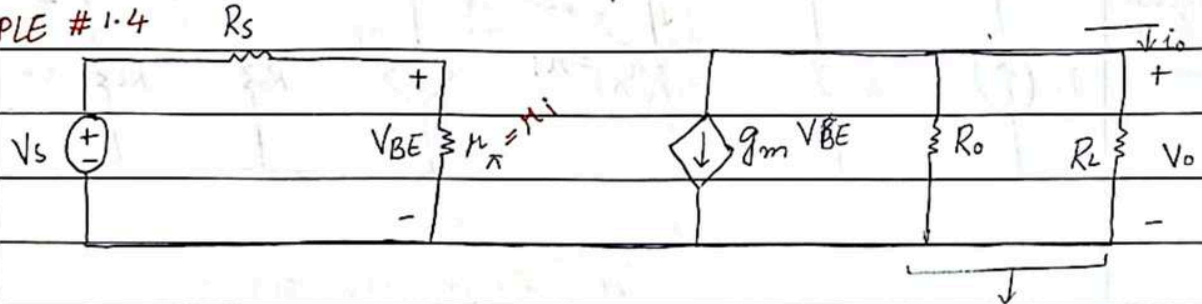
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BJT MODEL

(TRANS-CONDUCTANCE AMPLIFIER)

EXAMPLE #1.4

(1)



$$R_s = 5k\Omega$$

$$\mu_n = 2.5k\Omega$$

$$g_m = 40mA/V$$

$$R_o = 100k\Omega$$

$$R_L = 5k\Omega$$

$$\frac{V_o}{V_s} = ?$$

$$R_o \parallel R_L$$

$$i_o = \frac{R_o}{R_o + R_L} \times g_m V_{BE}$$

$$V_{BE} = \frac{\mu_n}{\mu_n + R_s} \times V_s \quad (i)$$

$$V_o = \frac{R_o R_L}{R_o + R_L} \times g_m V_{BE} \quad (ii)$$

$$V_s = \frac{V_o}{g_m} \quad (i) \text{ in } (ii)$$

$$V_o = \frac{R_o R_L}{R_o + R_L} \times g_m \left(\frac{\mu_n}{\mu_n + R_s} \right) \times V_s$$

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$$\frac{V_o}{V_s} = (-g_m) \left(\frac{\mu_{\pi}}{\mu_{\pi} + R_s} \right) \left(\frac{R_o R_L}{R_o + R_L} \right)$$

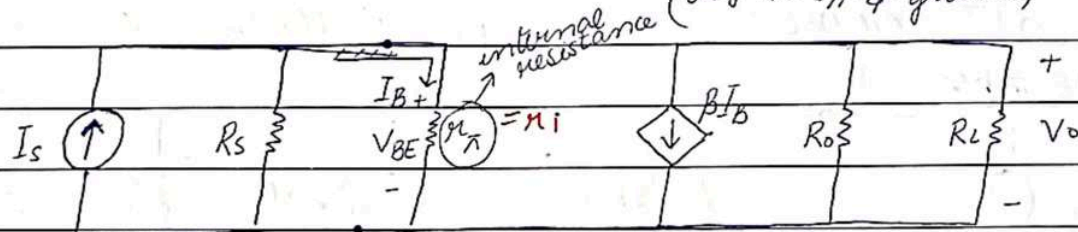
$$= (-40m) \left(\frac{2.5k}{(2.5k) + (5k)} \right) \left(\frac{(100k)(5k)}{(100k) + (5k)} \right)$$

$$G = \frac{V_o}{V_s} = -63.5 V/V$$

$$I_o/V_s = ?$$

eq #14 (2) BJT MODEL (CURRENT AMPLIFIER) (only b/w I/P & ground)

EXAMPLE #14



$$\beta = ?$$

values
use from previous Q.

$$\Rightarrow g_m V_{BE} = \beta I_B$$

$$\beta = g_m \left(\frac{V_{BE}}{I_B} \right)$$

$$= g_m (\mu_{\pi})$$

$$= (40m)(2.5k)$$

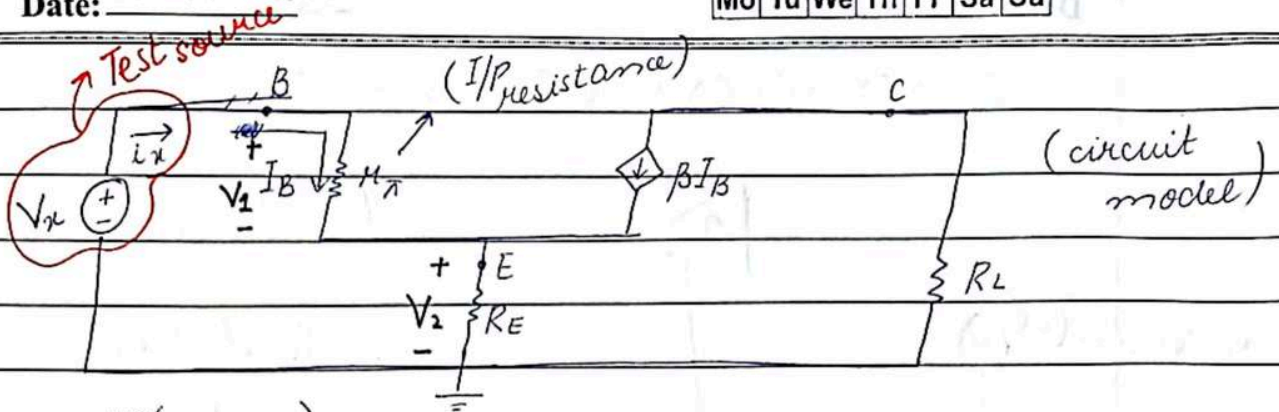
$$= 100 A/A$$

$$\mu_{\pi} = \frac{V_{BE}}{I_B}$$

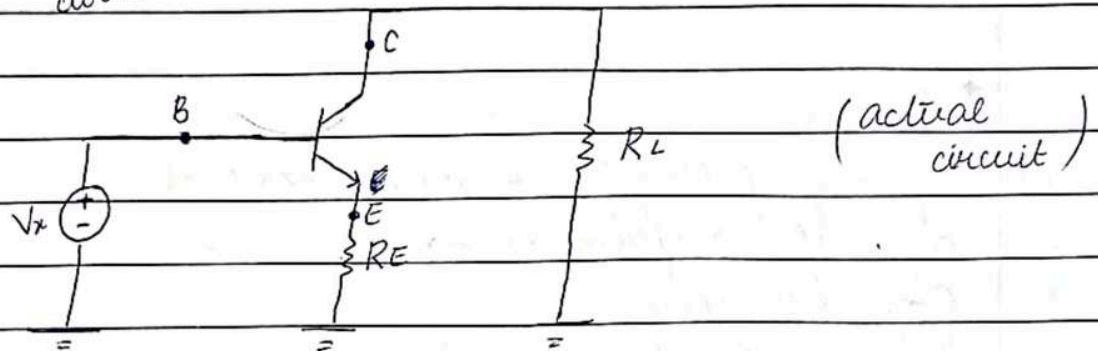
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Q.



V_x & i_x (unknowns - test voltage & current)
 $\Rightarrow$
 $\frac{V_x}{i_x} = R_{in}$



~~internal~~ internal resistance = $R_{in} = ?$

*

$$V_x = V_1 + V_2$$

$$V_x = (I_B)(r_{\pi}) + (I_B + \beta I_B)(R_E)$$

$$V_x = I_B \left(r_{\pi} + (1 + \beta) R_E \right)$$

$$\frac{V_x}{I_B} = r_{\pi} + (1 + \beta) R_E$$

$$\therefore i_x = I_B$$

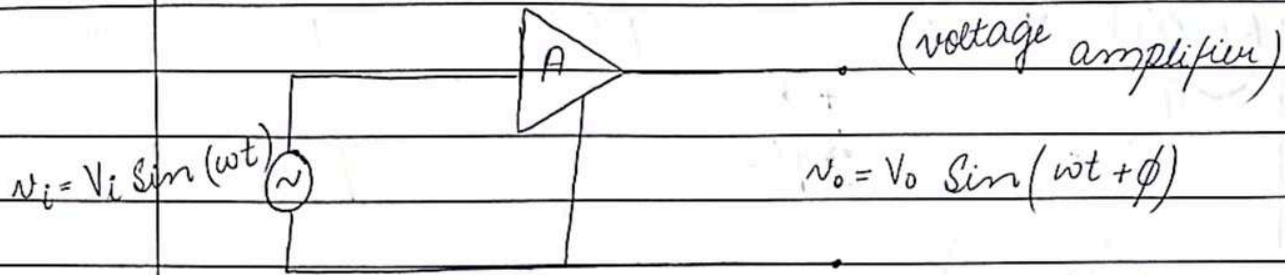
$$\Rightarrow \frac{V_x}{i_x} = r_{\pi} + (1 + \beta) R_E$$

$$R_{in} = r_{\pi} + (1 + \beta) R_E$$

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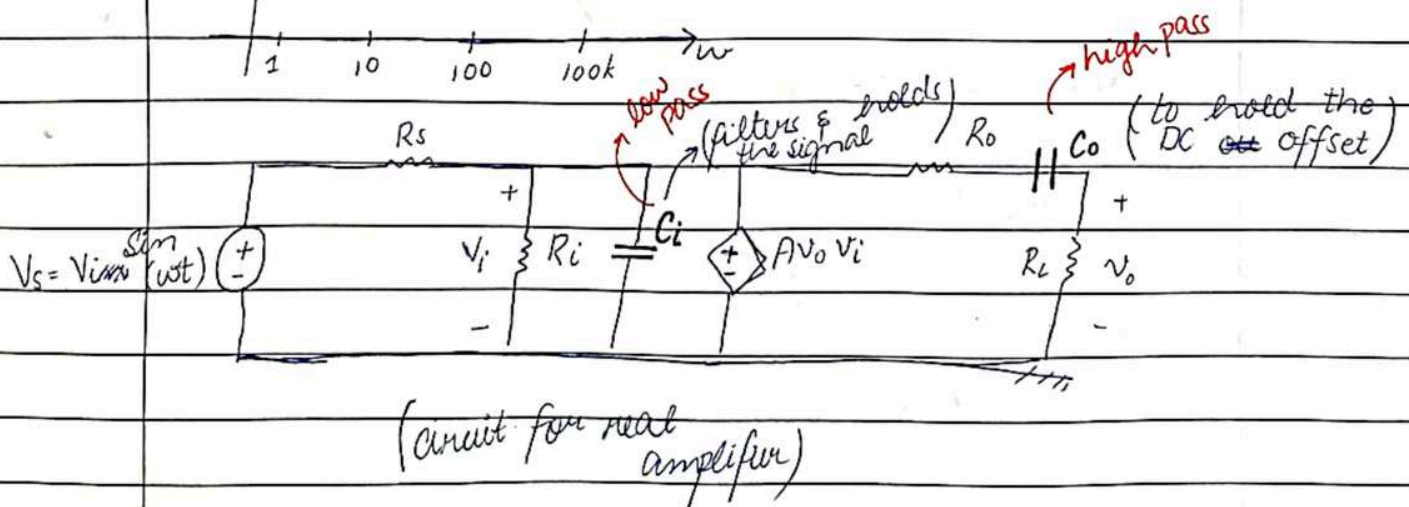
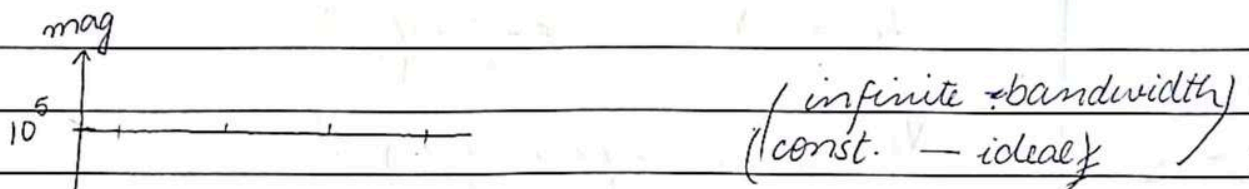
FREQUENCY RESPONSE OF AMPLIFIERS



* AXIOMS

- (i) frequency of I/p & O/p remains constant.
- (ii) Shape of waveform remains same
- (iii) Amplitude changes.
- (iv) Phase angle changes

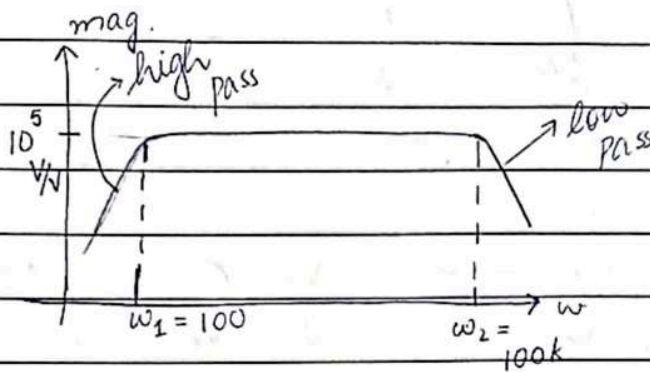
MAGNITUDE / GAIN OF AN 'IDEAL' VOLTAGE AMPLIFIER:



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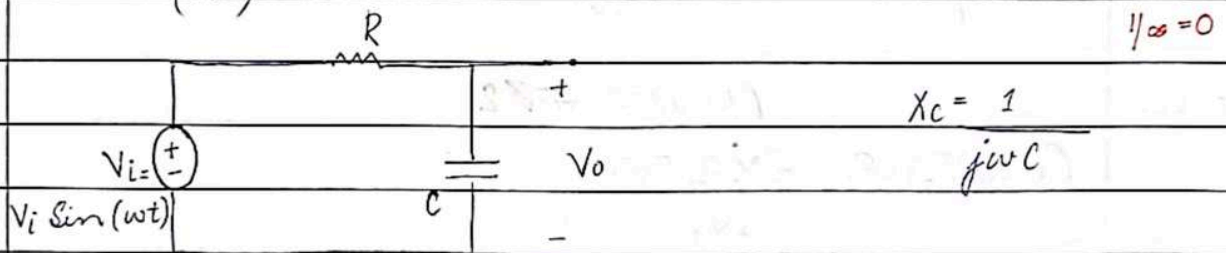
MAGNITUDE / GAIN OF A 'REAL' VOLTAGE AMPLIFIER



SINGLE TIME - CONSTANT CIRCUITS:

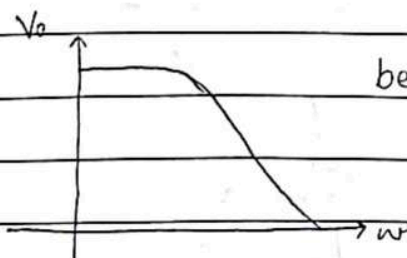
Those circuit which have only one time constant.

(RL) or (RC)



* when $\omega \rightarrow 0$; $X_c \rightarrow \infty$; $V_o = V_i$ (open circuit)
 very small

* when $\omega \rightarrow \infty$; $X_c \rightarrow 0$; $V_o = 0$ (short circuit)

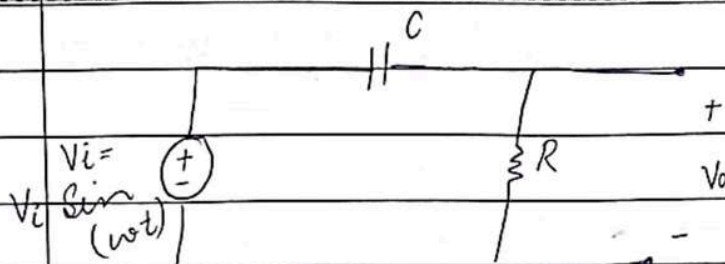


behaviour of a 'low pass' filter circuit

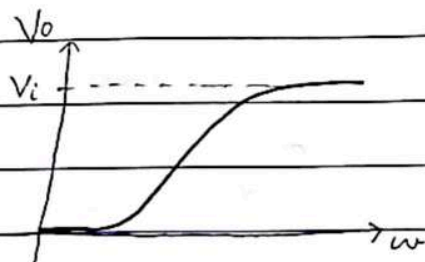
(Vo across capacitor)

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* when $\omega \rightarrow 0$; $X_C \rightarrow \infty$; $V_o = 0$ (small) (open circuit)
 when $\omega \rightarrow \infty$; $X_C \rightarrow 0$; $V_o = V_i$

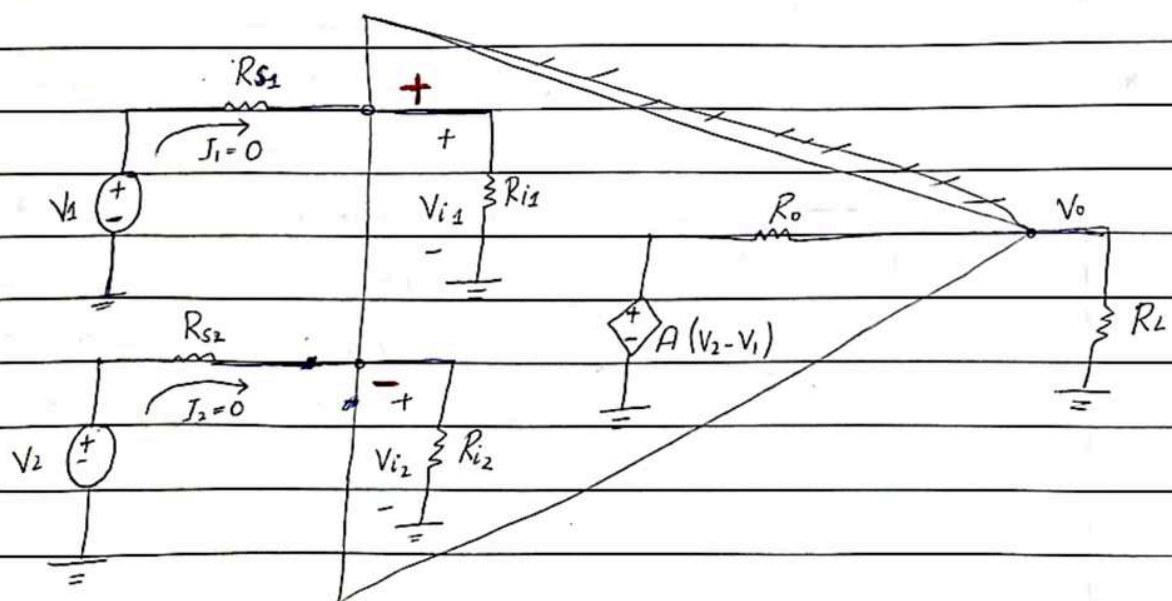
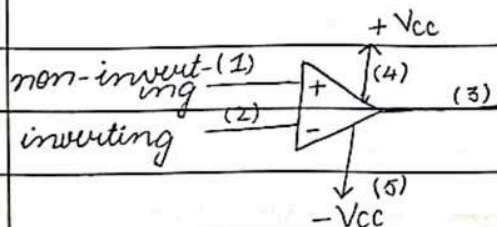


(high pass filter)
 V_o across resistor

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CHAPTER #2

OPERATIONAL AMPLIFIER:



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* An op-amp is primarily used as a voltage amplifier.

(i) An op-amp is used to amplify the difference b/w inverting and non-inverting voltage input (V_1 & V_2)

$$V_o = A(V_2 - V_1) \quad \text{or governing eq. of op-amp}$$

$$V_o = A(V_+ - V_-)$$

(ii) Input current from signal V_1 & V_2 is 'zero'. Op-amp does not draw any current from the sources (V_1 & V_2).

$R_{i1} \rightarrow \infty$ } internal (open circuit)
 $R_{i2} \rightarrow \infty$ } resistance

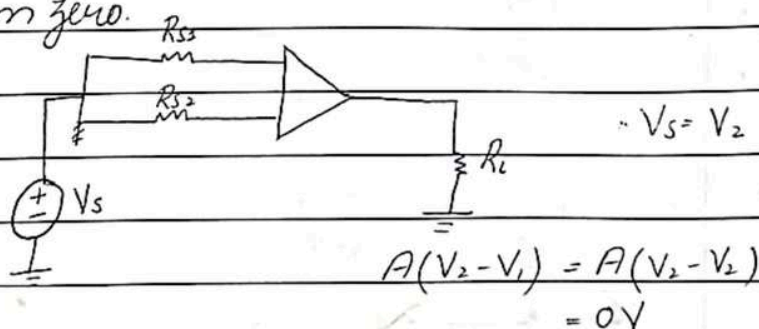
iii) $R_o = 0$

Common mode rejection: ∞

ideally $\rightarrow 0$

practically $\rightarrow$ non zero.

Common voltage



COMMON MODE GAIN (A_{cm}) — should be zero.

$$V_o = A(V_2 - V_1)$$

$$A_d = \frac{V_o}{V_2 - V_1}$$

(differential gain, A_d)

$$A_{cm} = \frac{V_o}{V_i}$$

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COMMON MODE REJECTION RATIO: (should be larger)

$$CMRR = 20 \log \left| \frac{A_D}{A_{CM}} \right|$$

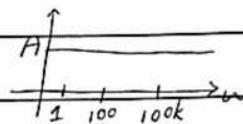
ideally $\rightarrow \infty$

(iv) C_i & C_o — DC coupled capacitors

(v) $CMRR \rightarrow \infty$

(vi) $A \rightarrow \infty$ (ideally) } gain
 $A = 10^5$ V/V (practically)

(vii) Infinite bandwidth

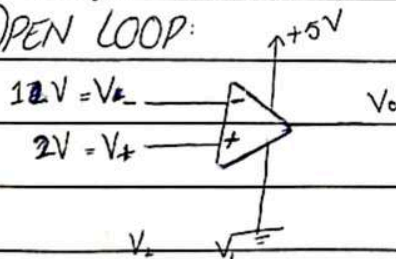


OP-AMP can be used in two ways:

- Open loop
- Closed loop

* V_o value can exceed V_{cc} value.

OPEN LOOP:



* if $V_+ > V_-$; $V_o = +V_{cc}$
 $V_+ < V_-$; $V_o = -V_{cc}$

$$V_o = A(V_+ - V_-)$$

$$A = 10^5$$

$$V_o = 10^5 (2 - 1)$$

$$= 1 \times 10^5 = 5V$$

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OR

$$V_o = A(V_+ - V_-)$$

$$= 10^5 (1 - 2)$$

$$= -10^5 = 0V$$

(ground)

(values ~~never~~ changed)
2 → 1, 1 → 2

$$V_- = 1\mu V$$

$$V_+ = 2\mu V$$

$$V_o = 10^5 (V_+ - V_-)$$

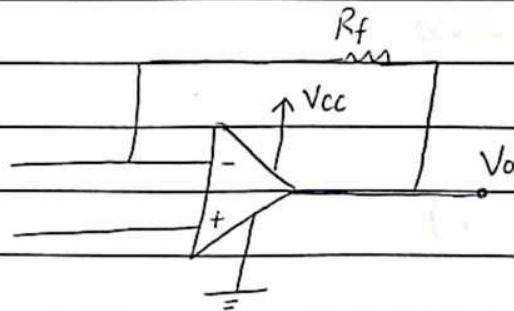
$$V_o = 10^5 (2\mu - 1\mu)$$

$$= 1\mu \times 10^5 = 1 \times 10^{-6} \times 10^5$$

$$= 0.1 V$$

CLOSED LOOP

(when O/P is connected to I/P)



* connect V_o with V_- always
non-inverting I/P

i) In open loop configuration, an op-amp can only be used as a comparator and you can not adjust the gain as per your choice.

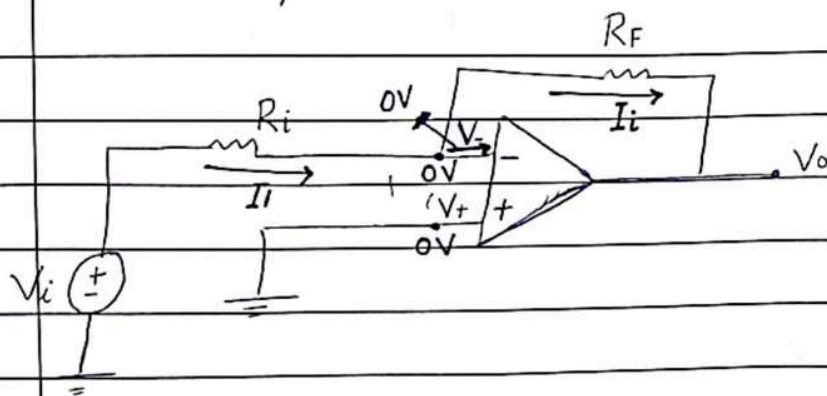
ii) In closed loop, the design becomes more flexible and you can change the gain as per your choice.

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INVERTING AMPLIFIER:

if you want gain = 10
put $R_F = 10k$
~~put $R_F = 10k$~~



adjust R_i & R_F

CASE - I: $(A \rightarrow \infty)$

$$V_o = A(V_+ - V_-)$$

$$V_+ - V_- = \frac{V_o}{A}$$

$$A \rightarrow \infty ; V_o/A \rightarrow 0$$

$$V_+ - V_- = 0 \Rightarrow V_+ - V_- = 0$$

$$V_+ = V_-$$

(Virtual short circuit)

if the loop is closed $\Rightarrow V_- = V_+$ (virtual short circuit)
using ohm's law.

$$V_+ = 0V$$

$$V_- = 0V$$

$$I_i = \frac{V_i - 0}{R_i} = \frac{V_i}{R_i} \quad (i)$$

$$0 - V_o = I_i R_F$$

$$V_o = - I_i R_F \quad (ii)$$

Date: _____

Substitute (ii) in (i)

$$V_o = -\left(\frac{V_i}{R_i}\right) \times R_F$$

$$\frac{V_o}{V_i} = -\frac{R_F}{R_i} = G$$

using node analysis

$$0 - \frac{V_i}{R_i} + 0 - \frac{V_o}{R_F} = 0$$

$$-\frac{V_i}{R_i} - \frac{V_o}{R_F} = 0$$

$$\frac{V_o}{R_F} = -\frac{V_i}{R_i}$$

$$\frac{V_o}{V_i} = -\frac{R_F}{R_i} = G$$

CASE - II: ($A \rightarrow \text{finite}$)

$$V_o = A(V_+ - V_-)$$

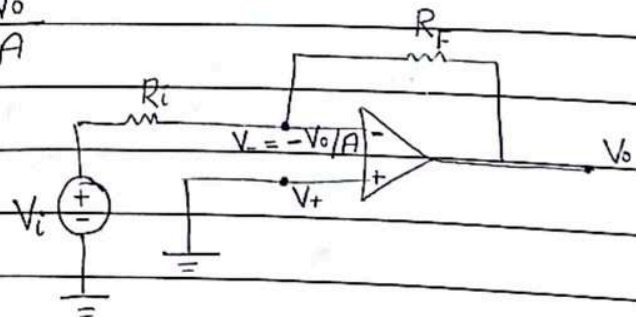
$$V_+ - V_- = V_o/A$$

$$A \rightarrow \text{finite}$$

$$V_+ = 0V \quad V_- = V_+ - \frac{V_o}{A}$$

$$V_+ = 0V$$

$$V_- = -V_o/A$$



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$$\frac{-V_o/A - V_i}{R_i} + \frac{(-V_o/A) - V_o}{R_F} = 0$$

$$\frac{-V_o}{AR_i} - \frac{V_i}{R_i} - \frac{V_o}{AR_F} - \frac{V_o}{R_F} = 0$$

$$\frac{-V_o}{AR_i} - \frac{V_o}{AR_F} - \frac{V_o}{R_F} = \frac{V_i}{R_i}$$

$$\frac{V_o}{AR_i} + \frac{V_o}{AR_F} + \frac{V_o}{R_F} = -\frac{V_i}{R_i}$$

$$V_o \left(\frac{1}{AR_i} + \frac{1}{AR_F} + \frac{1}{R_F} \right) = -\frac{V_i}{R_i}$$

$$V_o \left(\frac{1}{A} \left(\frac{1}{R_i} + \frac{1}{R_F} \right) + \frac{1}{R_F} \right) = -\frac{V_i}{R_i}$$

multiply R_F on both sides

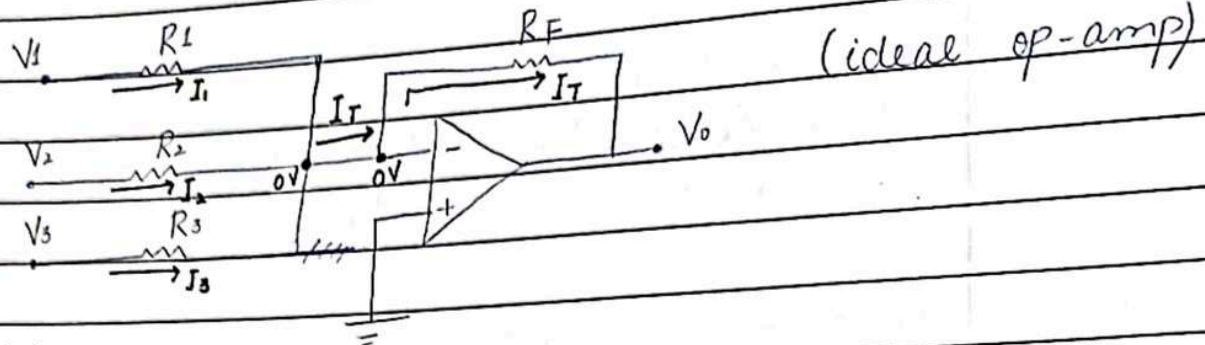
$$V_o \left(\frac{1}{A} \left(\frac{R_F}{R_i} + 1 \right) + 1 \right) = -\frac{R_F}{R_i} \times V_i$$

$$G = \frac{V_o}{V_{in}} = \frac{-R_F/R_i}{1 + 1/A(1 + (R_F/R_i))}$$

gain of an inverting amplifier

Date: 10/10

IMPLEMENTING ADDITION OF MULTIPLE SIGNALS WEIGHTED SUMMING AMPLIFIER



$$I_1 = V_1 / R_1$$

$$I_2 = V_2 / R_2$$

$$I_3 = V_3 / R_3$$

$$V_0 = -(V_1 + V_2 + V_3)$$

$$= -(V_1 + V_2 + V_3)$$

$$I_T = I_1 + I_2 + I_3$$

$$I_T = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \quad (i)$$

↓
remove -ve by
adding another
inverting amplifier

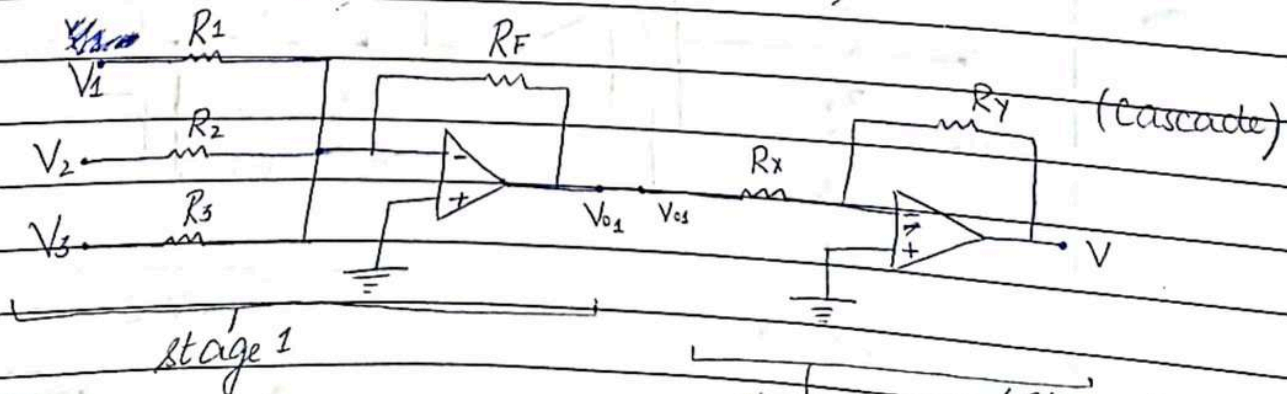
$$0 - V_0 = R_F I_T$$

$$V_0 = -I_T R_F \quad (ii)$$

Substitute (i) in (ii)

$$V_0 = -R_F \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

$$V_0 = - \left(\left[\frac{R_F}{R_1} \right] V_1 + \left[\frac{R_F}{R_2} \right] V_2 + \left[\frac{R_F}{R_3} \right] V_3 \right)$$



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To remove -ve sign (stage 2)

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$$V_{01} = - \left[\left(\frac{R_F}{R_1} \right) V_1 + \left(\frac{R_F}{R_2} \right) V_2 + \left(\frac{R_F}{R_3} \right) V_3 \right] \quad (iii)$$

$$V_0 = - \left(\frac{R_Y}{R_X} \right) V_{01} \quad (iv)$$

Substitute (iii) in (iv)

$$V_0 = + \left(\frac{R_Y}{R_X} \right) \left(\frac{R_F}{R_1} \right) V_1 + \left(\frac{R_Y}{R_X} \right) \left(\frac{R_F}{R_2} \right) V_2 + \left(\frac{R_Y}{R_X} \right) \left(\frac{R_F}{R_3} \right) V_3$$

IMPLEMENTING ADDITION/SUBTRACTION CIRCUIT.

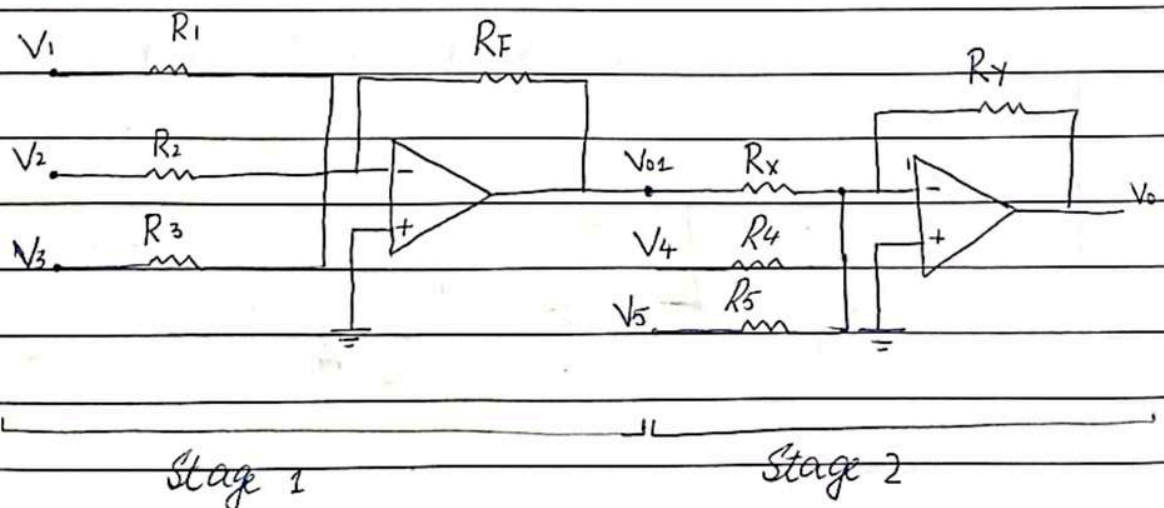
• WEIGHTED SUMMING AMPLIFIER:

$$V_0 = \alpha V_1 + \beta V_2 + \gamma V_3 + \epsilon V_4 + \phi V_5 \quad (i)$$

→ In case of addition & subtraction.

- 1) ~~all these~~ We use a two stage amplifier design
- 2) All those voltages that have to be added will be applied at the inverting terminal of stage 1.
- 3) All those I/p that have to be subtracted will be applied at the inverting terminal of stage 2 of op-amp.

use eq (i)



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$$V_{01} = -\left(\frac{R_F}{R_1}\right)V_1 - \left(\frac{R_F}{R_2}\right)V_2 - \left(\frac{R_F}{R_3}\right)V_3 \quad (ii)$$

$$V_0 = -\left(\frac{R_Y}{R_X}\right)V_{01} - \left(\frac{R_Y}{R_4}\right)V_4 - \left(\frac{R_Y}{R_5}\right)V_5 \quad (iii)$$

Substitute (ii) in (iii)

$$V_0 = -\left(\frac{R_Y}{R_X}\right)\left[-\left(\frac{R_F}{R_1}\right)V_1 - \left(\frac{R_F}{R_2}\right)V_2 - \left(\frac{R_F}{R_3}\right)V_3\right] - \left(\frac{R_Y}{R_4}\right)V_4 - \left(\frac{R_Y}{R_5}\right)V_5$$

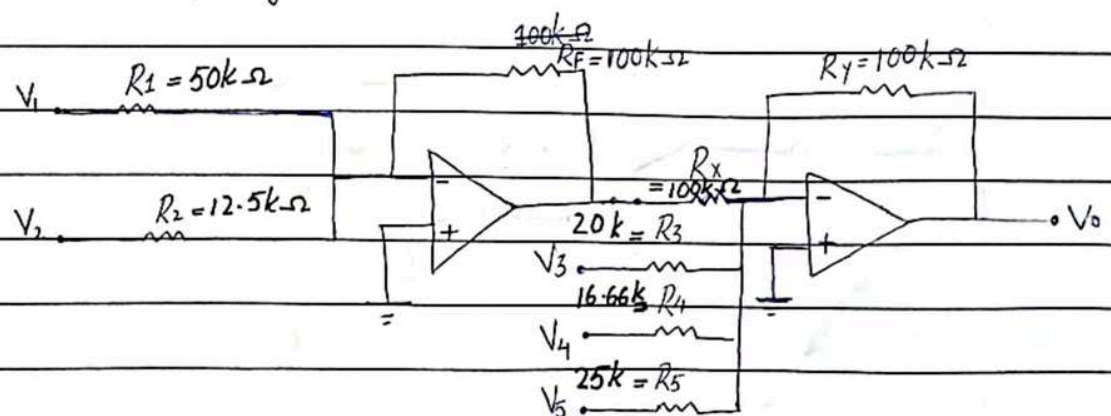
$$V_0 = +\left(\frac{R_Y R_F}{R_X R_1}\right)V_1 + \left(\frac{R_Y R_F}{R_X R_2}\right)V_2 + \left(\frac{R_Y R_F}{R_X R_3}\right)V_3 - \left(\frac{R_Y}{R_4}\right)V_4 - \left(\frac{R_Y}{R_5}\right)V_5$$

$$+ \alpha V_1 \quad + \beta V_2 \quad + \gamma V_3 \quad - \epsilon V_4 \quad - \phi V_5 \quad (\text{eq. (i)})$$

Q. Design a circuit that performs this eq.

$$V_0 = +2V_1 + 8V_2 - 5V_3 - 6V_4 - 4V_5$$

$$R_F = 100k\Omega \quad (\text{only})$$

* Use same resistance resistors in R_X & R_Y .

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$$= \left(\frac{R_Y R_F}{R_X R_1} \right) V_1 + \left(\frac{R_Y R_F}{R_X R_2} \right) V_2 - \left(\frac{R_Y R_F}{R_X R_3} \right) V_3 - \left(\frac{R_Y R_F}{R_X R_4} \right) V_4 - \left(\frac{R_Y R_F}{R_X R_5} \right) V_5$$

$$\frac{R_Y}{R_5} = 4$$

$$R_5$$

$$100k = 4 \Rightarrow R_5 = 25k\Omega$$

$$R_5$$

$$R_3 = 20k$$

$$\left(\frac{R_Y}{R_3} \right) = 5 \Rightarrow R_3 = 100k/5 = 20k$$

$$R_4 = 16.66k\Omega$$

$$\frac{R_Y \times R_F}{R_X R_2} = 8$$

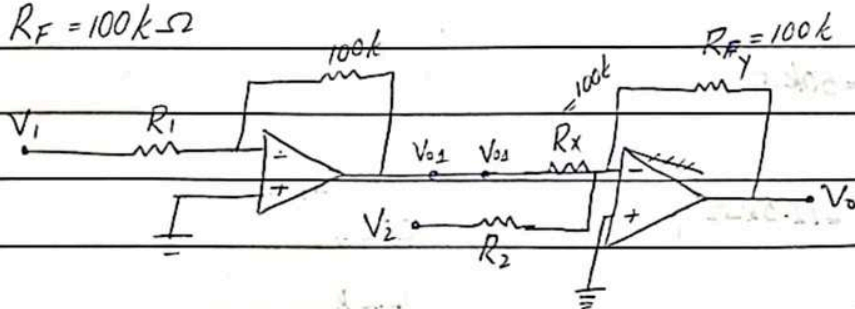
$$\frac{100k \times 100k}{100k R_2} = 8$$

$$R_2 = 12.5k\Omega$$

$$R_1 = 50k\Omega$$

Q. $V_0 = +V_2 - V_1$

$$R_F = 100k\Omega$$



$$R_1 = 100k\Omega$$

$$R_2 = 100k\Omega$$

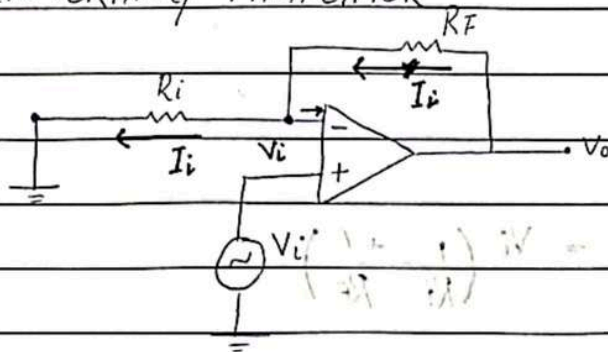
$$\left(\frac{R_Y}{R_X} \right) \left(\frac{R_F}{R_1} \right) = 1 \Rightarrow \frac{100k}{100k} \times \frac{100k}{R_1} = 1 \Rightarrow R_1 = 100k$$

* [voltage connected on (+) of op-amp $\rightarrow$ non inverting
ground on (-) $\rightarrow$ inverting]

Date: _____

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NON-INVERTING AMPLIFIER



Case (1)

$$A \rightarrow \infty$$

$$V_o = A(V_+ - V_-)$$

$$V_o = A(V_+ - V_-)$$

$$A$$

$$\therefore A \rightarrow \infty \Rightarrow V_+ - V_- = V_o / \infty$$

$$V_+ = V_-$$

$$V_+ = V_i$$

$$V_- = V_f = V_i$$

$$\frac{V_i - 0}{R_i} + \frac{V_i - V_o}{R_f} = 0$$

$$\frac{V_i}{R_i} + \frac{V_i}{R_f} = \frac{V_o}{R_f}$$

$$V_i \left(\frac{1}{R_i} + \frac{1}{R_f} \right) = \frac{V_o}{R_f}$$

$$V_o = \left(\frac{R_f}{R_i} + \frac{R_f}{R_f} \right) V_i$$

$$\boxed{\frac{V_o}{V_i} = 1 + \frac{R_f}{R_i}}$$

$$I_i = \frac{V_i - 0}{R_i} = \frac{V_i}{R_i}$$

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$$\frac{V_o - V_i}{R_F} = I_i$$

$$\frac{V_o}{R_F} - \frac{V_i}{R_F} = \frac{V_i}{R_i}$$

$$\frac{V_o}{R_F} = \frac{V_i}{R_i} + \frac{V_i}{R_F} = V_i \left(\frac{1}{R_i} + \frac{1}{R_F} \right)$$

Case-2:

$A \rightarrow \text{finite}$

$$V_o = A (V_+ - V_-)$$

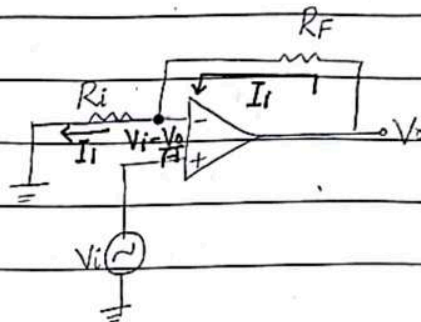
$$V_+ - V_- = \frac{V_o}{A}$$

$$\Rightarrow V_- = V_+ - \frac{V_o}{A} \quad (i)$$

∇

$$V_+ = V_i$$

$$V_- = \frac{V_i - V_o}{A}$$



$$I_i = \frac{V_i - (V_o/A) - 0}{R_i}$$

$$I_i = \frac{V_i}{R_i} - \frac{V_o}{A R_i}$$

$$\left[\frac{(V_i - \frac{V_o}{A}) - V_o}{R_F} \right] = I_i$$

$$\frac{V_o}{A} - \left(\frac{V_i - V_o}{A} \right) = I_i R_F$$

$$\frac{V_o}{A} - \frac{V_i}{A} + \frac{V_o}{A} = R_F \left(\frac{V_i}{R_i} - \frac{V_o}{A R_i} \right)$$

$$\frac{V_o - V_i + V_o}{A} = \left(\frac{R_F}{R_i} \right) V_i - \left(\frac{R_F}{R_i} \right) \left(\frac{V_o}{A} \right)$$

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$$V_o + \frac{V_o}{A} + \left(\frac{R_F}{R_i}\right)\left(\frac{V_o}{A}\right) = V_i + \left(\frac{R_F}{R_i}\right)V_i$$

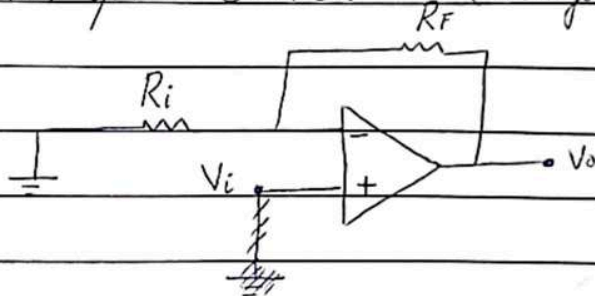
$$V_o \left(1 + \frac{1}{A} + \frac{1}{A} \left(\frac{R_F}{R_i}\right)\right) = V_i \left(1 + \frac{R_F}{R_i}\right)$$

$$V_o \left(1 + \frac{1}{A} \left(1 + \frac{R_F}{R_i}\right)\right) = V_i \left(1 + \frac{R_F}{R_i}\right)$$

*

$$\frac{V_o}{V_i} = \frac{1 + (R_F/R_i)}{1 + \frac{1}{A} (1 + (R_F/R_i))}$$

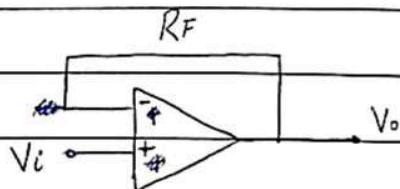
UNITY GAIN BUFFER: (Voltage follower)



$$\frac{V_o}{V_i} = 1 + \frac{R_F}{R_i}$$

(s.c) $R_F = 0 \Rightarrow V_o/V_i = 1$

(o.c) $R_i = \infty \Rightarrow V_o/V_i = 1$



(voltage gain follower)

$$V_o = A(V_+ - V_-)$$

$$V_+ - V_- = V_o/A$$

$$\therefore A \rightarrow \infty \Rightarrow V_+ - V_- = V_o/\infty \Rightarrow V_+ - V_- = 0$$

$$V_+ = V_-$$

$$V_+ = V_i$$

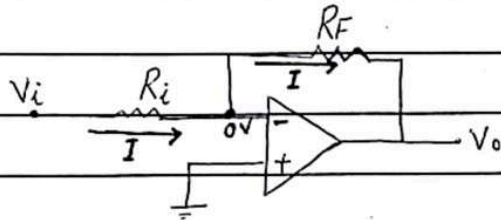
$$V_- = V_+ = V_i$$

$$V_- = V_o = V_i \Rightarrow \frac{V_o}{V_i} = 1$$

Date: 12/9

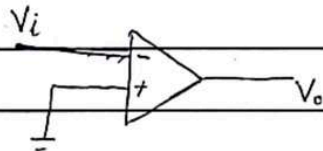
ELECTRONIC CIRCUIT DESIGN

A SPECIAL CASE OF INVERTING AMPLIFIER:



Here,
(Circuit resistance = R_i)

$$G = \frac{V_o}{V_i} = -\left(\frac{R_F}{R_i}\right)$$



(infinite resistance)

PROBLEM:

(use 1st circuit)

if $G = -1000$,
 $R_i = 100\text{ k}\Omega$
 $\Rightarrow R_F = G \times R_i = 100\text{ M}\Omega$
 (not possible)

$G = -1000$
 $R_F = 1\text{ M}\Omega$
 $\Rightarrow R_i = R_F / G = 1000\Omega$

OBJECTIVE:

- $\rightarrow R_i$ (no compromise) as large as possible
- $\rightarrow R_F$ (practical (feasible)) (as small as possible)
- $\rightarrow$ No compromise on gain

$$I = V_i / R$$

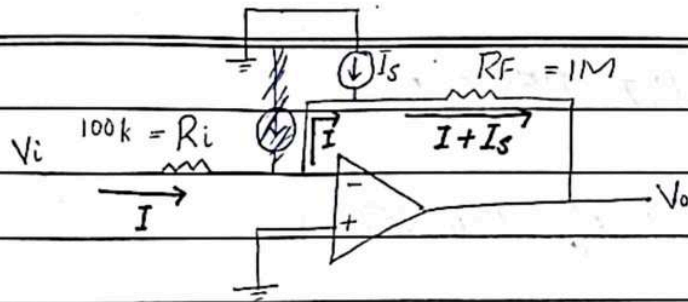
$$0 - V_o = I R_F$$

$$\frac{-V_o}{R_F} = I$$

$$V_o = -I R_F$$



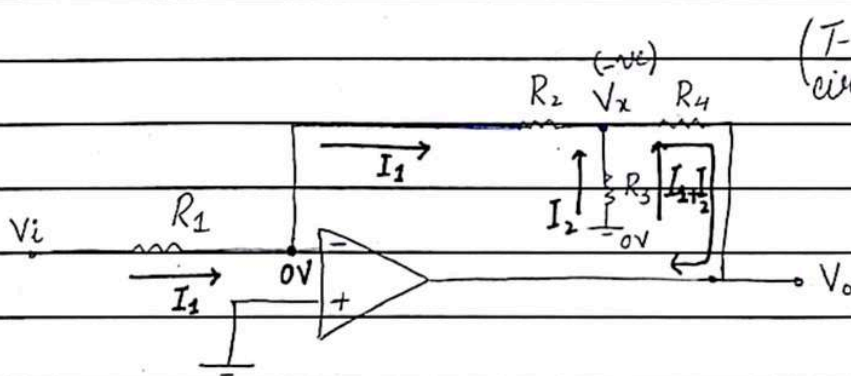
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* output also depends on current

$$V_o = -(I + I_s) R_f$$

Q.



(T-network circuit)

* Current flows from higher to lower potential

$$I_1 = \frac{V_i}{R_1} \quad (i)$$

$$0 - V_x = I_1 R_2$$

$$V_x = -I_1 R_2 \quad (ii)$$

$$0 - V_x = I_2 R_3$$

$$V_x = -I_2 R_3 \quad (iii)$$

$$V_x - V_o = (I_1 + I_2) R_4 \quad (iv)$$

Substitute (i) in (iii)

$$V_x = - \left(\frac{R_2}{R_3} \right) V_i$$



From (iii)

$$V_x = -I_2 R_3$$

$$I_2 = \frac{-V_x}{R_3}$$

$$= \left(\frac{R_2}{R_1 R_3} \right) V_i$$

$$V_x - V_o = (I_1 + I_2) R_4$$

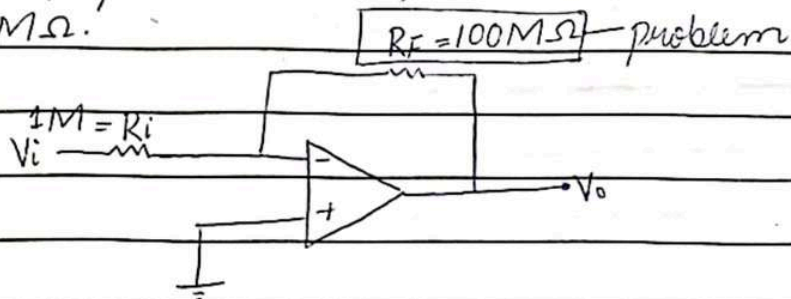
$$-\left(\frac{R_2}{R_1} \right) V_i - V_o = \left(\frac{V_i}{R_1} + \left(\frac{R_2}{R_1 R_3} \right) V_i \right) R_4$$

$$-V_o = \left(\frac{R_2}{R_1} \right) V_i + \left(\frac{R_4}{R_1} \right) V_i + \left(\frac{R_2 R_4}{R_1 R_3} \right) V_i$$

$$-V_o = \left[\frac{R_2}{R_1} + \frac{R_4}{R_1} + \frac{R_2 R_4}{R_1 R_3} \right] V_i$$

$$\frac{V_o}{V_i} = - \left[\frac{R_2}{R_1} + \frac{R_4}{R_1} + \frac{R_2 R_4}{R_1 R_3} \right]$$

Q. Design a normal inverting amplifier of gain $-100 (V/V)$
 $R_i = 1M\Omega$. To improve impracticality Design improved inverting amplifier (T-network). Resistances should be less than or equal to $1M\Omega$.



$$R_F = R_i \times G$$

$$= 100M\Omega$$



Date: _____

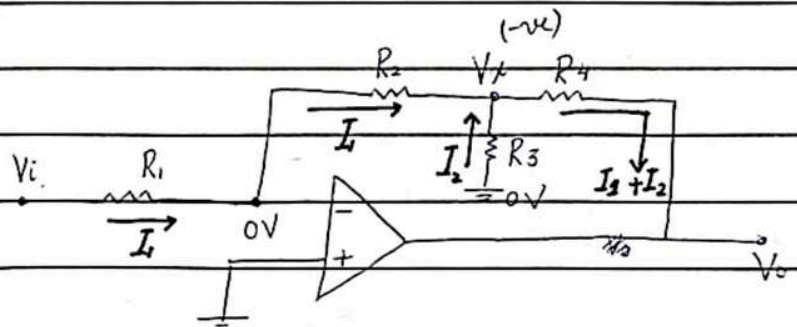
$$\frac{V_o}{V_i} = - \left[\frac{R_2}{R_1} + \frac{R_4}{R_1} + \frac{R_2 R_4}{R_1 R_3} \right]$$

$$+100 = - \left(\frac{1M}{1M} + \frac{1M}{1M} + \frac{1M \times 1M}{1M \times 1M \times R_3} \right)$$

$$100 = 1 + 1 + \frac{1M}{R_3}$$

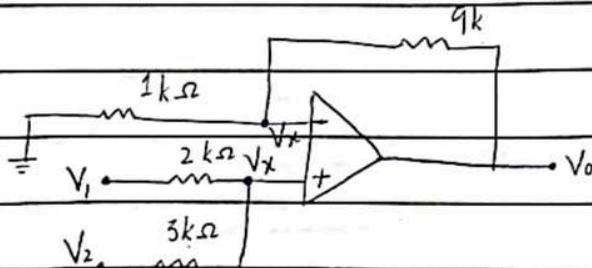
$$R_3 \cdot 98 = 1M$$

$$R_3 = 1.02 \text{ k}\Omega$$



Q.

$$V_o = +6V_1 + 4V_2 - 3V_3$$



closed -ve feedback

$$V_+ = V_- = V_x$$

$$\frac{V_x - V_1}{2k} + \frac{V_x - V_2}{3k} = 0$$

(non-inverting terminal)

(x 6k on bs)

$$3k V_x - 3k V_1 + 2k V_x - 2k V_2 = 0$$

$$5k V_x - 3k V_1 - 2k V_2 = 0$$

$$5V_x = 3V_1 + 2V_2$$

(i)

$$V_x = 3V_1 + 2V_2$$



Date: _____

$$\frac{V_x - 0}{1k} + \frac{V_x - V_0}{9k} = 0$$

(inverting terminal)

$$9V_x + V_x - V_0 = 0$$

$$10V_x = V_0 \quad (ii)$$

Put (i) in (iii)

$$V_0 = 10 \left(\frac{3V_1 + 2V_2}{5} \right)$$

$$\boxed{V_0 = 6V_1 + 4V_2}$$

$$\frac{V_x - V_3}{1k} + \frac{V_x - V_0}{9k} = 0$$

remove ground from inverting⁽⁻⁾ terminal and a voltage source (V_3) is connected.

$$9V_x - 9V_3 + V_x - V_0 = 0$$

$$V_x = V_0$$

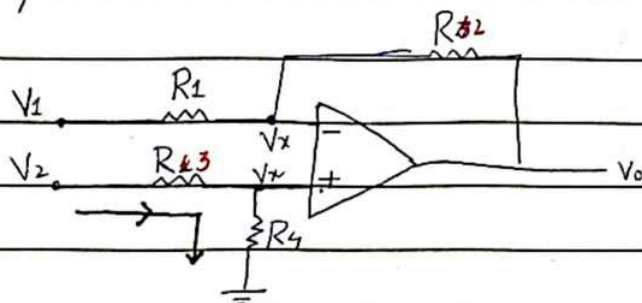
$$10V_x - 9V_3 - V_0 = 0$$

$$V_0 = 10V_x - 9V_3$$

$$V_0 = 10 \left(\frac{3V_1 + 2V_2}{5} \right) - 9V_3$$

$$V_0 = 6V_1 + 4V_2 - 9V_3$$

SINGLE OP-AMP DIFFERENCE AMPLIFIER :



Date: _____

$$V_+ = V_-$$

$$\frac{V_x - V_2}{R_3} + \frac{V_x - 0}{R_4} = 0$$

$$\frac{V_x - V_2}{R_3} + \frac{V_x}{R_4} = 0$$

$$V_x \left(\frac{1}{R_3} + \frac{1}{R_4} \right) = \frac{V_2}{R_3}$$

$$V_x \left(\frac{R_4 + R_3}{R_3 R_4} \right) = \frac{V_2}{R_3}$$

$$V_x = \left(\frac{R_4 + R_3}{R_3 R_4} \right) \left(\frac{V_2}{R_3} \right)$$

$$V_x = \left(\frac{R_4}{R_3 + R_4} \right) \left(\frac{V_2}{R_3} \right) \quad (i)$$

$$\frac{V_x - V_1}{R_1} + \frac{V_x - V_0}{R_2} = 0$$

$$\frac{V_x}{R_1} - \frac{V_1}{R_1} + \frac{V_x}{R_2} - \frac{V_0}{R_2} = 0$$

$$\frac{V_0}{R_2} = V_x \left(\frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_1}{R_1}$$



Date: _____

$$\frac{V_o}{R_2} = \left(\frac{R_1 + R_2}{R_1 R_2} \right) V_x - \frac{V_1}{R_1}$$

$$V_o = \frac{R_1 + R_2}{R_1} \times V_x - \frac{V_1 R_2}{R_1}$$

$$V_o = \left(1 + \frac{R_2}{R_1} \right) V_x - \left(\frac{R_2}{R_1} \right) V_1 \quad (ii)$$

Put (i) in (ii)

$$V_o = \left(1 + \frac{R_2}{R_1} \right) \left(\frac{R_4}{R_3 + R_4} \right) V_2 - \left(\frac{R_2}{R_1} \right) V_1$$

$$V_o = \alpha V_2 - \beta V_1$$

(Presently) } Problem

$$V_o = \beta (V_2 - V_1)$$

(expected)

Objective:

Sol.

$$\alpha = \beta$$

$$\left(1 + \frac{R_2}{R_1} \right) \left(\frac{R_4}{R_3 + R_4} \right) = \frac{R_2}{R_1}$$

$$\frac{R_4}{R_3 + R_4} + \frac{R_2}{R_1} \times \frac{R_4}{R_3 + R_4} = \frac{R_2}{R_1}$$

$$\frac{R_4}{R_3 + R_4} = \frac{R_2}{R_1} - \frac{R_2}{R_1} \left(\frac{R_4}{R_3 + R_4} \right)$$

$$\frac{R_4}{R_3 + R_4} = \frac{R_2}{R_1} \left(1 - \frac{R_4}{R_3 + R_4} \right)$$



Date: _____

$$\frac{R_4}{R_3 + R_4} = \frac{R_2}{R_1} \left(\frac{R_3}{R_3 + R_4} \right)$$

$$R_4 = \frac{R_2}{R_1} \times R_3$$

$$\star \quad \frac{R_4}{R_3} = \frac{R_2}{R_1}$$

$$\text{or} \quad \begin{aligned} R_4 &= R_2 \\ R_3 &= R_1 \end{aligned}$$

$$\text{Apply: } \frac{R_4}{R_3} = \frac{R_2}{R_1}$$

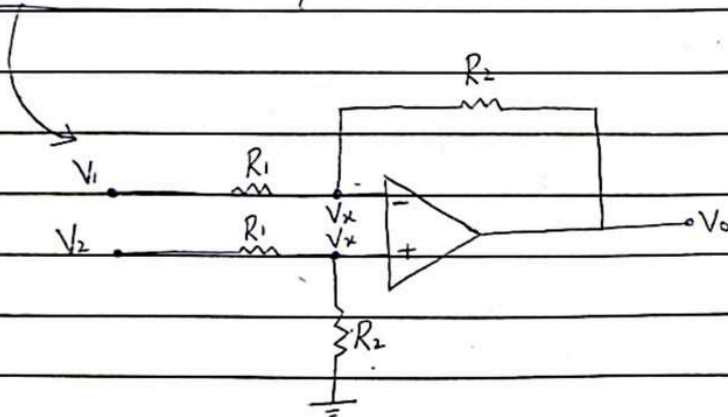
extra

$$\left(\frac{1+R_2}{R_1} \right) \times R_2$$

$$\frac{R_1/R_2 \times R_2}{R_1}$$

$$R_2/R_1$$

$$V_o = \frac{R_2}{R_1} (V_2 - V_1)$$

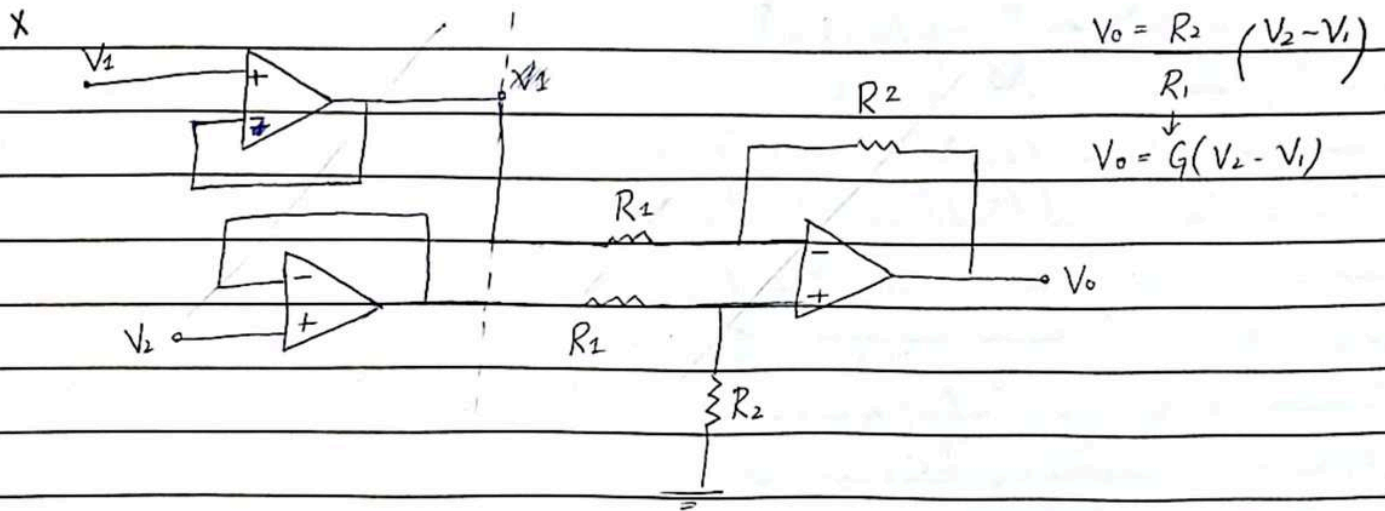


PROBLEMS

- (1) → if R_i would increase ^{then} gain becomes impractically large.
 → R_f should be feasible.
 → no compromise on gain.
- (2) R_1 & R_2 change precisely for circuit to ~~be~~ work properly.
 (according to the ratio)



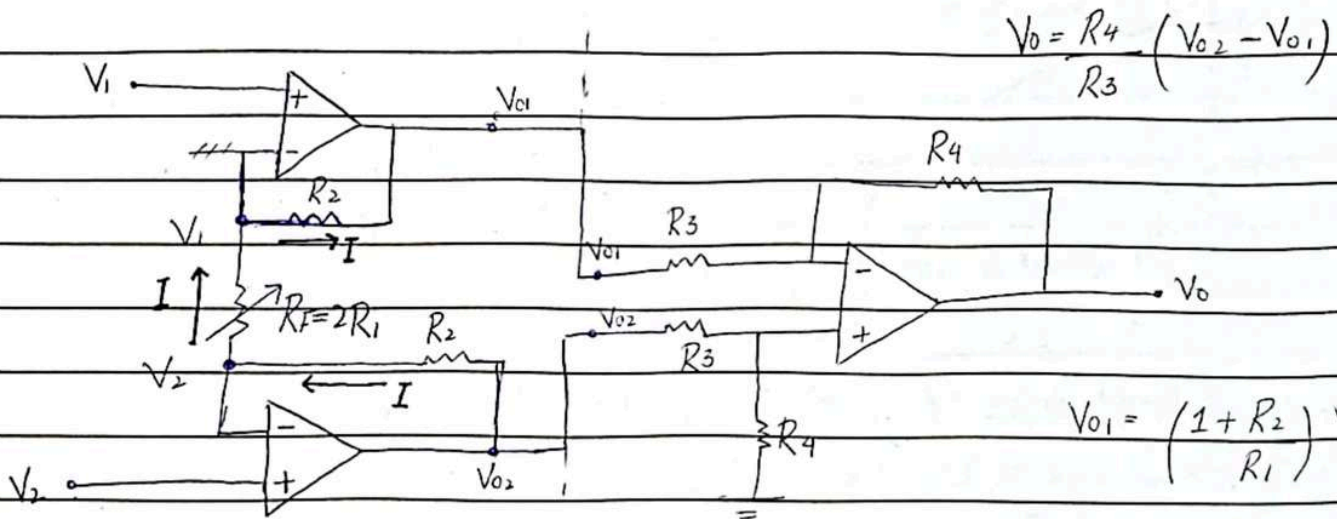
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$$V_0 = \frac{R_2}{R_1} (V_2 - V_1)$$

$$V_0 = G(V_2 - V_1)$$

INSTRUMENTATION AMPLIFIER:



$$V_0 = \frac{R_4}{R_3} (V_{02} - V_{01})$$

$$V_{01} = \left(1 + \frac{R_2}{R_1}\right) V_1$$

$$V_{02} = \left(1 + \frac{R_2}{R_1}\right) V_2$$

$$I = \frac{V_2 - V_1}{2R_1}$$

$$V_{02} - V_{01} = I(R_2) + I(2R_1) + I(R_2)$$

$$V_{02} - V_{01} = I(2R_2 + 2R_1)$$

$$V_{02} - V_{01} = \left(\frac{V_2 - V_1}{2R_1}\right)(2R_2 + 2R_1)$$

$$V_{02} - V_{01} = \left(1 + \frac{2R_2}{2R_1}\right)(V_2 - V_1)$$



Date: _____

$$V_o = \frac{R_4}{R_3} (V_{o2} - V_{o1})$$

* To increase:
increase R_2

$$V_o = \left(\frac{R_4}{R_3} \right) \left(1 + \frac{2R_2}{2R_1} \right) (V_2 - V_1)$$

$$= \underbrace{\left(\frac{R_4}{R_3} \right) \left(1 + \frac{2R_2}{2R_1} \right)}_{\text{Gain}} (V_2 - V_1)$$

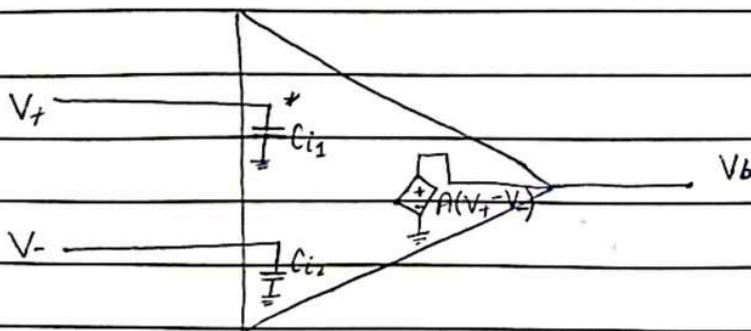
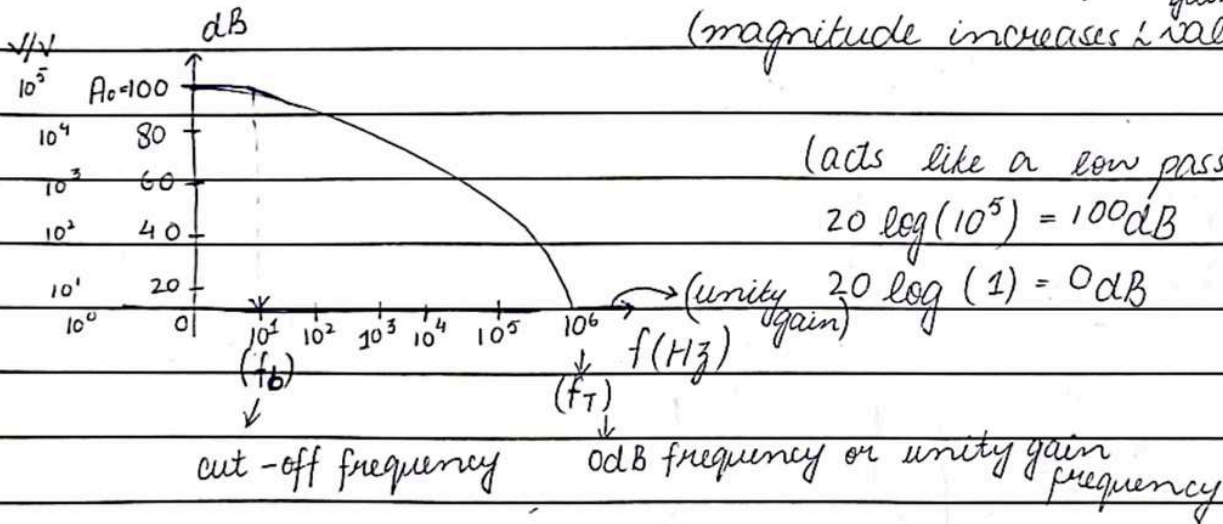
* To decrease
increase R_N

Gain

19/9

FREQUENCY DEPENDENCE OF OPEN-LOOP OP-AMP GAIN:

(magnitude increases & ^{gain} value decreases)



* if there no capacitor in the op-amp it will become an ideal op-amp.

ideal: $A(s) = A_o$ (constant gain throughout)

real: $A(s) = \frac{A_o}{1 + (s/\omega_b)}$ (i)

$= \frac{A_o}{1 + (s/\omega_b)} \times A_o$ (LPF) $\Rightarrow$

* Transfer function ($T(s)$):

• low pass = $\frac{\omega_b}{s + \omega_b}$ (take ω_b common
= 1
 $1(s/\omega_b) + 1$)

• high pass = $\frac{s}{s + \omega_b}$

$\Rightarrow$ As, $s = j\omega$

$\Rightarrow$ (i) $A(\omega) = \frac{A_0}{1 + j(\omega/\omega_b)}$

imaginary - depends on frequency itself

For large frequencies $\Rightarrow$ at large values of ' ω ' , $1 \ll \omega/\omega_b$

$A(\omega) = \frac{A_0}{j(\omega/\omega_b)}$

$A(\omega) = \frac{A_0 \omega_b}{j\omega}$

* $|A(\omega)| = \frac{A_0 \omega_b}{\omega}$ (magnitude of ^{open loop} amplifier gain) at any frequency ' ω '

Q. What is the amplifier open loop gain at unity gain frequency?
 $|A(\omega)| = 1$

Q. What is ^{f_T or ω_T} unity gain frequency?

$1 = \frac{A_0 \omega_b}{\omega_T} \Rightarrow \omega_T = A_0 \omega_b$

(frequency of ~~the~~ open loop gain)



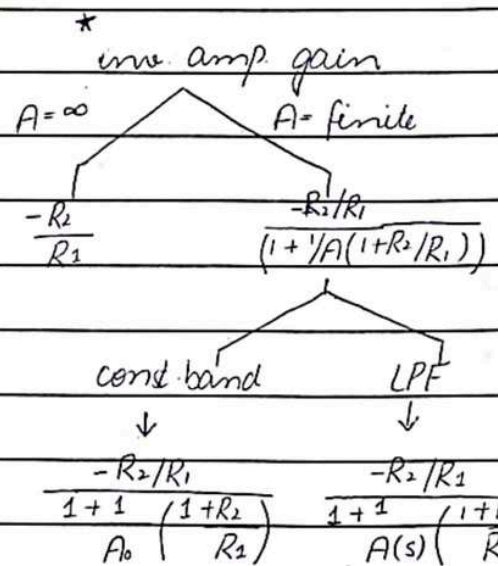
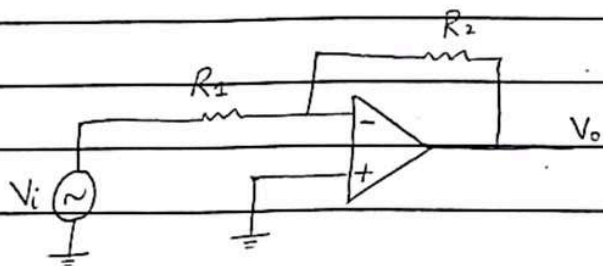
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$$\begin{aligned} f_T &= A_o f_b \\ &= 10^5 \times 10^1 \\ &= 10^6 \text{ Hz} \end{aligned}$$

$$\begin{aligned} \omega &= 2\pi f \\ *|A(\omega)| &\propto 1/\omega \end{aligned}$$

FREQUENCY DEPENDENCE OF CLOSED LOOP OP-AMP:

INVERTING AMPLIFIER:



$$G = \frac{V_o}{V_i} = \frac{-R_2/R_1}{1 + (1/A(s)(1 + R_2/R_1))}$$

$$\frac{V_o}{V_i} = \frac{-R_2/R_1}{1 + \left[\frac{(1 + R_2/R_1)}{A_o / (1 + (s/\omega_b))} \right]}$$

$$= \frac{-R_2/R_1}{1 + (1 + (R_2/R_1))((1 + s/\omega_b))/A_o}$$

$$= \frac{-R_2/R_1}{1 + (1 + R_2/R_1)(1/A_o + s/A_o \omega_b)}$$

$$= \frac{-R_2/R_1}{1 + 1/A_o (1 + (R_2/R_1)) + (s/A_o \omega_b) (1 + (R_2/R_1))}$$



Date: _____

$$V_o = \frac{-R_2/R_1}{1 + (1/A_o)(1 + (R_2/R_1)) + s / ((A_o \omega_b) / (1 + (R_2/R_1)))}$$

$$A_o = 10^5 \text{ (very large)}$$

$$\therefore \frac{V_o(s)}{V_i(s)} = \frac{-R_2/R_1}{1 + (s / ((A_o \omega_b) / (1 + (R_2/R_1))))}$$

* CLOSED LOOP OF OP-AMP (INVERTING)

$$G(s) = \frac{-R_2/R_1}{1 + (s / ((A_o \omega_b) / (1 + (R_2/R_1))))}$$

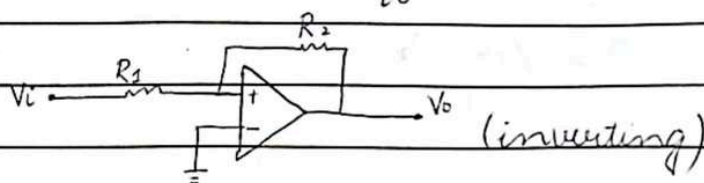
$$\omega_b' = \frac{A_o \omega_b}{1 + (R_2/R_1)} = \omega_R$$

cut-off frequency ω_R

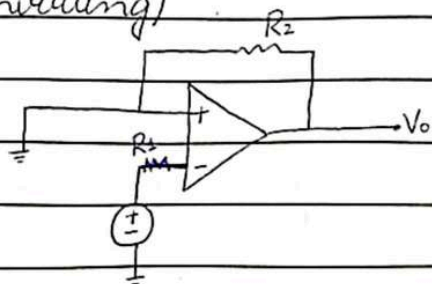
$$|G(\omega)| = \frac{(R_2/R_1)(A_o \omega_b)}{1 + (R_2/R_1)}$$

$$\omega_T' = \left(\frac{A_o \omega_b}{1 + (R_2/R_1)} \right) \left(\frac{R_2}{R_1} \right) = \omega_z$$

unity gain frequency



(non-inverting)



$$\frac{1 + R_2/R_1}{1 + 1/A (1 + R_2/R_1)}$$

$$G = \frac{1 + R_2/R_1}{1 + 1/A(s) (1 + R_2/R_1)} = \frac{1 + R_2/R_1}{1 + \frac{(1 + R_2/R_1)}{A_o / (1 + s/\omega_b)}}$$



Date: _____

$$= 1 + R_2/R_1$$

$$1 + (1 + R_2/R_1) (1/A_0 + s/A_0 \omega_b)$$

$$= 1 + R_2/R_1$$

$$1 + 1/A_0 (1 + R_2/R_1) + s/A_0 \omega_b (1 + R_2/R_1)$$

$$= 1 + R_2/R_1$$

$$1 + s / ((A_0 \omega_b) / (1 + (R_2/R_1)))$$

$$= 1 + R_2/R_1$$

$$1 + s / ((A_0 \omega_b) / (1 + (R_2/R_1)))$$

$$= G_0$$

$$1 + (s/\omega_k)$$

$$\therefore G_0 = 1 + R_2/R_1$$

$$; \omega_k = A_0 \omega_b / (1 + (R_2/R_1))$$

$$|G(\omega)| = \frac{G_0 \omega_k}{\omega} = \frac{(1 + (R_2/R_1)) (A_0 \omega_b / (1 + (R_2/R_1)))}{\omega}$$

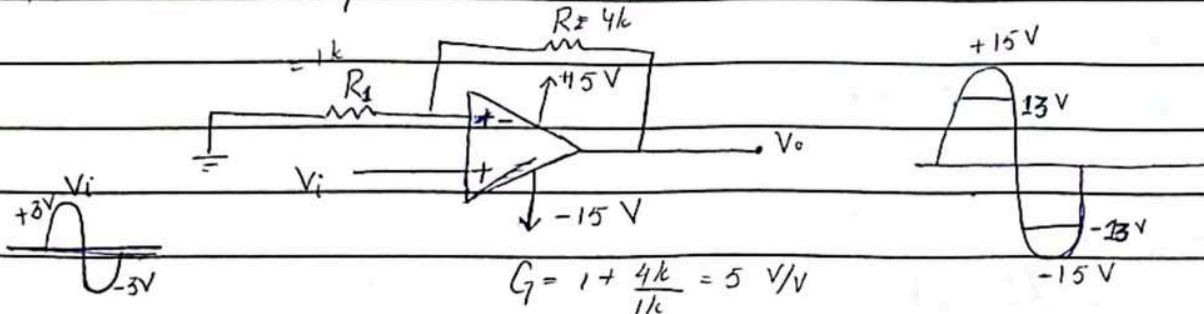
$$\text{cut-off frequency} \Rightarrow \omega_k = \frac{A_0 \omega_b}{1 + R_2/R_1}$$

$$\text{unity gain frequency} \Rightarrow \omega_y = G_0 \omega_k = \left(\frac{1 + R_2}{R_1} \right) \left(\frac{A_0 \omega_b}{1 + R_2/R_1} \right)$$



2.6 DC IMPERFECTIONS OF OP-AMP.

1) OUTPUT VOLTAGE SATURATION.

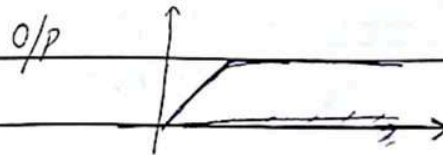
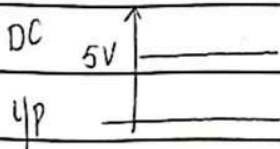
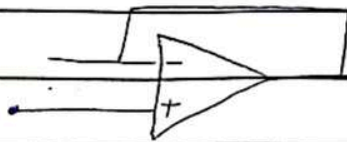


2) OUTPUT CURRENT LIMIT:

* Max. output current limited to $\pm 20 \text{ mA}$.

3) SLEW RATE: (SR)

if the i/p signal applied is changing / varying faster than the value of SR, the op-amp will not comply.



$$v_i = V_m \sin(\omega t)$$

$$\frac{dv_i}{dt} = V_m \cdot \omega \cdot \cos(\omega t)$$

$$SR \geq \omega V_m \quad (\text{safe})$$

$$\left. \frac{dv_i}{dt} \right|_{\max} = V_m \omega$$

$$SR \leq \omega V_m \quad (\text{unsafe})$$

$$SR \geq \left. \frac{dv_i}{dt} \right|_{\max}$$



Date: _____

$$SR = \omega_{\max} \cdot V_{I, \max}$$

$$SR = 2\pi f_{\max} \cdot V_{I, \max}$$

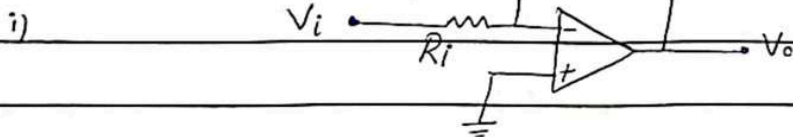
$$f_{\max} = \frac{SR}{2\pi \cdot V_{I, \max}}$$

$$V_{I, \max} = \frac{SR}{2\pi (f_{\max})}$$

26/9 DEPENDENT CURRENT & VOLTAGE SOURCES:

1) VCVS (voltage controlled voltage source)

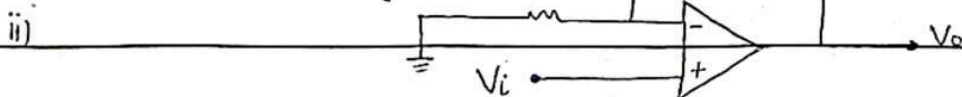
$$V_o = K V_i$$



$$V_o = -\left(\frac{R_F}{R_i}\right) \times V_i$$

(inverting)

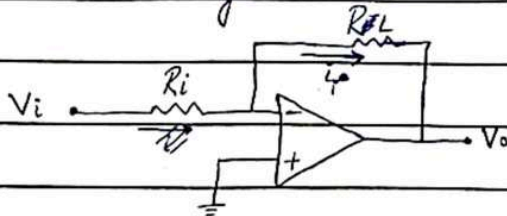
non-inverting:



$$V_o = \left(1 + \frac{R_F}{R_i}\right) \times V_i$$

2) VCCS (voltage controlled current source)

$$i_o = k V_i$$



$$i_i = \frac{V_i}{R_i}, \quad i_o = \frac{V_i}{R_i}$$

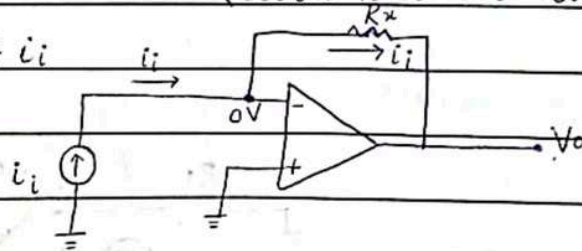
$$\Rightarrow i_o = \left(\frac{1}{R_i}\right) \times V_i$$



3) CCVS

(current controlled voltage source)

$$V_o = k i_i$$

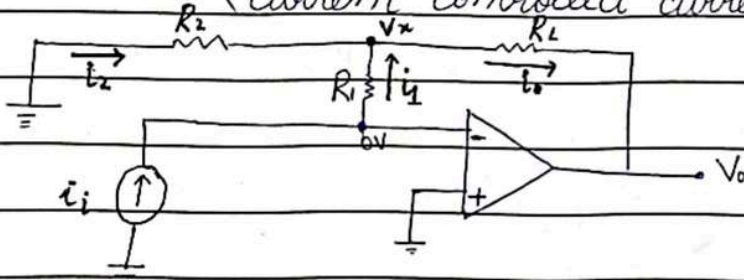


$$i_i = \frac{0 - V_o}{R_x} \Rightarrow i_i = -\frac{V_o}{R_x}$$

$$0 - V_o = (R_x)(i_i) \Rightarrow V_o = (-R_x)(i_i)$$

4) CCCS

(current controlled current source)



$$i_1 = i_i$$

$$i_o = k i_i$$

$$0 - V_x = i_1 R_1 \Rightarrow V_x = (-R_1)(i_1)$$

$$i_1 = (-V_x / R_1)$$

 i_1 in terms of V_x

$$0 - V_x = i_2 R_2$$

$$V_x = -i_2 R_2$$

$$i_2 = \frac{-V_x}{R_2}$$

$$i_1 + i_2 = i_o \quad (iii)$$

$$i_1 = (-V_x / R_1), \quad i_2 = (-V_x / R_2), \quad i_o = i_1 + i_2$$

$$+ i_1 R_1 = - i_2 R_2$$

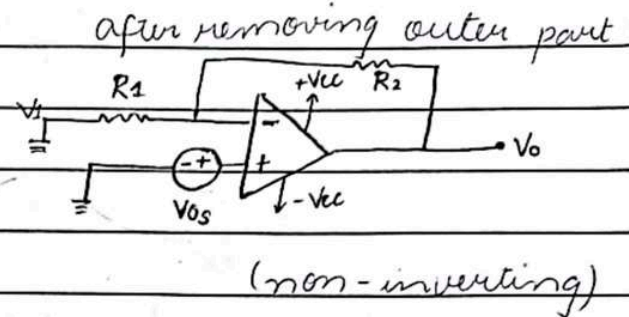
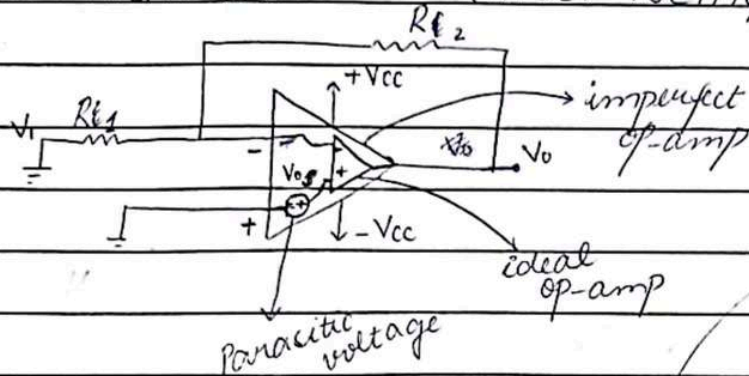
$$i_2 = \left(\frac{R_1}{R_2} \right) i_1 \quad (iv)$$

$$i_1 + \left(\frac{R_1}{R_2} \right) i_1 = i_o \Rightarrow \left(1 + \frac{R_1}{R_2} \right) i_1 = i_o \Rightarrow i_o = \left(\frac{1 + R_1}{R_2} \right) i_i$$

always DC signal

Date: 1/10/24

DC IMPERFECTIONS (OFFSET VOLTAGE)



$$V_o = \left(1 + \frac{R_2}{R_1} \right) V_i$$

$$V_i = V_{os}$$

$$V_o = \left(1 + \frac{R_2}{R_1} \right) V_{os}$$

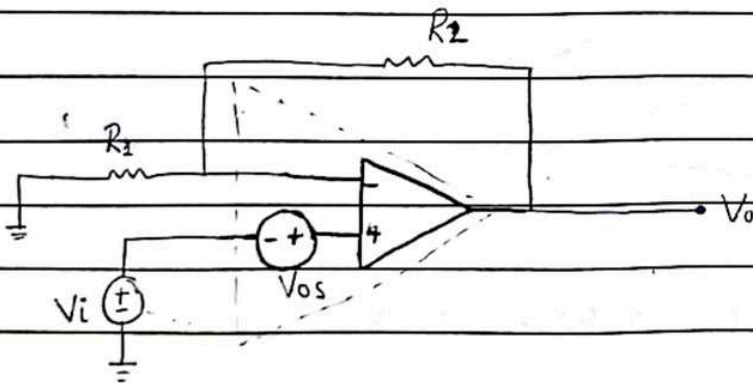
Let,

$$V_{os} = 5 \text{ mV}$$

$$R_2 = 1 \text{ M}\Omega, \quad R_1 = 1 \text{ k}\Omega$$

$$V_o = \left(1 + \frac{1 \text{ M}}{1 \text{ k}} \right) \times 5 \text{ m}$$

$$= 5.05 \text{ V}$$



$$V_o = \left(1 + \frac{R_2}{R_1} \right) (V_i + V_{os})$$

$$= \left(1 + \frac{R_2}{R_1} \right) V_i + \left(1 + \frac{R_2}{R_1} \right) V_{os}$$

$$V_i = 1 \text{ mV}$$

$$, V_{os} = 5 \text{ mV}$$

$$= 1000 (1 \text{ m})$$

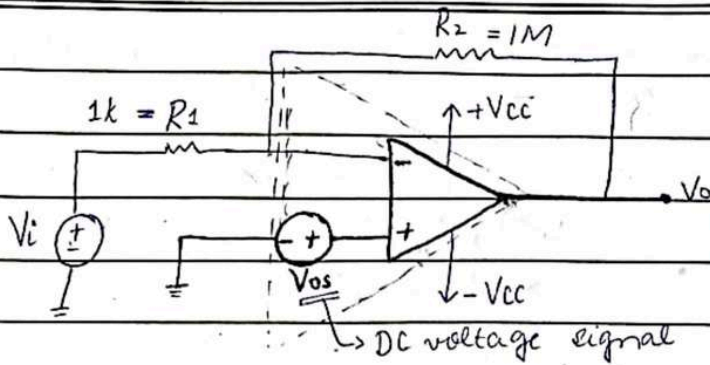
$$, = 1000 \times 5 \text{ m}$$

$$= 1.01 \text{ V}$$

$$= 5.05 \text{ V}$$



Date: _____



$$G = 1 + \frac{R_2}{R_1} = 1000$$

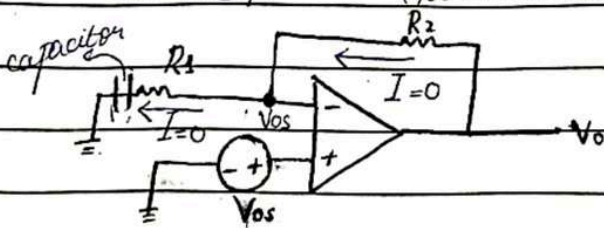
- 1) ignore V_i 2) ignore V_{os}
 ↓ (non-inverting) ↓ (inverting)

$$V_o = \underbrace{\left(1 + \frac{R_2}{R_1}\right) V_{os}}_{\text{draining term}} - \left(\frac{R_2}{R_1}\right) V_i$$

1st REMEDY: (Universal remedy)

- Use a potentiometer at 'offset' pin of ϕ -amp to introduce an equal but opposite voltage in series with V_{os} .
 (Pin 1 $\rightarrow +V_{cc}$, Pin 2 - offset, Pin 3 - ground)

2nd REMEDY: (Partial remedy / cost effective)



* only work if V_i is AC signal

V_{os} — DC signal $\Rightarrow I = DC$

if no current is flowing

$$V_o - V_{os} = IR \quad \therefore I = 0$$

$$V_o - V_{os} = 0$$

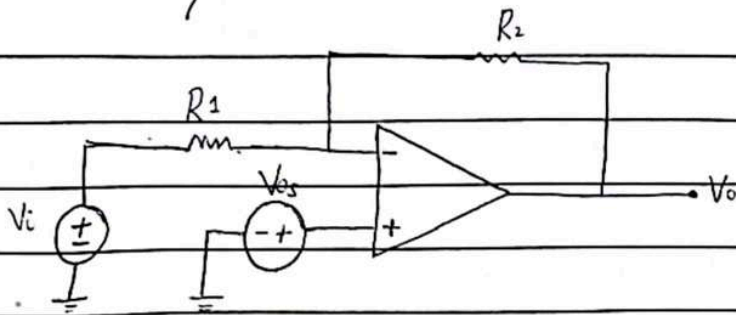
$$V_o = V_{os}$$



(w/o remedy)

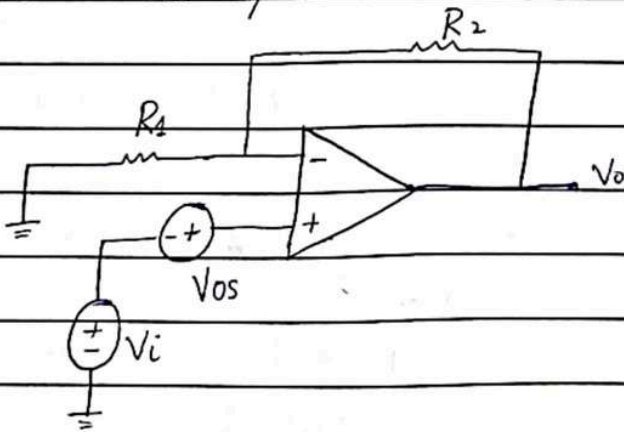
Date: _____

INVERTING WITH V_{os} :



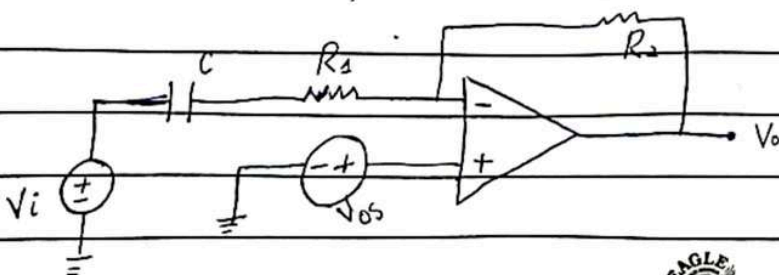
$$V_o = -\left(\frac{R_2}{R_1}\right)V_i + \left(1 + \frac{R_2}{R_1}\right)V_{os}$$

NON-INVERTING WITH V_{os}



$$V_o = \left(1 + \frac{R_2}{R_1}\right)V_i + \left(1 + \frac{R_2}{R_1}\right)V_{os}$$
$$= \left(1 + \frac{R_2}{R_1}\right)(V_i + V_{os})$$

INV. WITH V_{os} & PARTIAL REMEDY:

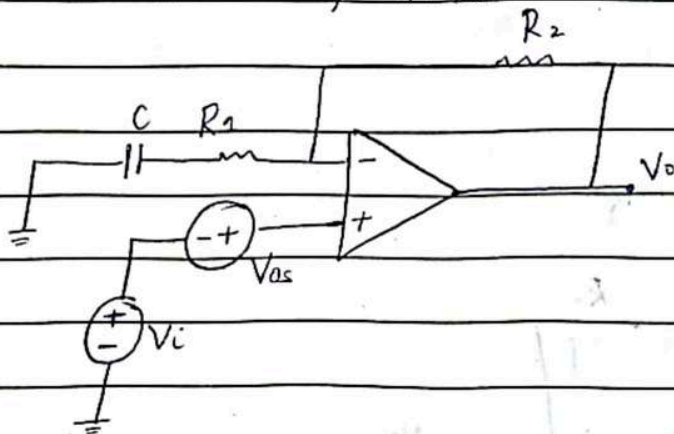


$$V_o = -\left(\frac{R_2}{R_1}\right)V_i + V_{os}$$



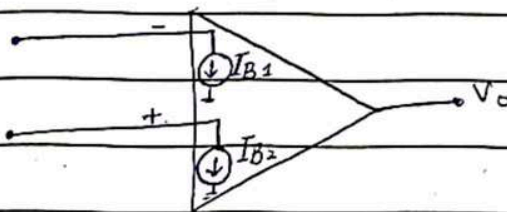
Date: _____

NON-INV. WITH V_{os} & PARTIAL REMEDY:



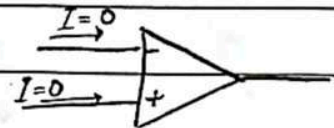
$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_i + V_{os}$$

DC IMPERFECTION (INPUT BIAS CURRENT)



average input bias current $\Rightarrow I_B = \frac{I_{B1} + I_{B2}}{2} \approx 100 \text{ nA}$

Input offset current $\Rightarrow I_{os} = |I_{B1} - I_{B2}| \approx 1 \text{ nA} (\sim 0)$



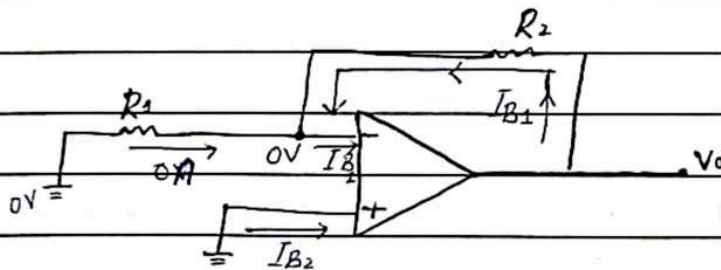
ideal op-amp
 $R_{in} \rightarrow \infty$



Date: _____

* In real op-amp some non-zero currents I_{B1} & I_{B2} tends to flow in the op-amp terminals. ~~Both~~ Both I_{B1} & I_{B2} are nano Amp current.

PROBLEM:



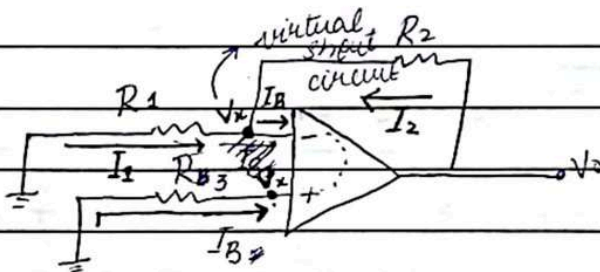
$$*(R_2 \uparrow \quad V_o \uparrow)$$

$$V_o - 0 = I_{B1} \times R_2$$

$$\boxed{V_o = I_{B1} \times R_2} \neq 0$$

$$\therefore I_{B1} = 10 \text{ nA}$$

REMEDY: (Put R_3 at non-inverting)



For simplicity:

$$I_{B1} = I_{B2} = I_B$$

$$0 - V_x = I_B R_{B3}$$

$$V_x = -I_B R_{B3}$$

$$\frac{I_1 \cdot 0 - V_x}{R_1} \Rightarrow I_1 = \frac{+I_B R_{B3}}{R_1}$$



By KCL:

$$I_1 + I_2 = I_B$$

$$I_2 = I_B - I_1$$

$$I_2 = I_B - \left(\frac{R_3}{R_1} \right) I_B$$

$$V_0 - V_X = I_2 R_2$$

$$V_0 = V_X + I_2 R_2$$

$$= -I_B R_3 + I_2 R_2$$

$$V_0 = -I_B R_3 + R_2 \left(I_B - \left(\frac{R_3}{R_1} \right) I_B \right)$$

$$V_0 = I_B \left(-R_3 + R_2 - \frac{R_2 R_3}{R_1} \right)$$

$$V_0 = I_B \left(R_2 - R_3 \left(1 + \frac{R_2}{R_1} \right) \right)$$

Let, Put $V_0 = 0$

$$0 = I_B \left(R_2 - R_3 \left(1 + \frac{R_2}{R_1} \right) \right)$$

$$R_2 - R_3 \left(1 + \frac{R_2}{R_1} \right) = 0$$

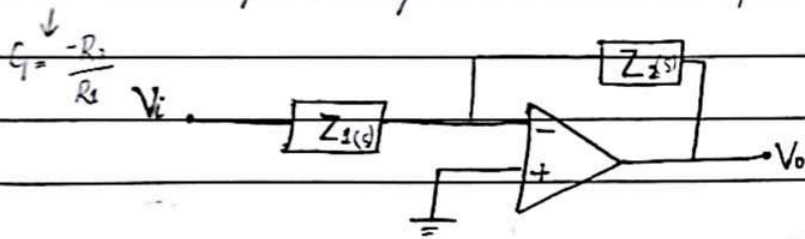
$$R_3 \left(\frac{R_1 + R_2}{R_1} \right) = R_2$$

$$R_3 = \frac{R_1 R_2}{R_1 + R_2} \Rightarrow R_3 = R_1 \parallel R_2$$



Date: 3/10

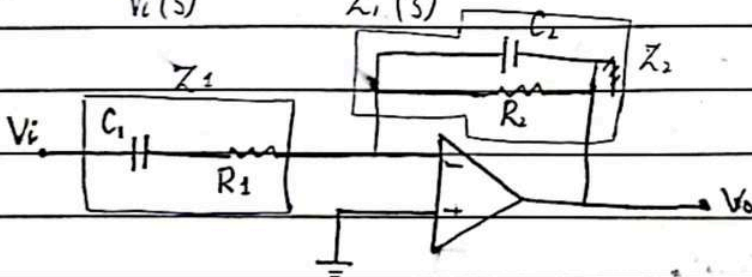
2.5.1 INVERTING CONFIGURATION WITH GENERAL IMPEDENCES



convert Z_1 or Z_2 to s domain
 Z_1 or Z_2 = Inductor / capacitor

Gain function / Transfer function.

$$G(s) = \frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$



$$\frac{V_o(s)}{V_i(s)} = -\frac{R_2 \parallel (1/C_2(s))}{R_1 + (1/C_1(s))}$$

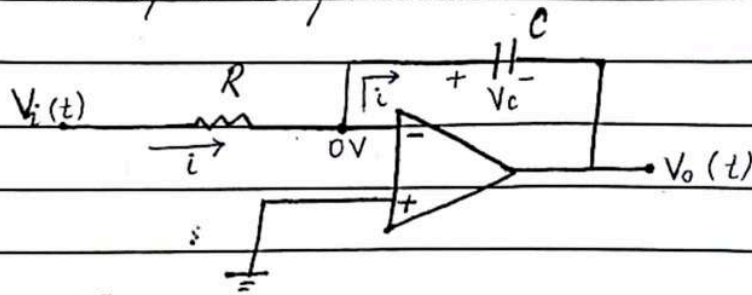
$$= -\frac{R_2 (1/C_2(s))}{R_1 + (1/C_1(s))}$$

$$= -\frac{R_2 / C_2(s)}{(R_1 + (1/C_1(s)))(R_2 + (1/C_2(s)))}$$

MATHEMATICAL OPERATIONS

- Summing
- Difference
- Integration
- Differentiation
- natural log
- log₁₀
- exponential
- multiplication
- division



2.5.2 INVERTING 'INTEGRATOR':

objective
 $V_o \propto \int V_i dt$

~~MESSAGE~~
 $i(t) = \frac{V_i(t)}{R}$

(i)

$$i(t) = C \frac{dv_c}{dt}$$

$$\frac{dv_c}{dt} = \frac{i(t)}{C}$$

$$\int \frac{dv_c}{dt} = \int \frac{i(t)}{C}$$

$$v_c(t) = \frac{1}{C} \int i(t) dt + v_c(0)$$

initially, when unchanged $\Rightarrow v_c(0) = 0$

$$v_c(t) = \frac{1}{C} \int i(t) dt$$

$\xrightarrow{\text{use eq (i)}}$

$$v_c(t) = \frac{1}{C} \int \frac{V_i(t)}{R} dt$$

$$v_c(t) = \frac{1}{RC} \int V_i(t) dt \quad (ii)$$

$$0 - V_o(t) = v_c(t)$$

$$V_o(t) = -v_c(t)$$

$$\Rightarrow (ii) \quad V_o(t) = -\frac{1}{RC} \int V_i(t) dt$$



Date: _____

METHOD-II TRANSFER FUNCTION:

$$\frac{V_o(s)}{V_i(s)} = \frac{-Z_2(s)}{Z_1(s)}$$
$$= \frac{-1/sC}{R}$$

$$C \rightarrow 1/sC$$

$$\frac{V_o(s)}{V_i(s)} = \frac{-1}{RCs} \quad (\text{iii})$$

$$V_o(s) = \frac{-1}{RC} \left(\frac{V_i(s)}{s} \right)$$

$$\begin{aligned} sF(s) &\xrightarrow{L^{-1}} \frac{d}{dt}(f(t)) \\ F(s) &\xrightarrow{L^{-1}} \int f(t) \end{aligned}$$

inverse Laplace (L^{-1}) on both sides

$$V_o(t) = \frac{-1}{RC} \int V_i(t) dt$$

Magnitude & phase:

$\Rightarrow$ (iii)

$$\frac{V_o(s)}{V_i(s)} = \frac{-1}{RCs}$$

$$G(s) = \frac{-1}{RCs}$$

$$s = j\omega$$
$$G(j\omega) = \frac{-1}{RCj\omega}$$

$$= \frac{j}{RC\omega}$$

$$G(\omega) = 0 + j \frac{1}{RC\omega}$$

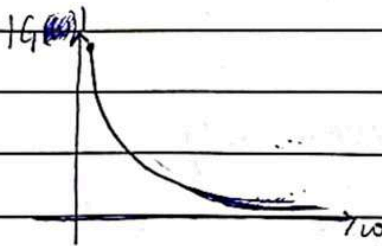


Date: _____

$$|G(\omega)| = \sqrt{0^2 + (1/RC\omega)^2} = 1/RC\omega$$

$$\angle G(\omega) = \tan^{-1} \left(\frac{1/RC\omega}{0} \right)$$

$$\theta = \pi/2 \text{ or } 90^\circ$$



Let,
 $RC=1$

$$\Rightarrow |G| = 1/\omega$$

$\omega = x\text{-axis}$

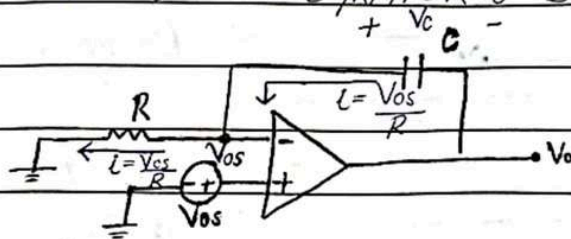
$|G| = y\text{-axis}$

$$\Rightarrow y = 1/x$$

$$G = \frac{1}{RC\omega} \angle 90^\circ$$

(low pass filter)
integrated circuit (removes noise)

2.6.3 IMPACT OF V_{os} ON INTEGRATOR'S BEHAVIOUR (offset voltage)



(virtual short circuit when closed loop).

$$i(t) = C \frac{dV_c}{dt}$$

$$i(t) = \frac{V_{os}}{R}$$

$$V_c = \frac{1}{C} \int i(t) dt$$

$$= \frac{1}{C} \int \frac{V_{os}}{R} dt$$

$$V_c = \frac{1}{RC} \int_0^t V_{os} dt$$



Date: _____

$$V_o - V_{os} = V_c$$

$$V_o = V_c + V_{os}$$

$$V_o = V_{os} + \frac{1}{RC} \int_0^t V_{os} dt$$

$$\left. \frac{1}{RC} \times V_{os} t \right|_0^t$$

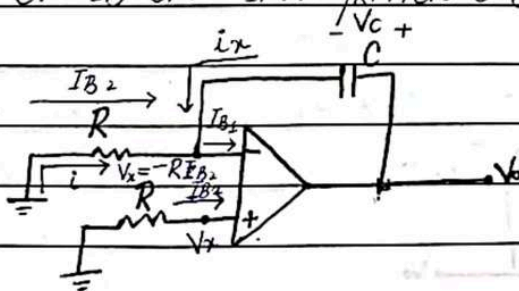
$$V_o = V_{os} + \frac{1}{RC} (V_{os} \times t)$$

$$* V_o = V_{os} + \left(\frac{V_{os}}{RC} \right) t$$

(time dependent)

↓
after a small time the circuit will saturate

IMPACT OF I_B ON INTEGRATOR'S BEHAVIOUR:



* capacitor blocks current (only single valued resistor)

current higher to lower
 $0 \rightarrow -ve$

$$0 - V_x = I_{B2} R$$

$$V_x = -I_{B2} R \quad (i)$$

$$i = \frac{0 - (R I_{B2})}{R}$$

$$i = \frac{R I_{B2}}{R}$$

$$i = I_{B2}$$

$$i_x + I_{B2} = I_{B1}$$

$$i_x = I_{B1} - I_{B2}$$



$$V_c = \frac{1}{C} \int_0^t i_x(t) dt$$

$$V_c = \frac{1}{C} \int_0^t \underbrace{(I_{B1} - I_{B2})}_{\text{constant}} dt$$

$$V_c = \left(\frac{I_{B1} - I_{B2}}{C} \right) \times t \quad (ii)$$

$$V_c = V_o - V_x$$

$$V_o = V_x + V_c$$

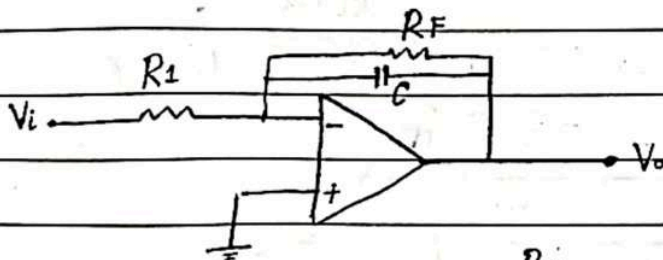
⇒ ~~value of~~ V_c & V_x

$$* V_o = -I_{B2} R + \left(\frac{I_{B1} - I_{B2}}{C} \right) t$$

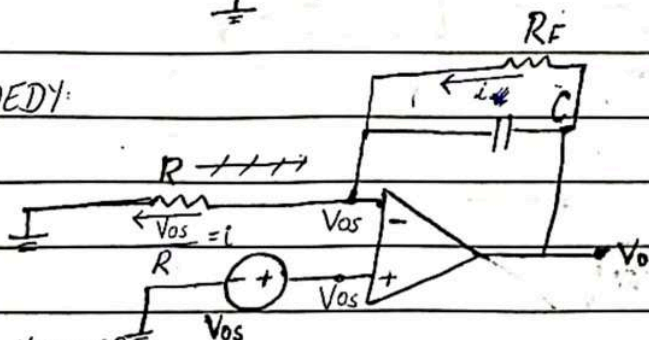
(Bias current)

(INVERTING 'DIFFERENTIATION')

if we want to stop saturation due to effect on any reason



PARTIAL REMEDY:



(b/w capacitor & resistor)
current always goes through R_F because impedance of capacitor is higher

$$V_o - V_{os} = i R_F$$

$$V_o = V_{os} + i R_F$$

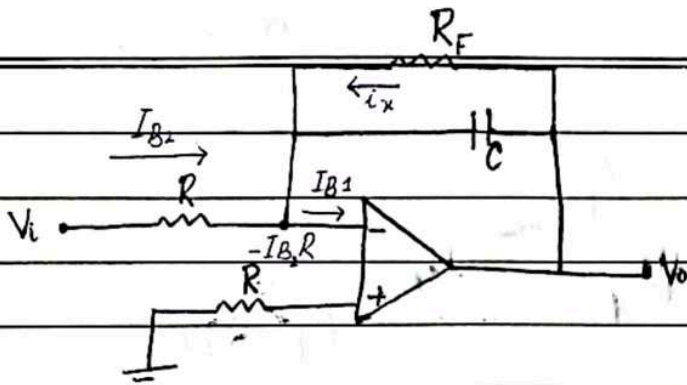
$$V_o = V_{os} + \left(\frac{V_{os}}{R} \right) R_F \Rightarrow V_o = \left(1 + \frac{R_F}{R} \right) V_{os}$$

(not increasing linearly with time)

$$V_o = V_{os}$$



Date: _____



$$V_o = -(-I_{B2}R) = Ri_x$$

$$V_o = +I_{B2}R + Ri_x$$

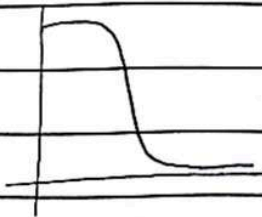
$$* \boxed{V_o = I_{B2}R + R(I_{B1} - I_{B2})}$$

$$G(s) = \frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

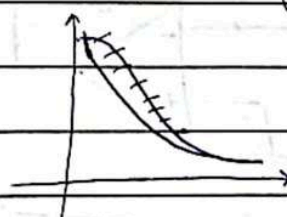
$$= -\frac{(1/Cs) \parallel R_F}{R_1}$$

$$-(1/Cs) \parallel R_F = \frac{(-1/Cs)(R_F)}{(-1/Cs) + (R_F)}$$

$$G(s) = \frac{-R_F/R_1}{1 + s(CR_F)}$$



$$G(s) = \frac{-1}{RC(s)}$$



(ideal low pass filter)

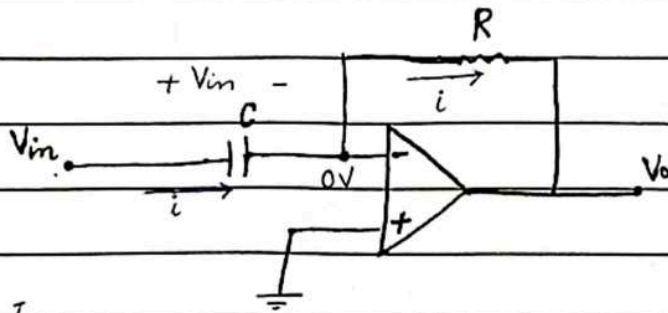


Date: 8/10

OP-AMP DIFFERENTIATOR

* $s \rightarrow t$

~~L~~ Laplace inverse



METHOD-I

$$i = C \frac{dv_c}{dt}$$

$$\therefore \frac{C}{t \sqrt{v_c}}$$

$$i = C \frac{dV_{in}}{dt} \quad (i)$$

$$0 - V_o = iR$$

$$V_o = -iR \quad (ii)$$

$$\Rightarrow (i) \text{ in } (ii) \quad V_o = -C \frac{dV_{in}}{dt} \times R$$

$$V_o = -RC \frac{dV_{in}}{dt}$$

Let, $R = 1M\Omega$, $C = 1\mu F \approx RC = 1$

$$V_o = \frac{dV_{in}}{dt}$$

⊗

METHOD-II

(TRANSFER FUNCTION)

$$\frac{V_o(s)}{V_{in}(s)} = -\frac{Z_2}{Z_1}$$

$$V_{in}(s) \quad Z_1$$

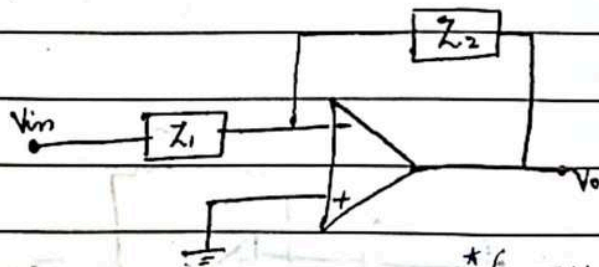
$$V_o(s) = -R$$

$$V_i(s) \quad 1/Cs$$

$$V_o(s) = -RCs$$

$$V_i(s)$$

$$V_o(s) = -RC (V_i(s) s)$$



$$\begin{aligned} \int v_i(t) dt &\xrightarrow{L} V_i(s)/s \\ \frac{dv_i(t)}{dt} &\xrightarrow{L} s V_i(s) \end{aligned}$$



Date: _____

Laplace inverse

$$v_o(t) = -RC \left(\frac{dV_{in}}{dt} \right)$$

$$V_o(s) = -RCs$$

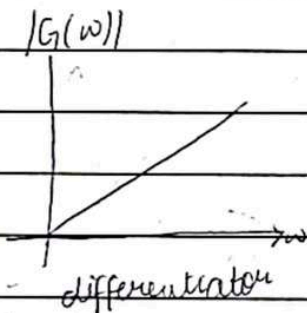
$$V_i(s)$$

$$s = j\omega$$

$$G(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = -RC(j\omega) = 0 - j(RC\omega)$$

$$|G(\omega)| = RC\omega$$

$$\angle G(\omega) = \tan^{-1} \left(\frac{-RC}{0} \right) = -90^\circ \text{ or } -\pi/2 \quad \tan^{-1}(\text{img/real})$$



(shorter behaviour)

$$y = mx$$

$$|G(\omega)| = (RC) \omega$$

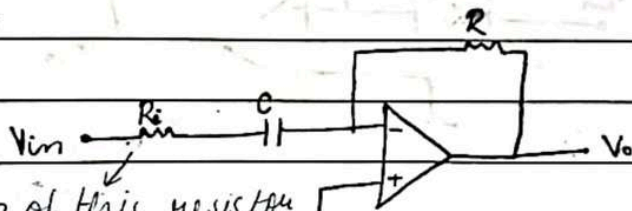
↓
constant

* This is a noise amplifier because it works as a high pass filter.

$$\frac{dV_i}{dt} = \frac{V_i(t) - V_i(t-1)}{\Delta T}$$

↑ final ↑ initial

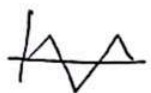
REMEDY:



(inverting)

by allocation of this resistor
— it works like high pass filter



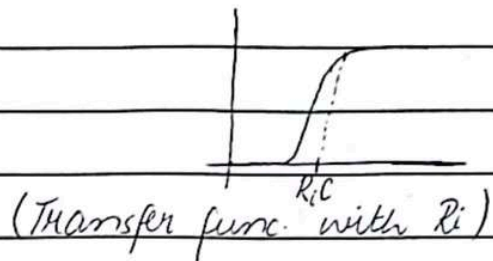
 (ramp signal)

 (sawtooth signal)

Date: _____

$$\frac{V_o}{V_i} = - \frac{R}{R_i + (1/sC)}$$

$$= - \frac{R/R_i}{1 + (1/sRC)}$$

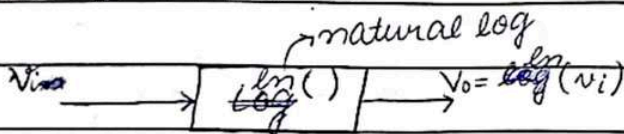


~~V_i~~

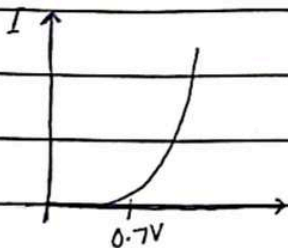
NON-LINEAR OPERATION OF OP-AMP:

• LOGARITHMIC AMPLIFIER:

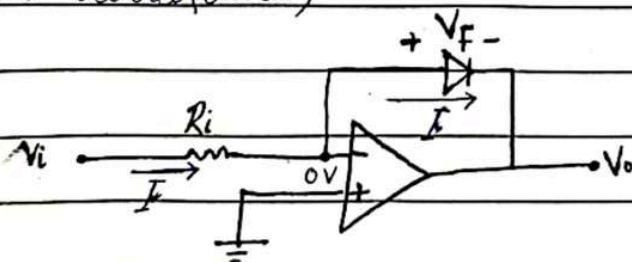
$$V_o \propto \ln(V_i)$$



All the P-n junction / semi-conductor devices exhibit an exponential behaviour ~~due~~ because it has depletion region. When depletion region reaches ..



* Log amp is an inverting amp, the R_f is replaced with any pn junction device (diode)



Date: _____

$$I = \frac{V_i}{R_i}$$

Diode i-v equation:

$$I = I_s \left(e^{\frac{V_F}{\eta V_T}} - 1 \right)$$

I = forward current

(depends on physical conditions)
(const - given)

I_s = reverse saturation current

V_F = voltage across diode

$\eta = 1$ for silicon & 2 for germanium (const)

V_T = Thermal voltage (temp. dependent)

Assume ;

$$V_F \gg \eta V_T \rightarrow e^{\frac{V_F}{\eta V_T}} \gg 1$$

$\Rightarrow$ Neglect '1'

$$\therefore I = I_s e^{\frac{V_F}{\eta V_T}}$$

Take $\ln()$ on both sides

$$\ln(I) = \ln \left(I_s e^{\frac{V_F}{\eta V_T}} \right)$$

$$\ln(I) = \ln I_s + \ln \left(e^{\frac{V_F}{\eta V_T}} \right)$$

$$\ln I = \ln I_s + \frac{V_F}{\eta V_T}$$

$$V_F = \eta V_T (\ln I - \ln I_s)$$

$$V_F = \eta V_T \left(\ln \left(\frac{V_i}{R_i} \right) - \ln I_s \right)$$



$$0 - V_B = V_F$$

$$V_0 = -V_F$$

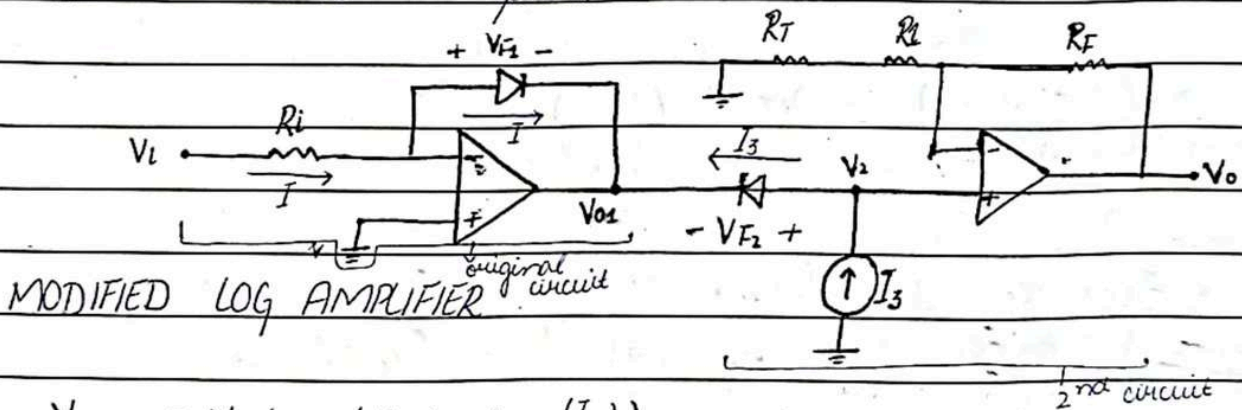
$$* V_0 = -\eta V_T \left[\ln \left(\frac{V_i}{R_i} \right) - \ln I_s \right]$$

PROBLEMS:

1) $\ln(I_s)$ needs to be removed

2) $V_0 \propto \eta V_T$ (if temp. changes - V_0 changes)

will use thermistor to reciprocate



$$V_{01} = -\eta V_T \left(\ln \left(\frac{V_i}{R_i} \right) - \ln I_s \right) \quad (i)$$

$$\therefore V_F = \eta V_T (\ln I - \ln I_s)$$

$$\Rightarrow V_{F2} = \eta V_T (\ln I_3 - \ln I_s) \quad (ii)$$

* diode's forward voltage - +ve

$$V_{F2} = V_2 - V_{01}$$

put values from (i) & (ii)

$$V_2 = V_{F2} + V_{01}$$

$$V_2 = \eta V_T (\ln I_3 - \ln I_s) - \eta V_T \left[\ln \left(\frac{V_i}{R_i} \right) - \ln I_s \right]$$

$$V_2 = \eta V_T \ln I_3 - \eta V_T \ln I_s - \eta V_T \ln \left(\frac{V_i}{R_i} \right) + \eta V_T \ln I_s$$

$$= \eta V_T \ln I_3 - \eta V_T \ln \left(\frac{V_i}{R_i} \right)$$



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$$V_2 = \eta V_T \left(\ln I_3 - \ln \frac{V_i}{R_i} \right)$$

$$= -\eta V_T \left[\ln \left(\frac{V_i}{R_i} \right) - \ln(I_3) \right]$$

$$V_2 = -\eta V_T \left[\ln \left(\frac{V_i}{R_i} \right) \right]$$

$\therefore I_3$ is constant

$$V_o = \left(\frac{1 + R_F}{R_i + R_T} \right) V_2$$

(use 2nd circuit)

$$= \left(\frac{1 + R_F}{R_i + R_T} \right) \times -\eta V_T \ln \left(\frac{V_i}{I_3 R_i} \right)$$

$$* \boxed{V_o = - \left(\frac{1 + R_F}{R_i + R_T} \right) \times \eta V_T \ln \left(\frac{V_i}{I_3 R_i} \right)}$$

R_i has smaller value

* I_3 & R_i are constant terms

$$\left(\frac{1 + R_F}{R_i + R_T} \right) \eta V_T$$

let $\eta = 1$ & $R_i = 0.1 \Omega$ (ignore)

$$\Rightarrow \left(\frac{1 + R_F}{R_T} \right) V_T$$

ignore

let, $R_F = 100k$, $R_T = 1k$

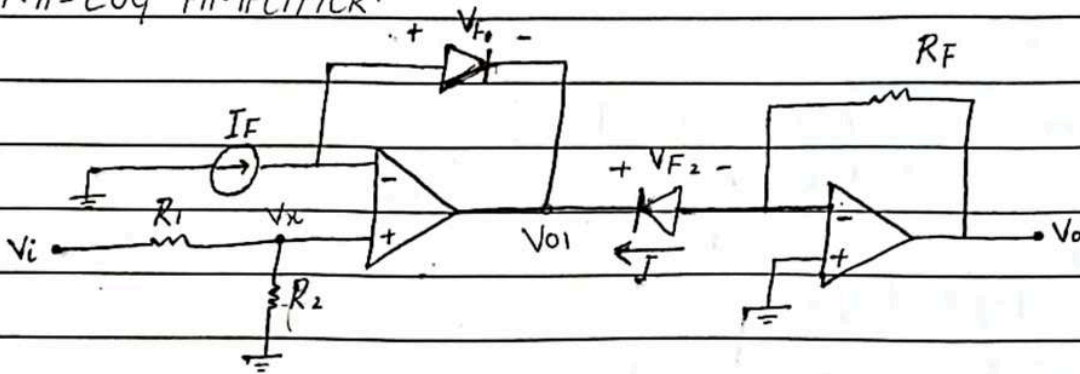
$$\Rightarrow \left(\frac{R_F}{R_T} \right) V_T$$

$$\Rightarrow R_F \left(\frac{V_T}{R_T} \right)$$



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ANTI-LOG AMPLIFIER:



$$V_x = \left(\frac{R_2}{R_1 + R_2} \right) \times V_i$$

$$I = V_0 / R$$

$$V_{F1} = +\eta V_T (\ln I_F - \ln I_S)$$

$$V_x - V_{01} = V_{F1}$$

$$V_{01} = V_x - V_{F1}$$

$$V_{01} = \left(\frac{R_2}{R_1 + R_2} \right) \times V_i - \eta V_T (\ln I_F - \ln I_S) \quad (i)$$

$$0 - V_{01} = V_{F2} \Rightarrow V_{01} = -V_{F2} \quad (ii)$$

$$V_{F2} = +\eta V_T (\ln I - \ln I_S)$$

Substitute (i) in (ii)

$$-V_{F2} = V_x - V_{F1}$$

$$-\eta V_T (\ln I - \ln I_S) = \left(\frac{R_2}{R_1 + R_2} \right) V_i - \eta V_T (\ln I_F - \ln I_S)$$

$$-\eta V_T \ln I + \eta V_T \ln I_S + \eta V_T \ln I_F = \left(\frac{R_2}{R_1 + R_2} \right) V_i - \eta V_T \ln I_S + \eta V_T \ln I_S$$

$$-\eta V_T (\ln I - \ln I_F) = \left(\frac{R_2}{R_1 + R_2} \right) \times V_i$$

$$\left(\frac{R_2}{R_1 + R_2} \right) V_i = \eta V_T (\ln I_F - \ln I)$$

$$\ln \left(\frac{I_F}{I} \right) = \frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2} \right) \times V_i$$

$$\ln\left(\frac{I}{I_F}\right) = \frac{-1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) \times V_i$$

$$I/I_F = \ln^{-1} \left(\frac{-1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) \times V_i \right)$$

$$I = I_F \ln^{-1} \left(\frac{-1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) \times V_i \right)$$

$$\frac{V_o}{R_F} = I_F \ln^{-1} \left(\frac{-1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) \times V_i \right)$$

$$V_o = I_F R_F \ln^{-1} \left(\frac{-1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) V_i \right)$$

$$R_2 = R_T$$

$$\frac{1}{\eta V_T} \left(\frac{R_T}{R_1 + R_T}\right)$$

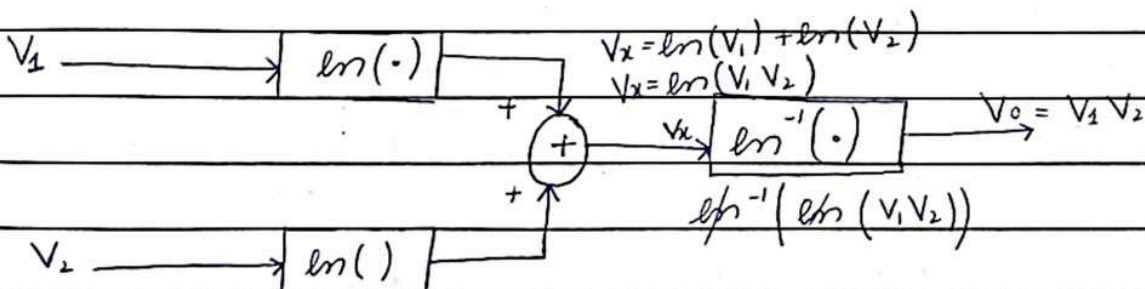
$$R_1 \gg R_T$$

$$\frac{1}{\eta V_T} (R_T/R_1)$$

$$\frac{1}{\eta R_1} (R_T/V_i)$$

Replace R_2 with R_T

$$V_o = I_F R_F \ln^{-1} \left(\frac{-V_i}{R_1} \right)$$

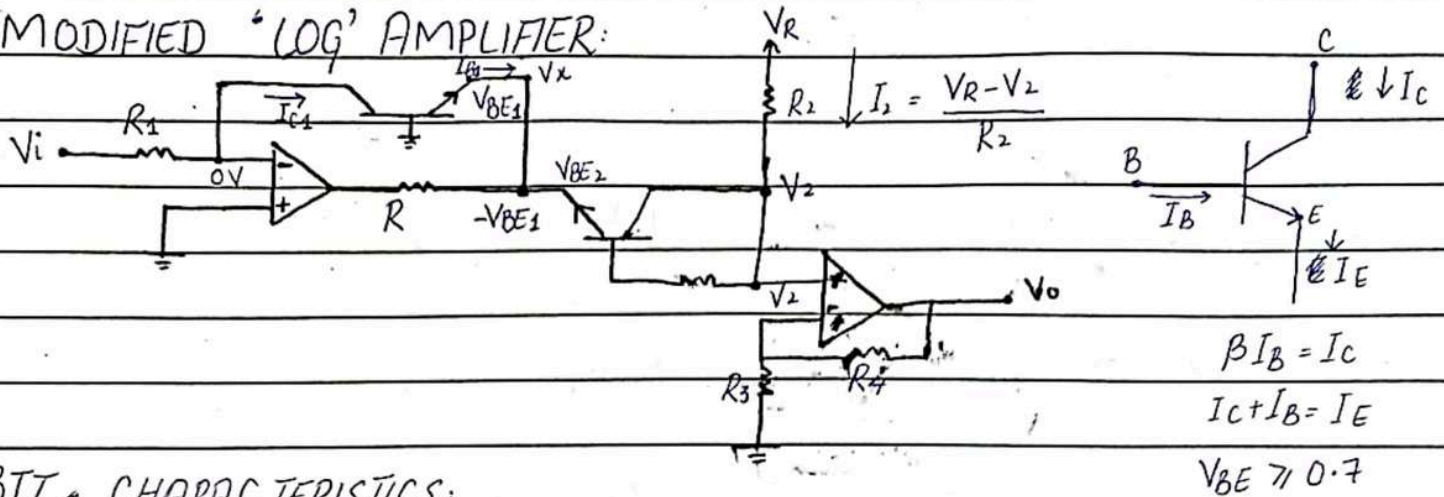


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MODIFIED 'LOG' AMPLIFIER:



BJT CHARACTERISTICS:

$$I_E = I_S e^{V_{BE}/V_T}$$

$$V_{BE} = V_T \ln(I_E / I_S)$$

$$I_{C1} = \frac{V_i}{R_1} \approx I_{E1}$$

$$V_{BE1} = V_T \ln(I_{B1} / I_S)$$

$$V_{BE2} = V_T \ln(I_{E2} / I_S)$$

$$V_2 - (-V_{BE1}) = V_{BE2}$$

$$V_2 + V_{BE1} = V_{BE2}$$

$$V_2 = V_{BE2} - V_{BE1}$$

$$V_2 = V_T \ln(I_{E2} / I_S) - V_T \ln(I_{E1} / I_S)$$

$$V_2 = V_T \ln \left(\frac{I_{E2}}{I_{E1}} \right)$$

* Generally V_2 is small that we neglected
 $V_2 \ll V_R$

$$I_{C2} = I_2 = V_R / R_2$$

$$I_{E2} \approx I_{C2}$$

$$I_{E2} = V_R / R_2$$

$$V_2 = V_T \ln \left(\frac{V_R / R_2}{V_i / R_1} \right)$$



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$$V_2 = V_T \ln \left(\frac{V_R R_1}{V_i R_2} \right)$$

$$V_2 = -V_T \ln \left(\frac{V_i R_2}{R V_R R_1} \right)$$

$$V_2 = V_T \ln(V_i)$$

From the last amplifier

$$V_o = \left(1 + \frac{R_4}{R_3} \right) \times V_2$$

$$V_o = - \left(1 + \frac{R_4}{R_3} \right) \times V_T \ln(V_i)$$

$$R_4/R_3 \gg 1$$

$$V_o = - \left(\frac{R_4}{R_3} \right) V_T \ln(V_i)$$

$$V_o = -R_4 \left(V_T/R_3 \right) \ln(V_i)$$

$$V_o = -k \ln(V_i)$$

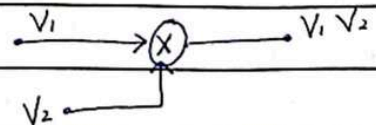
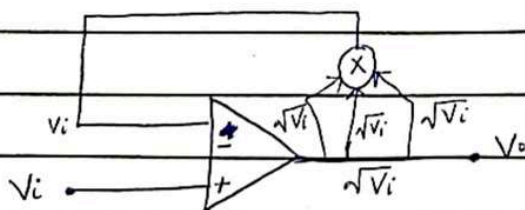
Suppose,

$$V_R = 10V, R_1 = 10k$$

$$R_2 = 100k\Omega$$

SQUARE ROOT EXTRACTOR:

$$V_i \rightarrow \sqrt{(\cdot)} \rightarrow V_o = \sqrt{V_i}$$



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CHAPTER

SIGNAL GENERATOR

SINE WAVE GENERATOR

(OSCILLATOR)

OSCILLATORS:

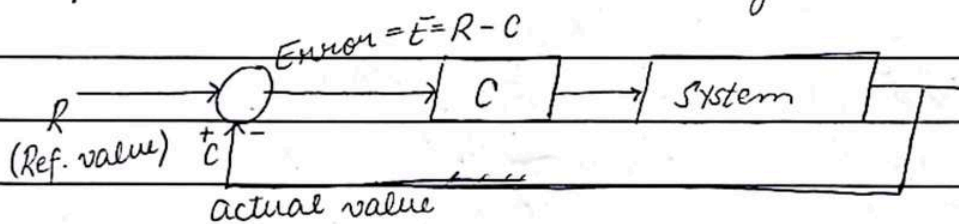
Everything having oscillator - bi-directional signal

Oscillators generates a finite output signal by using an amplifier followed by a frequency selective circuit connected in pos feedback loop with NO input signal

freq. selective circuit
⇒ filter

NEGATIVE FEEDBACK LOOP

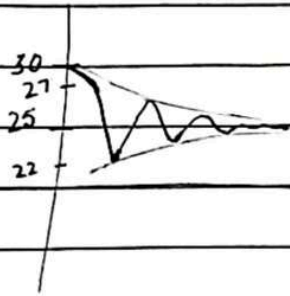
(it is preferred because error converges to zero - stable system)



REF TEMP. (R)	ACTUAL TEMP. (C)	ERROR (E)
25°	30°	-5°
25°	27°	-2°
25°	22°	+3°
25°	26°	-1°
25°	24.5°	+0.5°
25°	25.2°	-0.2°
25°	25°	0
25°	25°	0

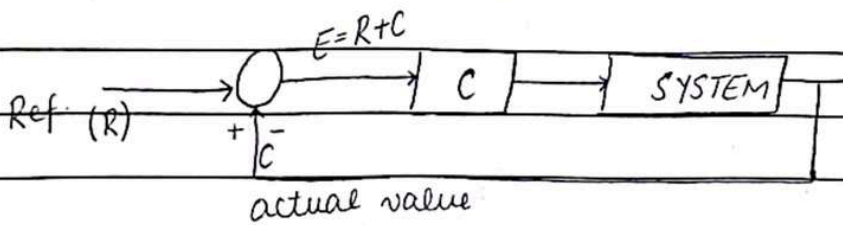


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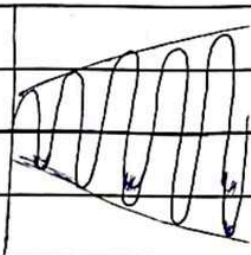


error converging to zero

POSITIVE FEEDBACK SYSTEM

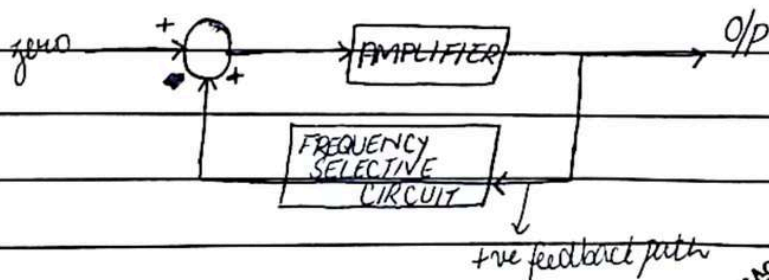


REF. (R)	ACTUAL TEMP (C)	ERROR (E=R+C)
25	30	+55
25	35	+60
		∞



(exponentially increasing)
error diverging - unstable system

SINUSOIDAL OSCILLATOR:



the feedback path



SUPPOSE:

$$N_i = x_s$$

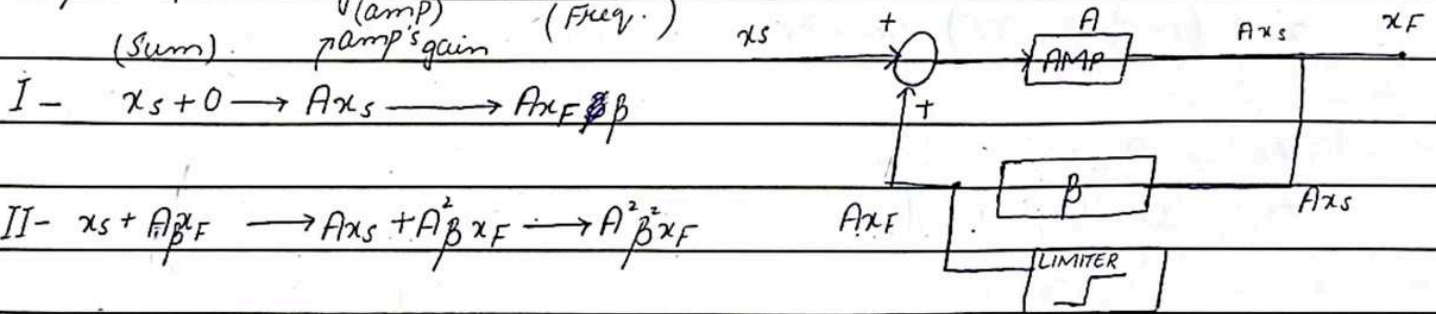
x_s — resonator of infinite sinusoidal frequency. $\rightarrow x_F$
 $M_f \sin(fwt)$

$$x_s = M_1 \sin(\omega t) + M_2 \sin(2\omega t) + M_3 \sin(3\omega t) + \dots + M_n \sin(n\omega t)$$

* magnitude of $M_1, M_2, M_3, \dots, M_n$ — is extremely small of the order 10^{-30}

amplifier's gain — A , freq. selective circuit — β

O/p = finite signal = x_F
 (Sum) $\uparrow$ amp's gain (Freq.)



I — $x_s + 0 \rightarrow A x_s \rightarrow A x_F \beta$

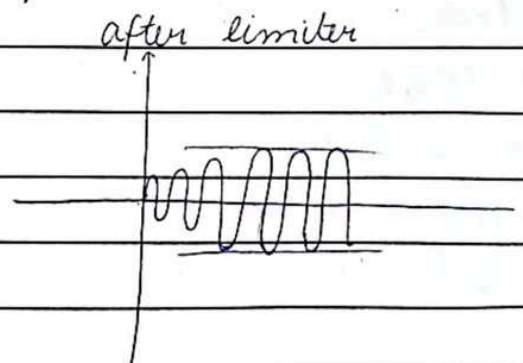
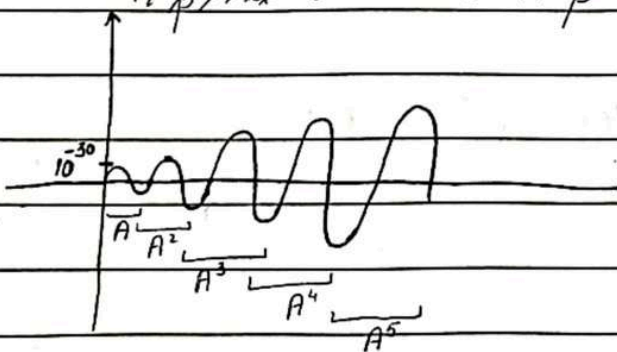
II — $x_s + A \beta x_F \rightarrow A x_s + A^2 \beta x_F \rightarrow A^2 \beta^2 x_F$

III — $x_s + A^2 \beta^2 x_F \rightarrow A x_s + A^3 \beta^2 x_F \rightarrow A^3 \beta^3 x_F$ " $A = 10^5, \beta = 1$ "

IV — $x_s + A^3 \beta^3 x_F \rightarrow A x_s + A^4 \beta^3 x_F \rightarrow A^4 \beta^4 x_F$

$\downarrow$ $\downarrow$

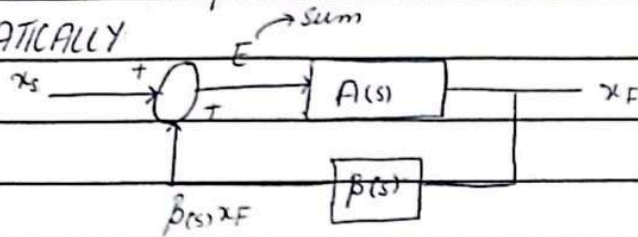
$$A^n \beta^n x_F \rightarrow A x_s + A^n \beta^{n-1} x_F \rightarrow A^n \beta^n x_F$$



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Thermal noise of the resistor is used

MATHEMATICALLY



$$E = x_s + \beta(s) x_F \quad (i)$$

$$x_F = A(s) E \quad (ii)$$

Put (i) in (ii)

$$x_F = A(s) (x_s + \beta(s) x_F)$$

$$x_F (1 - A(s) \beta(s)) = A(s) x_s$$

$\frac{x_F}{x_s} = \frac{A(s)}{1 - A(s) \beta(s)}$
--

When NO I/P is connected

$$\frac{x_F}{x_s} = \frac{x_F}{0} = \infty$$

$$\frac{A(s)}{1 - A(s) \beta(s)} = \infty$$

(when denominator = 0)

$$\Rightarrow 1 - A(s) \beta(s) = 0$$

$$A(s) \beta(s) = 1$$

(BERK HAUSAN'S CRITERIA)

$$A(\omega_0) \beta(\omega_0) = 1$$

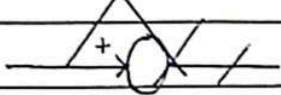
$$\omega_0 = 2\pi f$$



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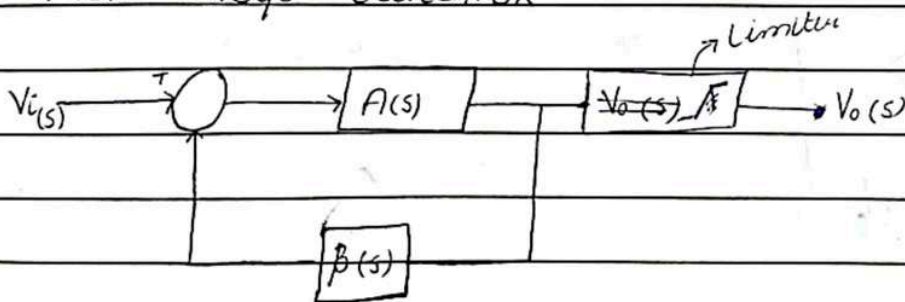
So, if we apply zero input, the output would be a finite signal of frequency (f_0) if and only if $A(\omega_0)\beta(\omega_0)=1$; $\omega_0 = 2\pi f$ holds at the frequency (f_0)

WEIN-BRIDGE OSCILLATOR:



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WEIN-BRIDGE OSCILLATOR



BARKHOUSAN CRITERIA:

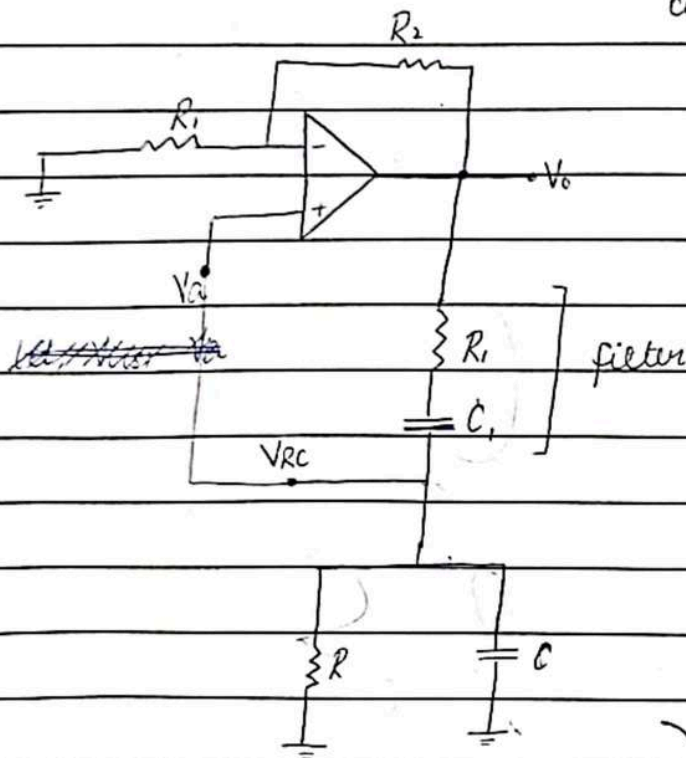
To get a finite sine signal at $f = f_0$
 $A(\omega_0)\beta(\omega_0)=1$



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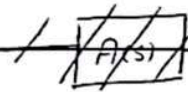
circuit when limiter is not connected (as previous)

-ve feedback (o/p connected to -ve terminal)



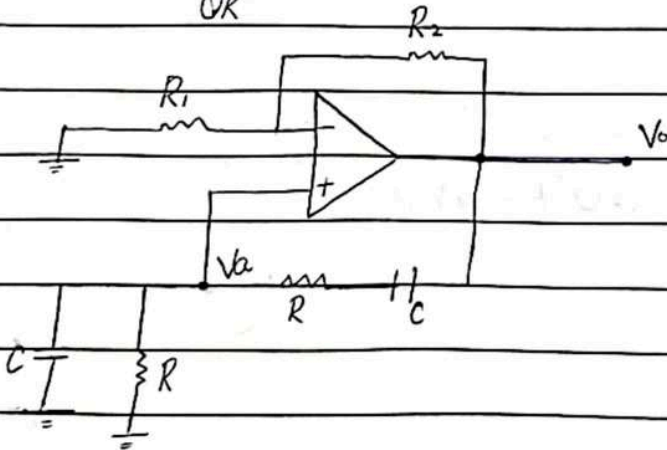
Let, $V_i(s) = V_a$

$\Rightarrow$



Remove 'Va' & 'VRC' after joining names

OR



(non-inv.)

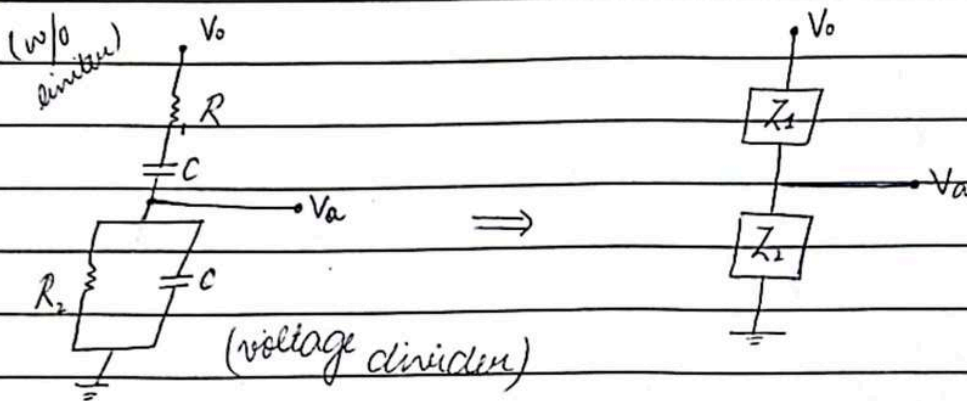
$$V_o(s) = \left(1 + \frac{R_2}{R_1}\right) V_a(s)$$

$$\frac{V_o(s)}{V_a(s)} = \left(1 + \frac{R_2}{R_1}\right)$$

$$\therefore A(s) = 1 + \frac{R_2}{R_1}$$

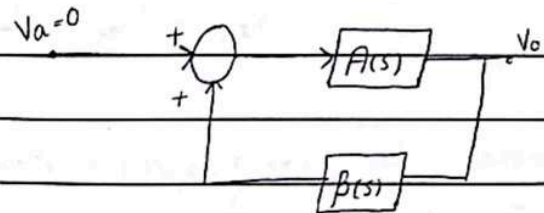
$$\frac{V_o(s)}{V_a(s)} = A(s)$$





$$\beta(s) = \frac{Z_2}{Z_1 + Z_2}$$

$$= \frac{R_2 \parallel (1/sC)}{(R_1 + 1/sC) + (R_2 \parallel 1/sC)}$$



$$= \frac{1 + (R_2/R_1)}{3 + (RC)s + \frac{1}{(RC)s}}$$

$$A(s) = \frac{1 + R_2}{R_1}$$

* $A(s) \beta(s) = 1$

$$A(s) \beta(s) = \frac{1 + (R_2/R_1)}{3 + (RC)s + \frac{1}{(RC)s}}$$

Put $s = j\omega$

$$A(\omega) \beta(\omega) = \frac{1 + (R_2/R_1)}{3 + j(RC\omega) + \frac{1}{j(RC\omega)}}$$



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$$= 1 + (R_2/R_1) \\ 3 + j(RC\omega - \frac{1}{RC\omega})$$

BARK HOUSAN CRITERIA To get a ^{REAL} sine wave of freq. f_0 ,
 $A(\omega)B(\omega) = 1$ at $f = f_0$

$$A(\omega_0)B(\omega_0) = \frac{1 + (R_2/R_1)}{3 + j(RC\omega_0 - \frac{1}{RC\omega_0})}$$

If we make this imaginary part = 0, then this whole thing becomes REAL.

TO MAKE SYSTEM REAL:

$$RC\omega_0 - \frac{1}{RC\omega_0} = 0$$

$$RC\omega_0 = \frac{1}{RC\omega_0}$$

$$2\pi f_0 = \omega_0$$

$$\omega_0^2 = \frac{1}{(RC)^2}$$

$$\Rightarrow \boxed{\omega_0 = \frac{1}{RC}}$$

(filter design)

FOR R_1 & R_2

$$A(\omega_0)B(\omega_0) = 1$$

$$\frac{1 + (R_2/R_1)}{3 + j(RC\omega_0 - \frac{1}{RC\omega_0})} = 1$$

$$1 + (R_2/R_1) = 1$$

$\therefore$ at $\omega_0 = \frac{1}{RC}$

$$\frac{1 + (R_2/R_1)}{3} = 1$$



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$$1 + \frac{R_2}{R_1} = 3$$

$$\frac{R_2}{R_1} = 2$$

$$\frac{R_2}{R_1} \gg 2$$

Assume, $R_1 = 100k$

$$R_2 = 200k$$

should be greater (practicality)

$$\Rightarrow R_2 = 201k\Omega$$

* FILTER:

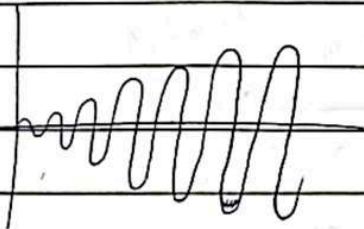
$$\omega_0 = 1/RC$$

$$\omega_0 = 2\pi f_0$$

$f_0 \rightarrow$ given

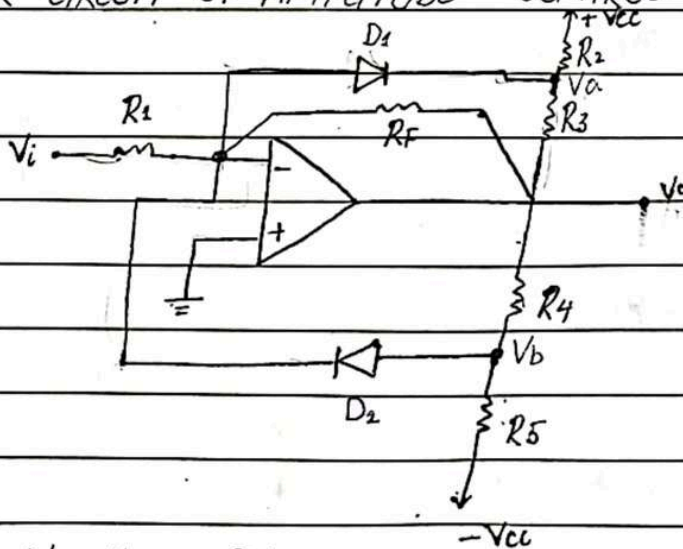
AMP:

$$R_2/R_1 \gg 2$$



(w/o limiter)

LIMITER CIRCUIT OF AMPLITUDE CONTROL:



FOR +VE HALF CYCLE

• V_i is very small (close to zero) but +ve.

$$V_o = -\left(\frac{R_F}{R_1}\right) \times V_i \quad (\text{inverting amplifier})$$



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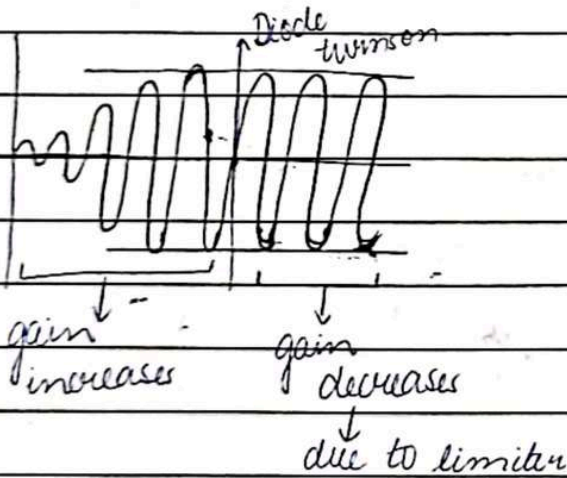
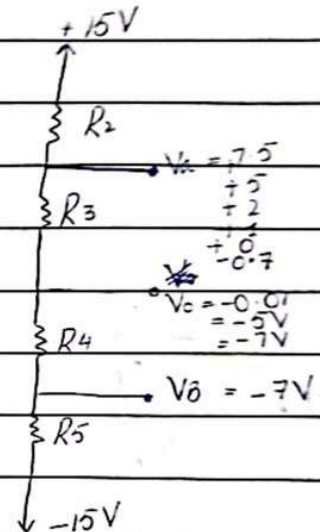
- V_o is also very small and -ve ≈ -0.01
- * $V_{ANODE} > V_{CATHODE}$ (short circuit) $\rightarrow$ forward
- $V_{ANODE} < V_{CATHODE}$ (open circuit) — reverse biased

- V_i will gradually increase
- V_o will move & more -ve (decrease) ($\uparrow$)

$$\Rightarrow V_o = - \left(\frac{R_F}{R_1} \right) \times V_i$$

- V_A will continue to decrease in magnitude
- Until $V_A = -0.7V \rightarrow$ at this point 'Diode turns on'
- It brings R_3 in parallel with R_F

$$\bullet V_o = - \left(\frac{R_3 \parallel R_F}{R_1} \right) \times V_i$$



$$* \frac{R_F}{R_1} \gg \frac{R_3 \parallel R_F}{R_1}$$

$$G_1 \gg G_2$$

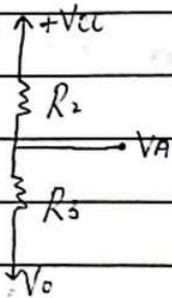
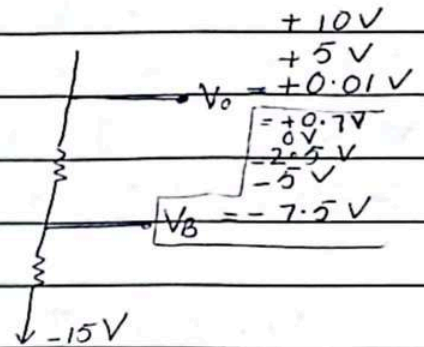


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FOR -VE CYCLE:

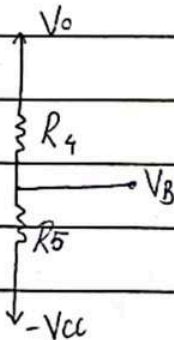
- V_i becomes more '-ve'
- V_o becomes more '+ve'
- V_B becomes less '-ve'

$$V_o = \left(\frac{R_F \parallel R_4}{R_1} \right) \times V_i \quad (\text{diode short})$$



$$V_A = \left(\frac{R_3}{R_2 + R_3} \right) V_{cc} + \left(\frac{R_2}{R_2 + R_3} \right) V_o \quad (i)$$

$\downarrow$ $V_o \rightarrow \text{ground}$ $\downarrow$ $V_{cc} \rightarrow \text{+}$



$$V_B = \left(\frac{R_4}{R_4 + R_5} \right) (-V_{cc}) + \left(\frac{R_5}{R_4 + R_5} \right) \times V_o \quad (ii)$$

TRANSFER CHARACTERISTICS

for (i)

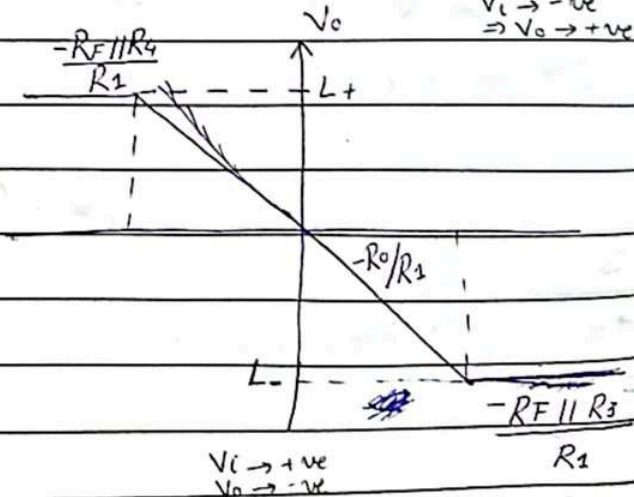
$$V_A = -V_D, \quad V_o = L_-$$

$$\Rightarrow (i) \quad L_- = -V_{cc} \times \frac{R_3}{R_2} - V_D \left(1 + \frac{R_3}{R_2} \right)$$

for (ii)

$$V_B = +V_D, \quad V_o = L_+$$

$$\Rightarrow (ii) \quad L_+ = +V_{cc} \left(\frac{R_4}{R_5} \right) + V_D \left(1 + \frac{R_4}{R_5} \right)$$



Date: 22/10

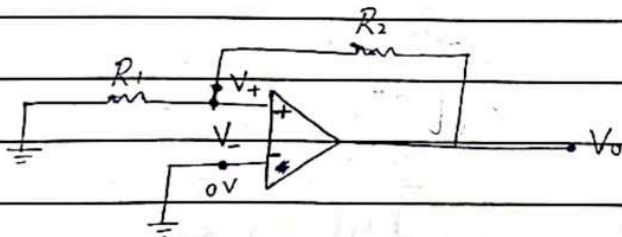
MULTI-VIBRATOR CCT's FOR SIGNAL GENERATION

used to generate sq. wave

- Bistable MVB (2 stable states)
- Mono-stable MVB (1 stable state) (used in alarms)
- A-stable MVB (No stable state)

→ BISTABLE MVB:

- Bistability: DC amplifier in +ve feedback loop (no filter)
- * Open loop gain: $A\beta > 1$
- Feedback contains a voltage divider of R_1 & R_2



• $V_- = 0V$

- No excited input voltage
- Electrical noise in ckt introduces a small +ve voltage at V_+

Hence,

$$V_o^1 = A(V_+ - V_-)$$

$$\Rightarrow V_o^1 > 0$$

$V_o^1 \rightarrow 1^{st}$ situation

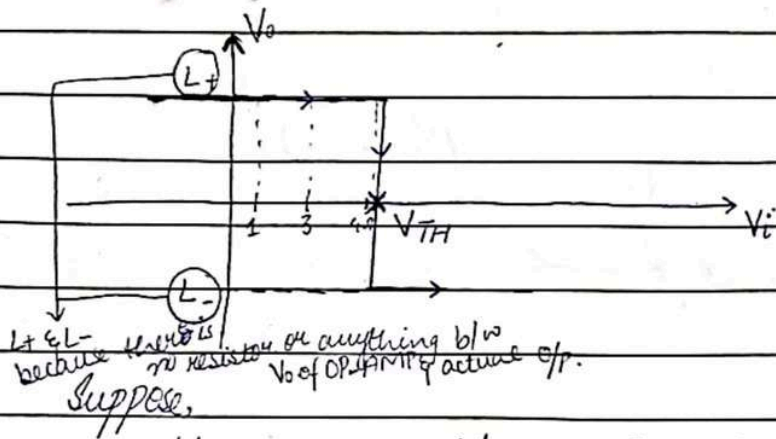
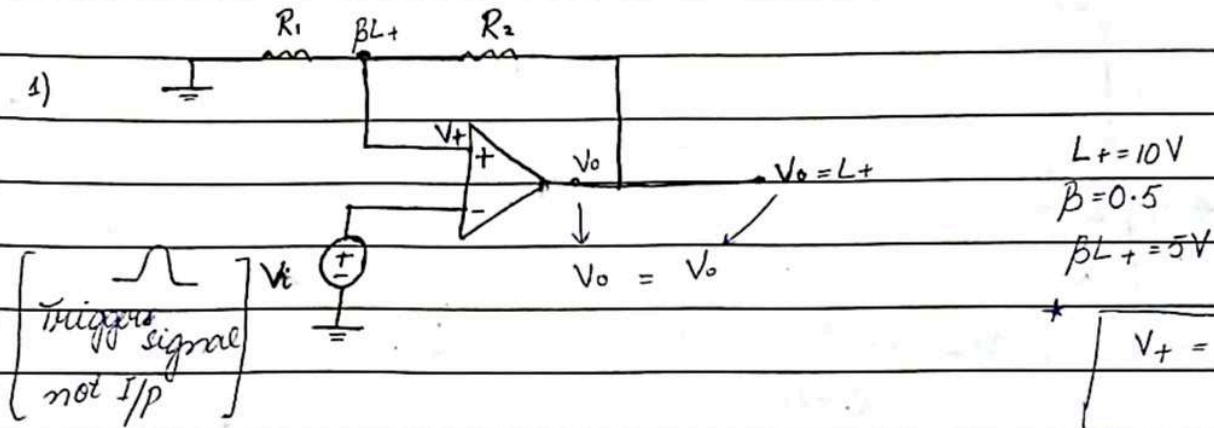
O/P is fed back at $V_+ = \left(\frac{R_1}{R_1 + R_2} \right) \times V_o^1$

$$V_+ = \beta V_o^1$$

* saturation happens in pico sec. $\Rightarrow V_o$ - increases



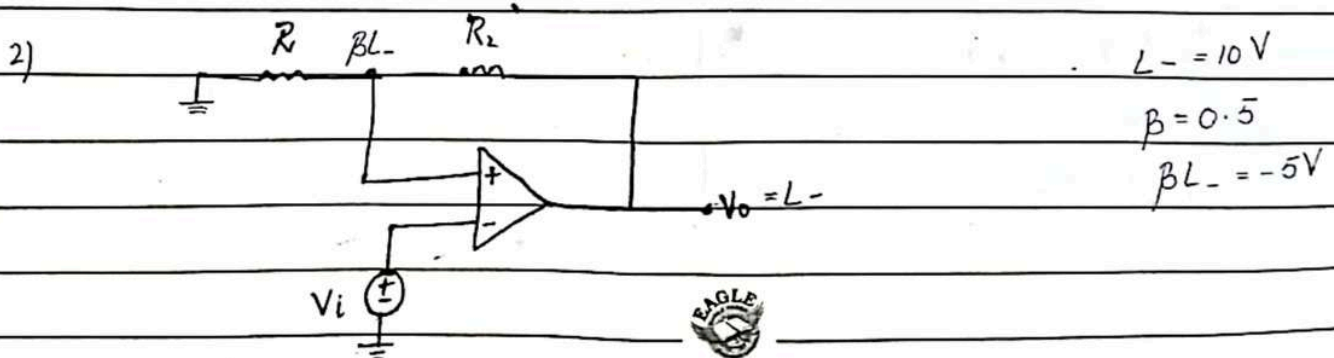
TRANSFER CHARACTERISTICS OF BISTABLE MVB



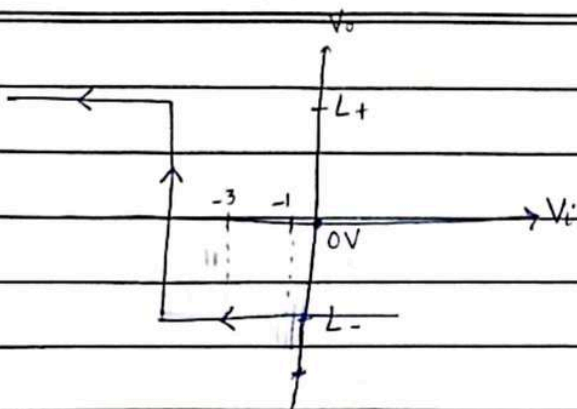
Suppose,

$$V_o = L_+ ; \quad V_+ = \left(\frac{R_1}{R_1 + R_2} \right) L_+ = \beta L_+$$

- $V_+ = \beta L_+$
- $V_- = V_i$
- AMP; $V_o = A(L_+ - L_-)$ (Vi $\gg \beta L_+$)
- initially, $V_- = 0V$
- We start increasing V_i gradually



Date: _____

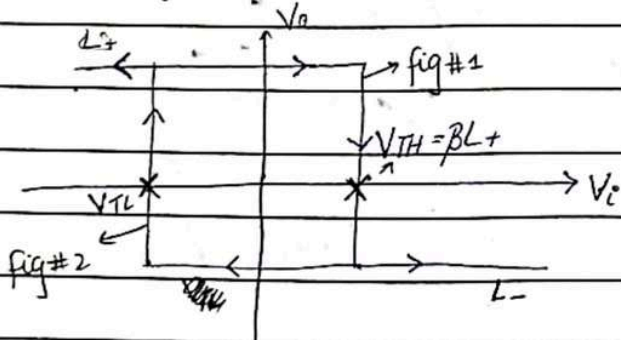


(Fig #2)

Suppose, $V_o = L_-$; $V_+ = \frac{R_1}{R_1 + R_2} \times L_- = \beta L_-$

- $V_+ = +\beta L_-$
- $V_- = V_i = 0$
- AMP ; $V_o = A(V_+ - V_-)$ ($V_i \leq \beta L_-$)
- initially, $V_- = 0V$
- we start decrease V_i gradually

Combine Fig #1 & fig #2



(Hysteresis comparator)

V_{TH} - V Threshold (high to low)

V_{TL} - V threshold low (low to high)

* inverting bistable MVB
or

hysteresis comparator

*
$$\begin{cases} V_{TH} = \frac{R_1}{R_1 + R_2} \times L_+ \\ V_{TL} = \frac{R_1}{R_1 + R_2} \times L_- \end{cases}$$



Date: _____

- * if trigger (V_i) $> \beta L_+$ — V_o goes from L_+ to L_- (V_{TH})
 if trigger (V_i) $< \beta L_-$ — V_o goes from L_- to L_+ (V_{TL})

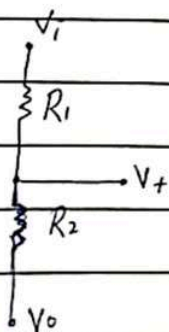
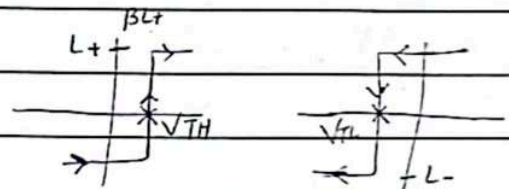
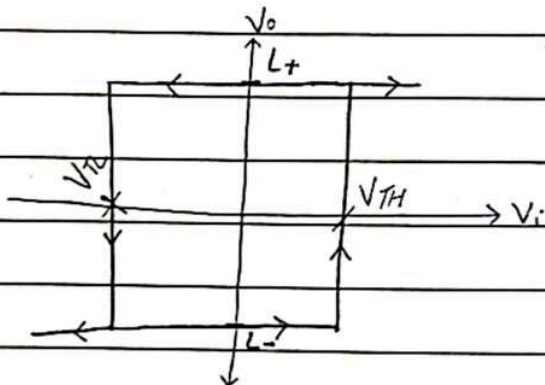
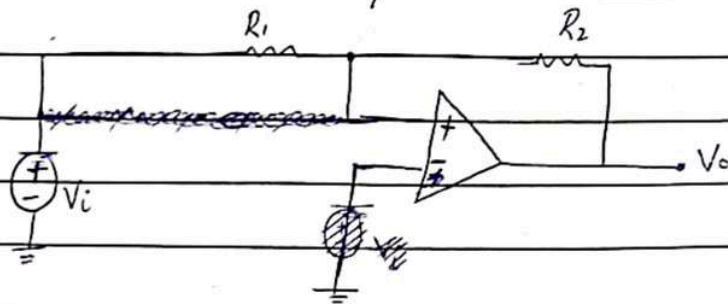
~~We have noise at ripple near zero crossing~~

Hysteresis comparator:

helps to eliminate the ripples near the zero crossing of the signal wave form.

INVERTING BISTABLE MUB

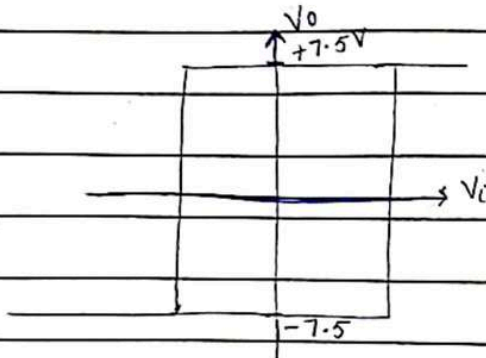
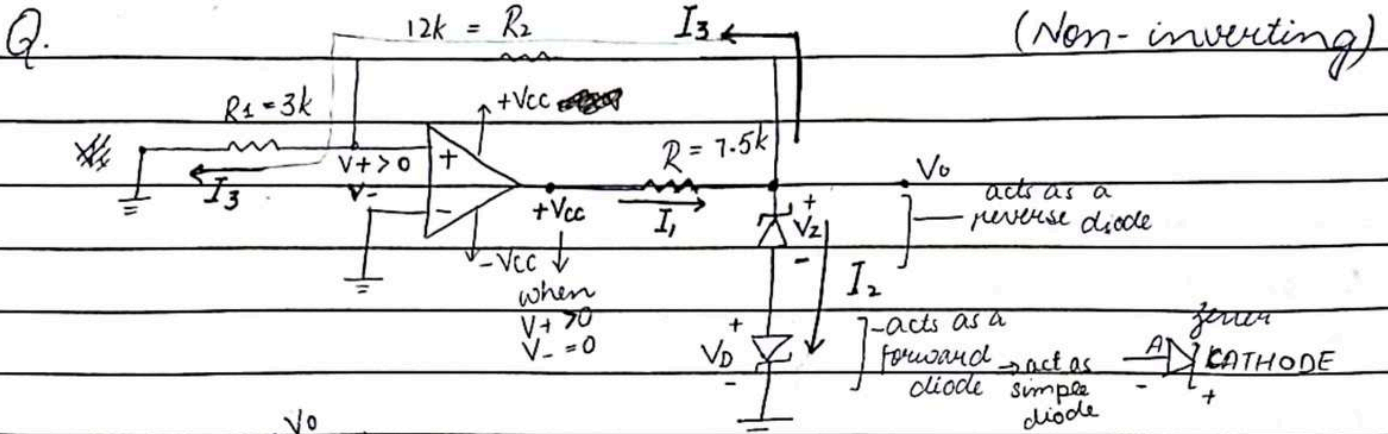
24/10 NON-INVERTING BISTABLE MUB



$$* \left[\begin{aligned} V_+ &= \left(\frac{R_1}{R_1 + R_2} \right) V_o + \left(\frac{R_2}{R_1 + R_2} \right) V_i \\ * \left[\begin{aligned} V_{TH} &= - \left(\frac{R_1}{R_2} \right) \times L_- \\ V_{TL} &= - \left(\frac{R_1}{R_2} \right) \times L_+ \end{aligned} \right. \end{aligned} \right.$$



Date: _____



→ forward current in Zener:
it acts like an ordinary diode

$V_{AK} = 0$
anode cathode

→ reverse current:

it acts like a Zener diode

$V = V_Z$
Zener

Let $V_Z = 6.8V$, $V_D = 0.7V$ (given)

$V_O = V_{DZ} + V_D$
 $= 6.8 + 0.7$

$V_O = 7.5V$

(L+)

for $V_+ > 0$
 $V_- = 0$

0 - $V_O = V_Z + V_D$

$-V_O = 6.8 + 0.7$

$V_O = -7.5V$

(L-)

for $V_+ < 0$
 $V_- = 0$

V_O replaced by $-V_{CC}$
(direction of currents changes)

Let, $R = 1k\Omega$, $V_{CC} = 15V$

$R_1 = 12k$, $R_2 = 3k$

$I_1 = \frac{V_{CC} - 7.5}{7.5k} = 10mA$

$I_3 = \frac{7.5 - 0}{12k + 3k} = 0.5mA$

$12k R_2 + 3k R_1$



$$I_2 = I_3 - I_1$$

$$= 0.5 \text{ mA}$$

$$\therefore I_3 \neq I_1 + I_2$$

$$V_+ = \frac{R_1}{R_1 + R_2} \times V_0$$

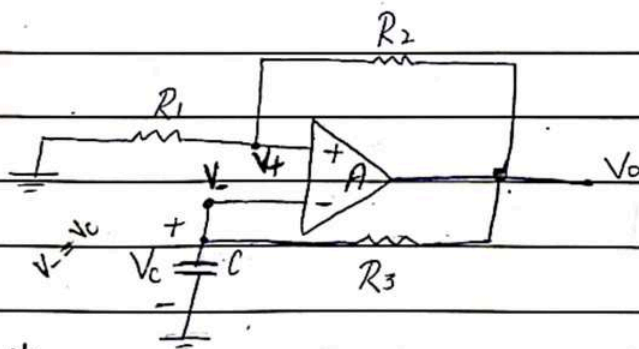
$$= \frac{3k}{3k + 15k} \times -7.5$$

$$V_+ = -1.25 \text{ V}$$

29/10

ASTABLE MVB (using inverting bistable config)
(sq. or rect. wave generation)

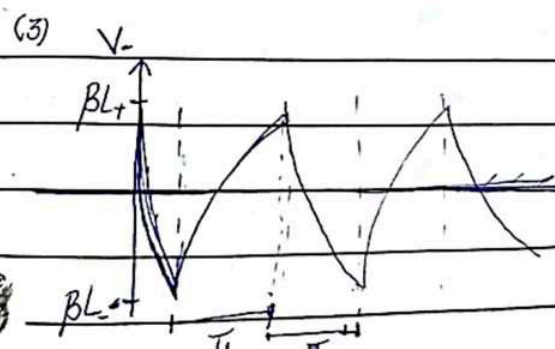
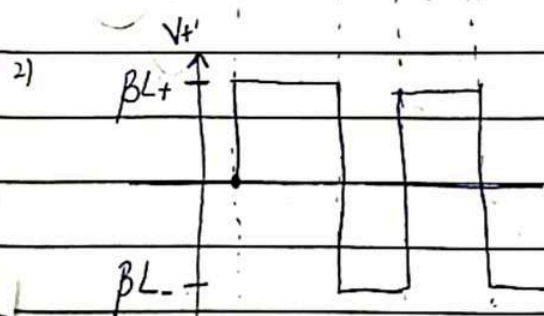
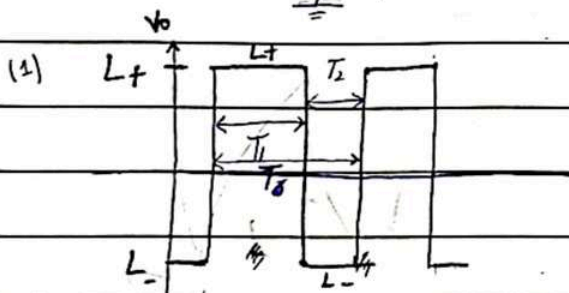
- No stable state
- Keeps toggling between L_+ & L_-
- Used to generate sq. waves & rect. waves.
- Uses bistable MVB circuit with RC circuit in ~~the~~ ^{-ve} feedback.



$$V_0 = A(V_+ - V_-)$$

$$V_- = V_0$$

(Sq. or rect wave)



Date: _____

Initially:

$$V_0 = L_+ , V_+ = \beta L_+$$

- V_c will start rising exponentially $\therefore V_c = V_-$
- V_- will start rising exponentially
- Because $V_0 = L_+$ is fixed, V_+ is also ~~fixed~~ ~~constant~~ fixed
- $V_0 = A (V_+ - V_-)$

TIME PERIOD:

$$v_c(t) = v_c(\infty) - (v_c(\infty) - v_c(0)) e^{-t/RC}$$

for T_1 : $v_c(0) = \beta L_-$

$$v_c(\infty) = L_+$$

$$\therefore \beta = \frac{R_1}{R_1 + R_2}$$

at $t = T_1$, $v_c(t) = \beta L_+$

$$\beta L_+ = L_+ - (L_+ - \beta L_-) e^{-T_1/RC}$$

$$\ln \left(\frac{\beta L_+ - L_+}{-(L_+ - \beta L_-)} \right) = \ln \left(\frac{-T_1}{RC} \right)$$

$$* T_1 = RC \ln \left(\frac{1 - \beta(L_-/L_+)}{1 - \beta} \right)$$

For symmetry power supplies ; $L_+ = -L_-$

$$* T_1 = RC \ln \left(\frac{1 + \beta}{1 - \beta} \right)$$

for T_2 :

$$v_c(0) = \beta L_+$$

$$v_c(\infty) = L_-$$

at $t = T_2$, $v_c(t) = \beta L_-$



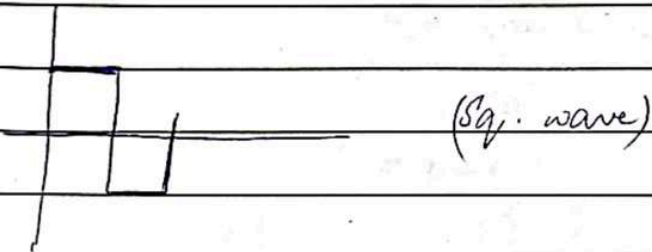
Date: _____

$$\beta L_- = L_- - (L_- - \beta L_+) e^{-T_2/RC}$$

$$\times T_2 = RC \ln \left(\frac{1 - \beta (L_+/L_-)}{1 - \beta} \right)$$

$$\star T_2 = RC \ln \left(\frac{1 + \beta}{1 - \beta} \right)$$

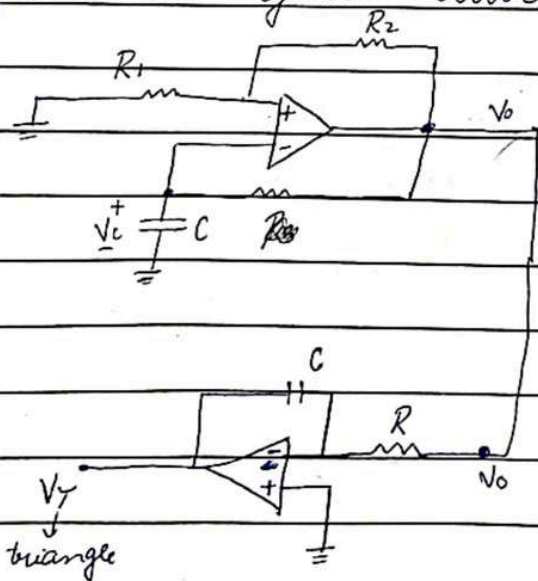
* for symmetrical power supplies $T_1 = T_2$



$$\star \beta = \frac{R_1 + R_2}{R_1 + R_2}$$

$$\star T = T_1 + T_2$$

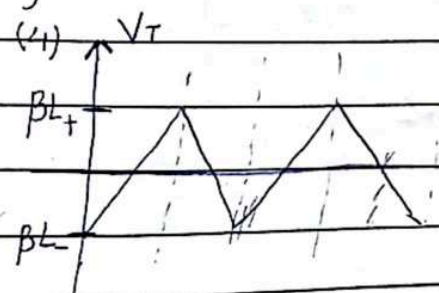
To get a triangular wave:

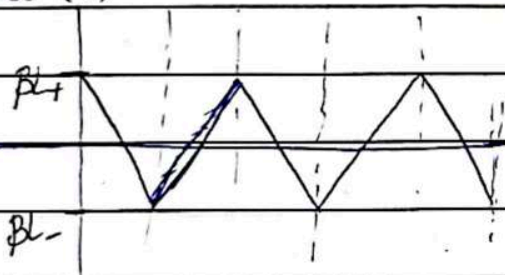
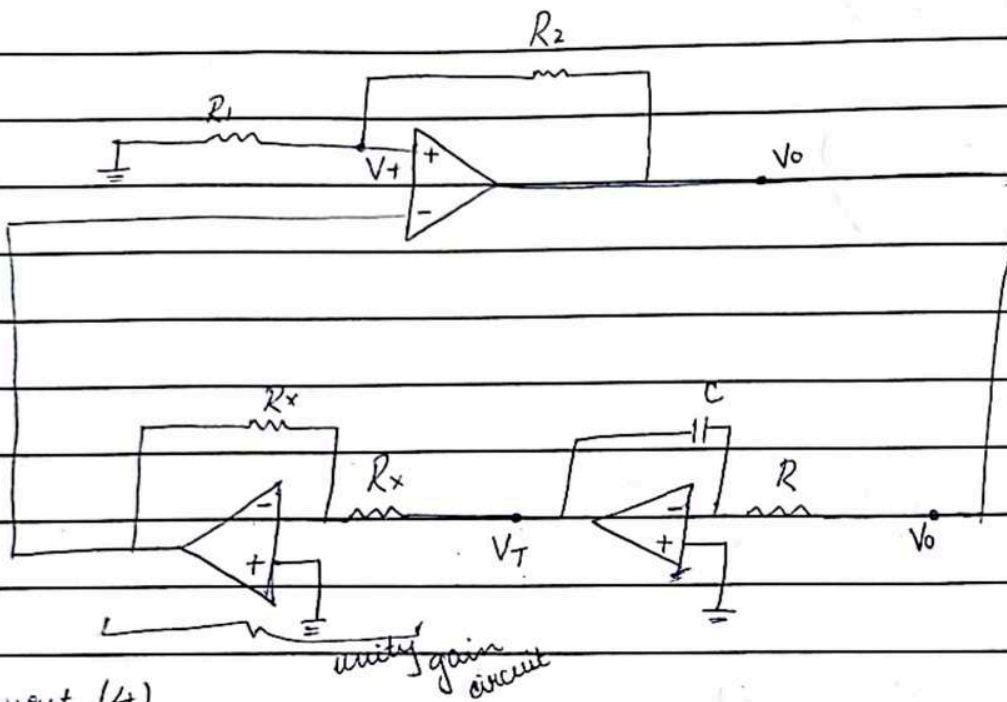


$$V_T = -\frac{1}{RC} \int V_o dt \quad \text{assume } V_o = 10$$

$$= -\frac{1}{RC} \int +10 dt$$

$$RC = 1$$





Function generator (Sq. + Δ wave ...)

For sym. power supplies; $L+ = -L-$

$$T_1 = RC \ln \left(\frac{V_{TH} - V_{TL}}{L+} \right)$$

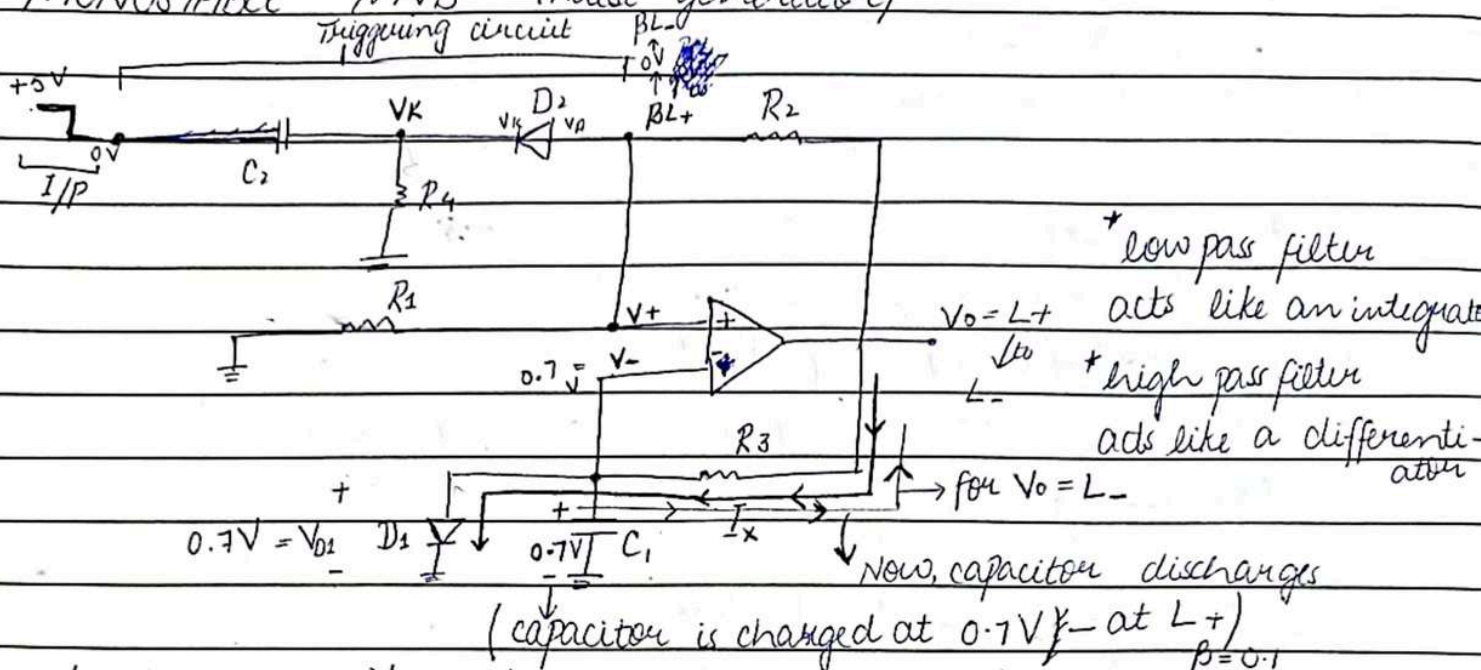
$$T_2 = RC \ln \left(\frac{V_{TH} - V_{TL}}{-L-} \right)$$

$$\therefore V_{TH} = BL+$$

$$\therefore V_{TL} = BL-$$



MONOSTABLE MVB (Pulse generator)



• $V_o = L+$, $V_+ = \beta L+$

• Because $V_K > \beta L+$ ($0.5V > 1.2$)

$\Rightarrow D_2$ is reverse biased

(open circuit)

• For I_x flow, D_1 is forward biased

$\Rightarrow D_1$ is short circuit

• $V_o = A(V_+ - V_-)$

$= A(\beta L+ - 0.7)$

• When apply a trigger; $V_K = 0V$

$\Rightarrow D_2$ becomes forward biased (short circuit)

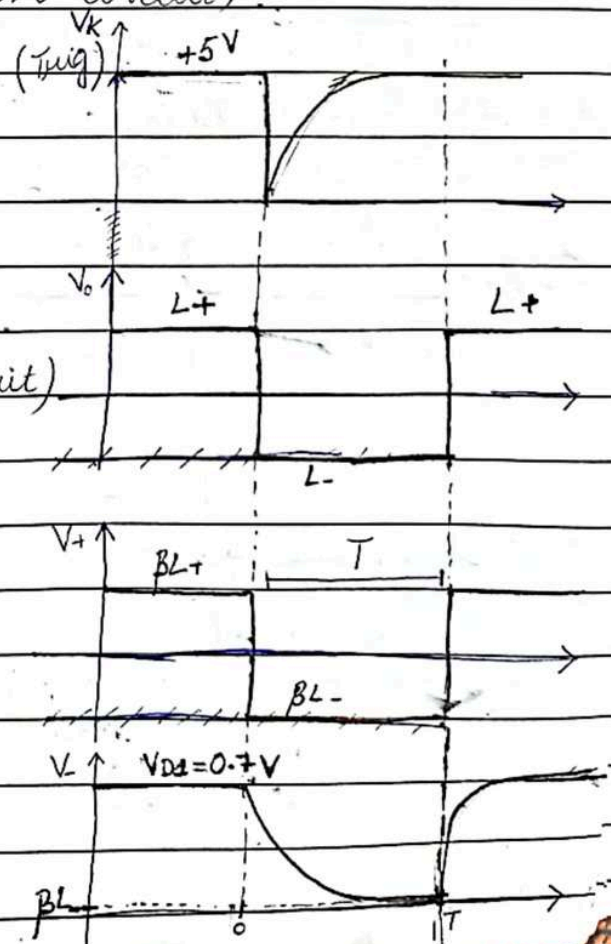
• D_2 is short circuit, $0V$ at V_K will travel ahead & force itself at V_+

• $V_o = A(V_+ - V_-)$

$= A(\beta L+ - 0.7)$

• In this duration, V_K becomes +ve again

$\Rightarrow D_2$ is again open



$$V_0 = A(V_+ - V_-)$$

$$= A(\beta L_- - V_-)$$

$$-1.2 - (-1.2001)$$

$$y(t) = y(\infty)$$

TIME PERIOD:

$$v_c(t) = v_c(\infty) - (v_c(\infty) - v_c(0)) e^{-t/R_3 C_1}$$

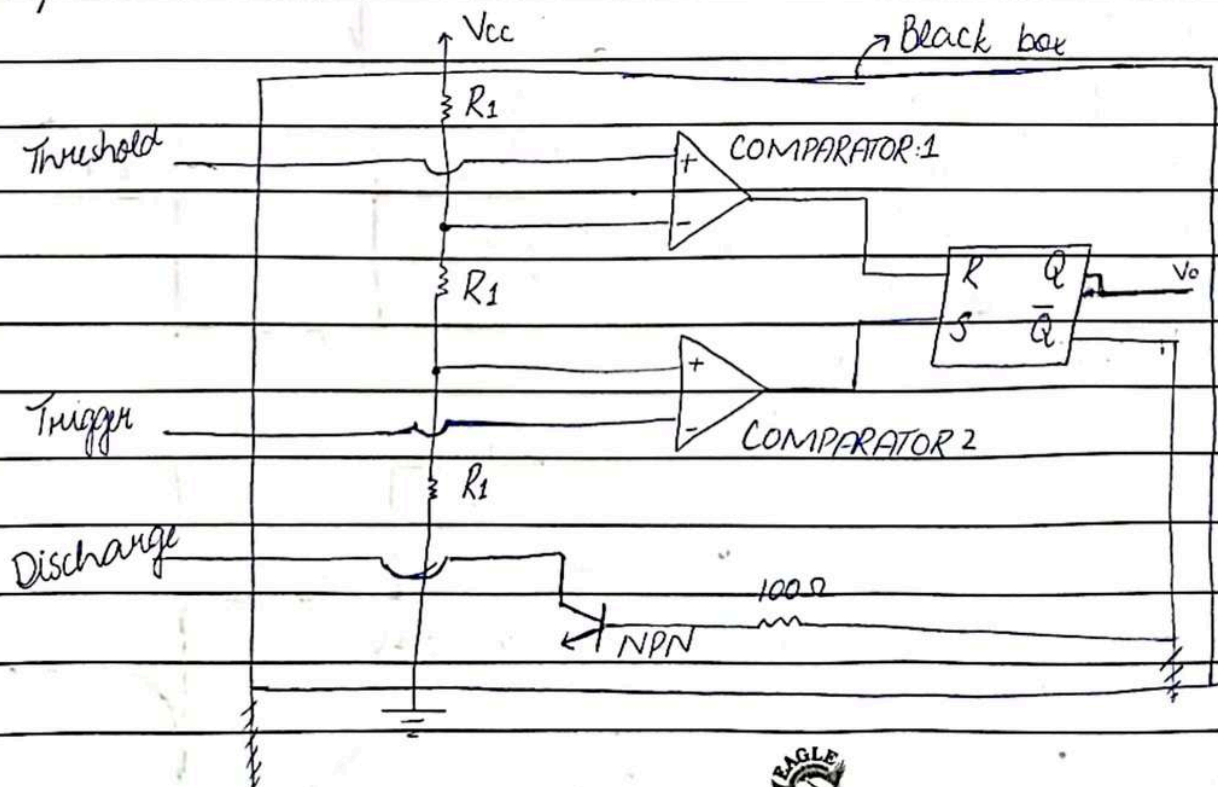
$$v_c(0) = V_{D1} \quad \text{at } t=T \quad ; \quad v_c(T) = \beta L_-$$

$$v_c(\infty) = L_-$$

$$\beta L_- = L_- - (L_- - V_{D1}) e^{-T/R_3 C_1}$$

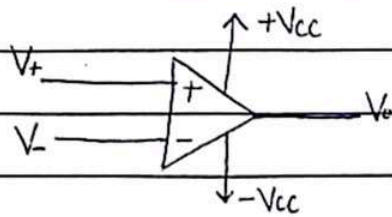
$$\Rightarrow T = R_3 C_1 \ln \left(\frac{V_{D1} - L_-}{\beta L_- - L_-} \right)$$

7/11 555 TIMER CIRCUIT:



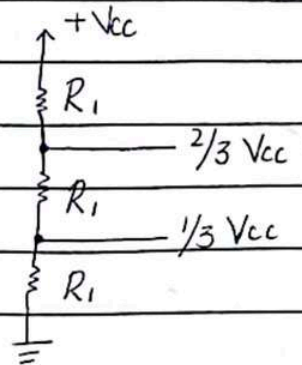
Date: _____

COMPARATOR



$$V_+ > V_- \Rightarrow V_0 = +V_{cc}$$

$$V_+ < V_- \Rightarrow V_0 = GND$$



R	S	Q	$\bar{Q}$
0	0	Previous state	
0	1	1	0 (Set)
1	0	0	1 (reset)
1	1	undefined	

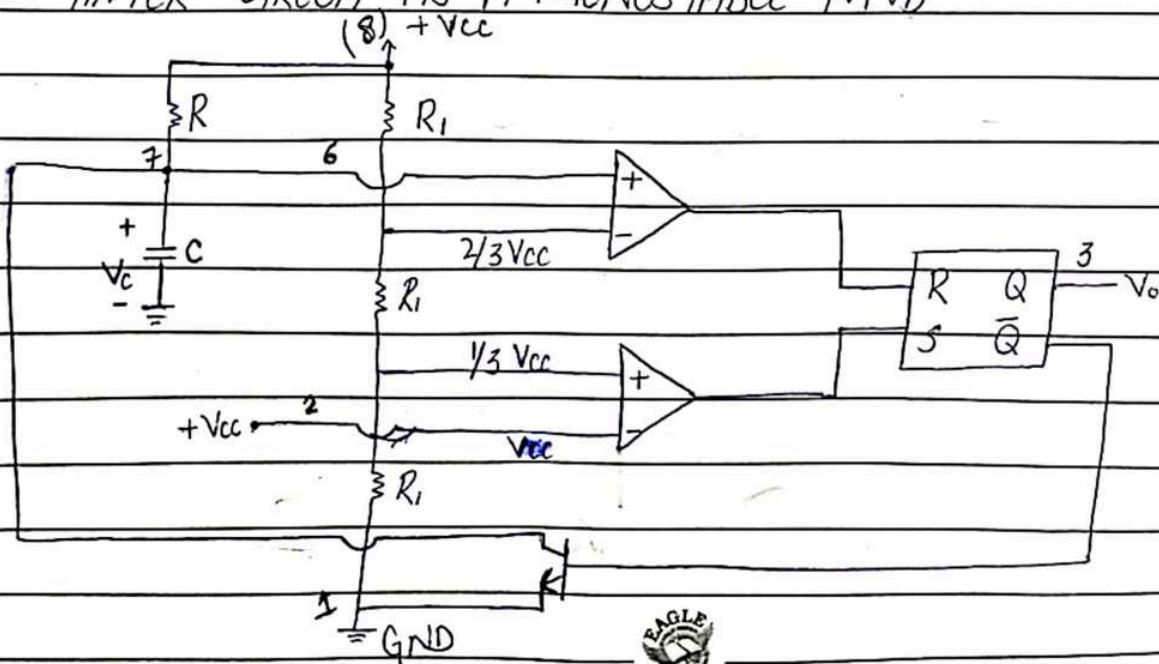
R	Q
S	$\bar{Q}$

RS flip flop

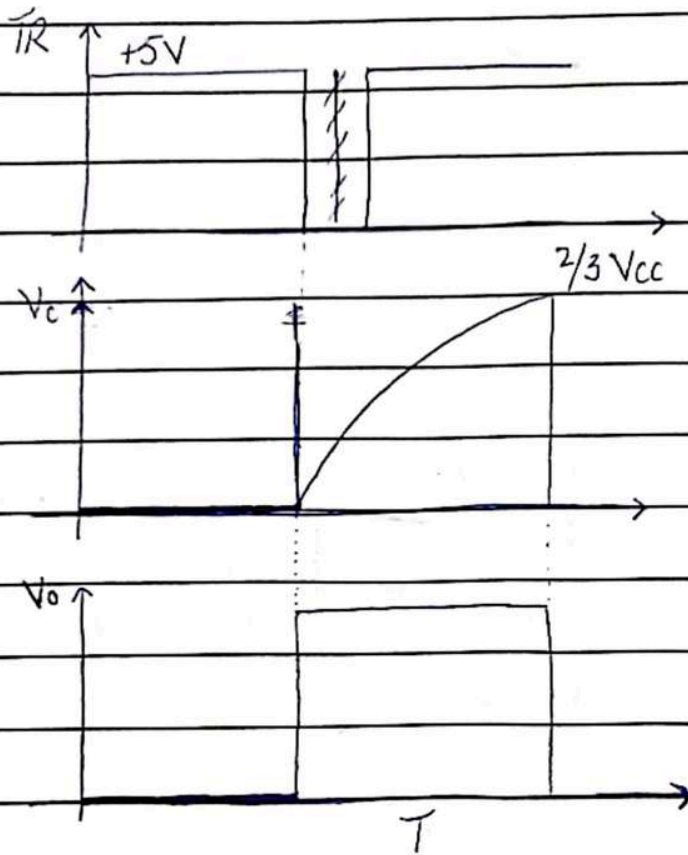
when $\bar{Q} = 0$ NPN open (off)

when $\bar{Q} = 1$ NPN short (ON)

555 TIMER CIRCUIT AS A MONOSTABLE MVB



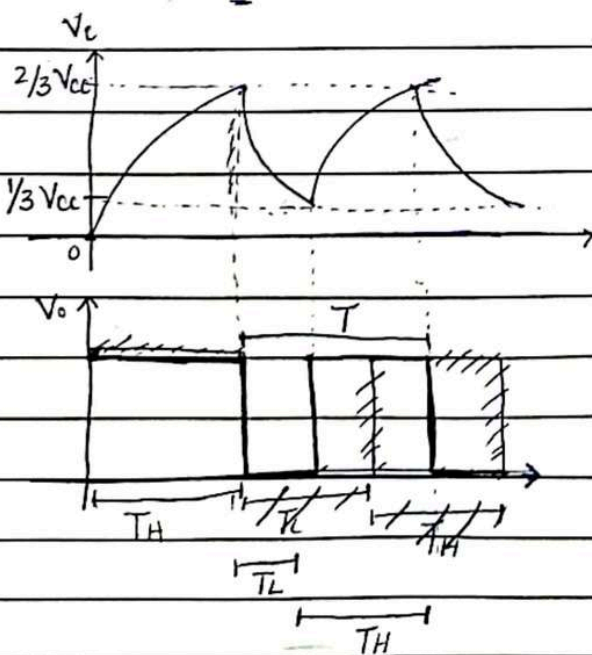
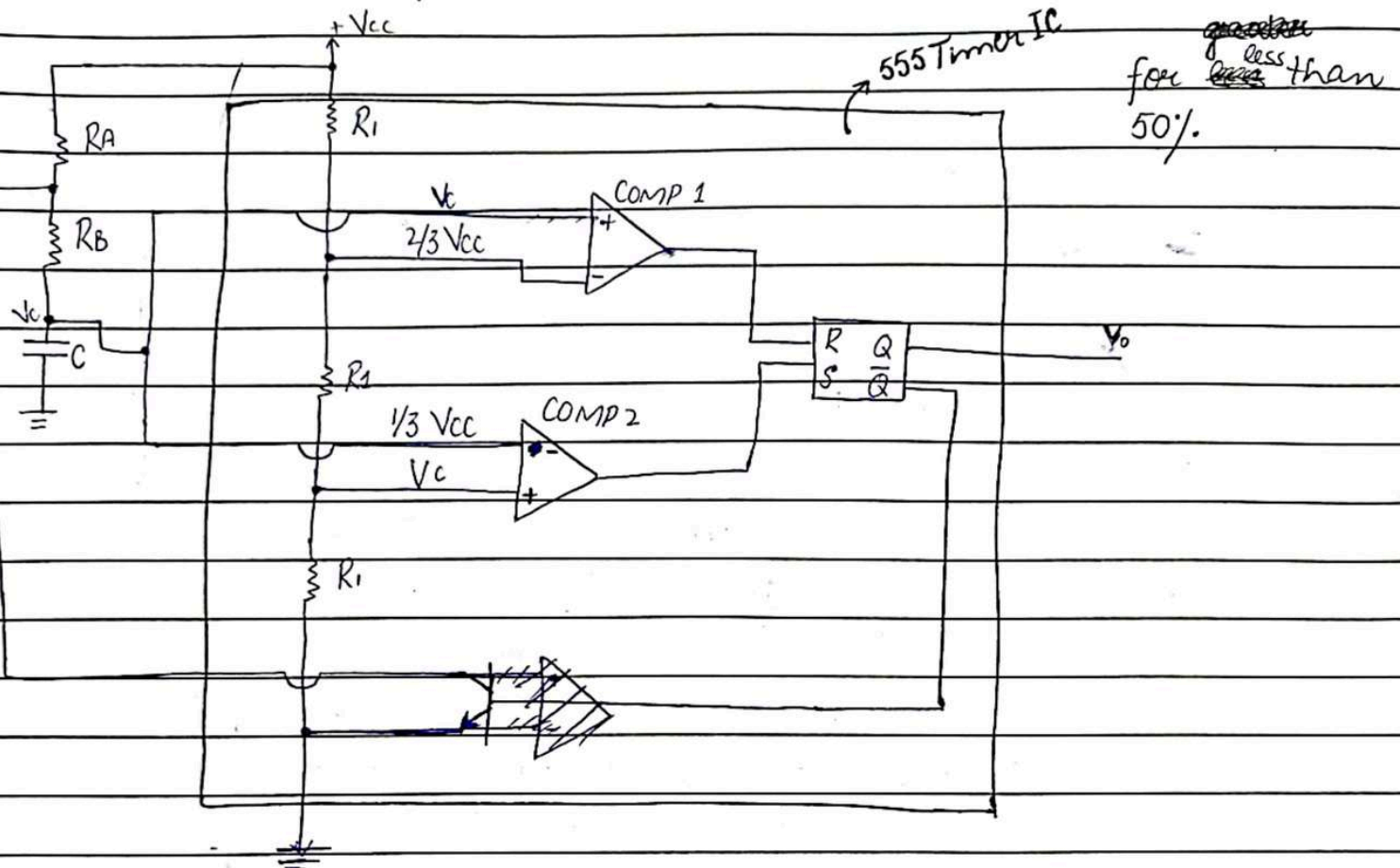
Trigger (TR)



$$\begin{aligned} * \int \quad T &= \ln 3 RC \\ T &= 1.1 RC \end{aligned}$$

Date: 12/11

ASTABLE MULTIVIBRATOR USING 555 TIMER



flip flop

R	S	Q
0	0	Prev
0	1	1
1	0	0
1	1	∞

$$\tau_c = (R_A + R_B)C$$

changing $\tau_c = R_B \times C$

↓

discharging



Date: _____

$$V_c(t) = V_c(\infty) - (V_c(\infty) - V_c(0)) e^{-t/RC}$$

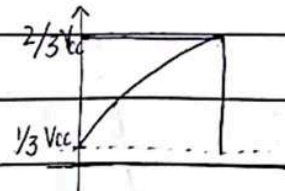
FOR CHARGING $V_c(\infty) = V_{CC}$

$$V_c(0) = \text{initial value} = 1/3 V_{CC}$$

at $t = T_H$

$$\Rightarrow V_c = 2/3 V_{CC}$$

$$\frac{2}{3} V_{CC} = V_{CC} - (V_{CC} - 1/3 V_{CC}) e^{-T_H/(R_A + R_B)C}$$



* $T_H = \ln 2 (R_A + R_B) C$

FOR DISCHARGING

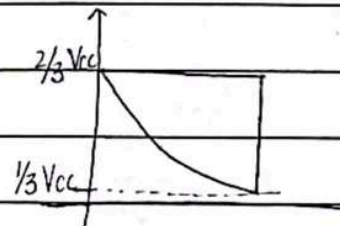
$$V_c(\infty) = 0$$

$$V_c(0) = 2/3 V_{CC}$$

at $t = T_L$

$$\Rightarrow V_c = 1/3 V_{CC}$$

$$1/3 V_{CC} = 0 - (0 - 2/3 V_{CC}) e^{-T_L/R_B C}$$



* $T_L = \ln 2 (R_B) C$

$$T = T_H + T_L$$

* $T = \ln 2 (R_A + 2R_B) C$

* $D = T_H / (T_H + T_L) = T_H / T$

* $D = \frac{R_A + R_B}{R_A + 2R_B}$ (duty cycle)

$D \geq 51\%$
(max)

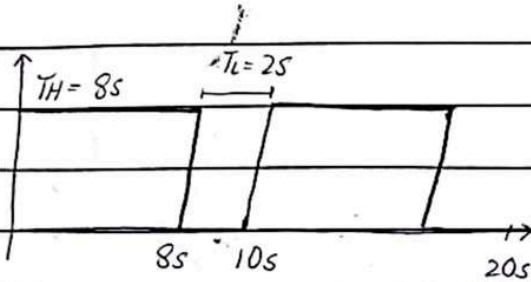


Date: _____

DUTY CYCLE:

The percentage ^{OR ratio} of the total time period for which the signal is logic high OR logic '1'.

EXAMPLE:



$$D = 8/10 = 0.8 \quad (80\%)$$

$$* D = \frac{T_H}{T_H + T_L} \times 100\%$$

DRAWBACKS

- $T_H \neq T_L$ (never)

In current settings, T_H can never be equal to T_L . So, you can't achieve a square wave and a 50% duty cycle.

FOR 50% DUTY CYCLE: (square wave)

$$T_H = \ln 2 (R_A) C$$

$$T_L = \ln 2 (R_B) C$$

$$T = T_H + T_L$$

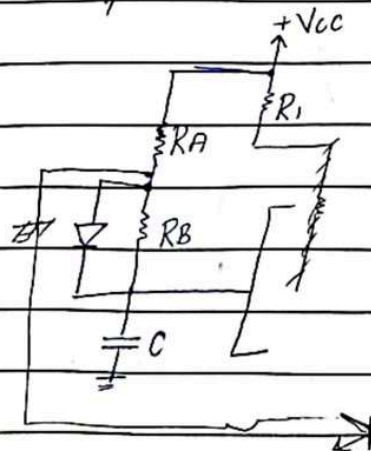
$$T = \ln 2 (R_A + R_B) C$$

$$D = \frac{T_H}{T_H + T_L} = \frac{T_H}{T}$$

$$= \frac{R_A}{R_A + R_B}$$

$$R_A = R_B$$

$$\therefore R_A = R_B$$



Date: 14/11

FOR DUTY CYCLE $\geq 50\%$

greater

$$T_1 = \ln 2 (R_A + R_B) C$$

$$T_2 = \ln 2 (R_B) C$$

$$T = \ln 2 (R_A + 2R_B) C$$

$$f = 1/T$$

$$D = \frac{R_A + R_B}{R_A + 2R_B} \times 100\%$$

FOR DUTY CYCLE $\leq 50\%$

(less than)

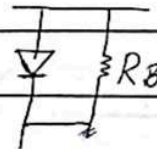
$$T_1 = \ln 2 (R_A) C$$

$$T_2 = \ln 2 (R_B) C$$

$$T = \ln 2 (R_A + R_B) C$$

$$f = 1/T$$

$$D = \frac{R_A}{R_A + R_B}$$



Q. Use 1nF capacitor in a 555 timer circuit, find the value of R that generates a pulse of 10μs.

$$T = (\ln 3) RC$$

$$\frac{10\mu}{(\ln 3)(10n)} = R$$

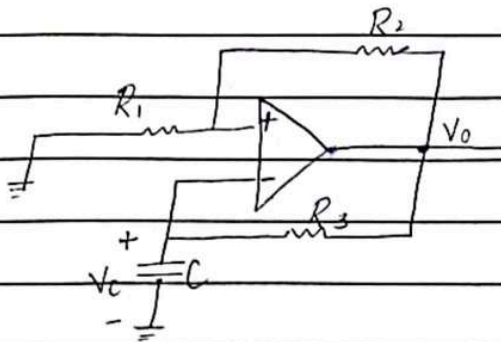
$$R = 9.1 k\Omega$$

Q. Design 2 different ~~a wave~~ circuits that can generate sq. wave. (50% duty cycle)



Date: _____

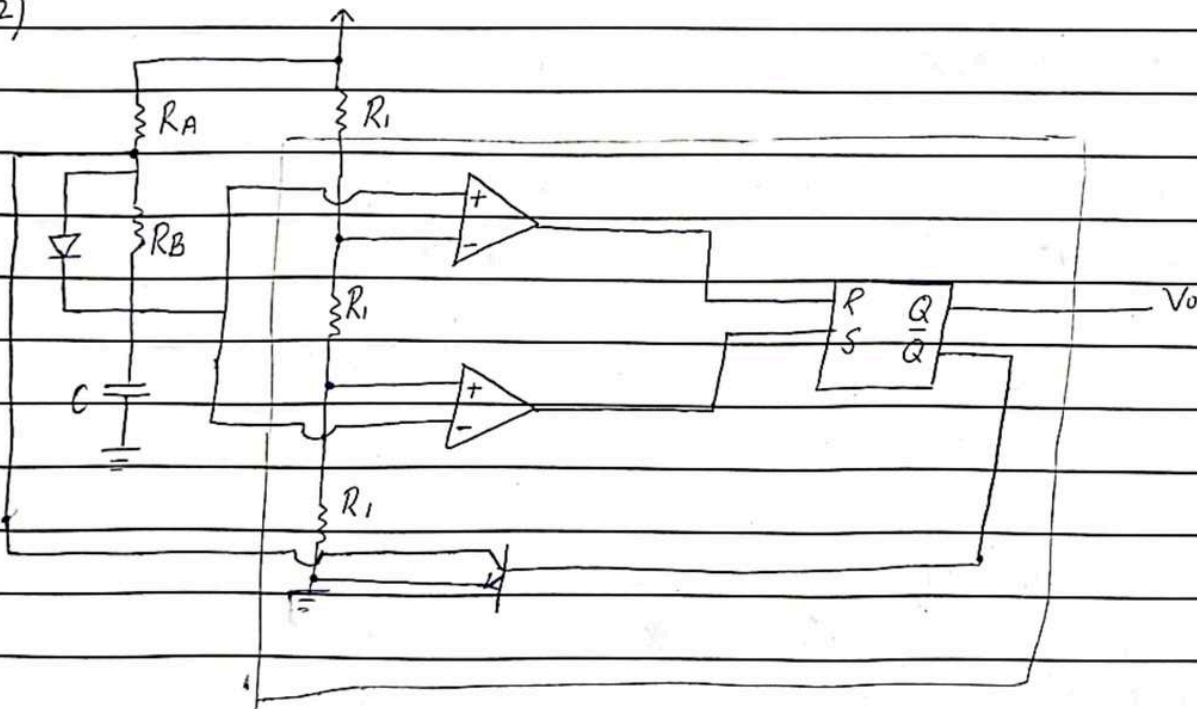
1)



$$T = RC \ln \left(\frac{1+\beta}{1-\beta} \right) \quad \text{for symmetrical}$$

$$L_+ = -L_-$$

2)



Q. Use a 680pF capacitor, design a circuit using 555 timer that generates a rectangular signal of 75% duty cycle & 50kHz frequency find R_A & R_B .

→ Astable MVB using 555 timer

$$T = \frac{1}{f} = \frac{1}{50\text{k}}$$

$$= 20\mu\text{s}$$



Date: _____

$$D = \frac{R_A + R_B}{R_A + 2R_B}$$

$$\therefore 75\% = 0.75 = \frac{3}{4}$$

$$\frac{3}{4} = \frac{R_A + R_B}{R_A + 2R_B}$$

$$4R_A + 4R_B = 3R_A + 3R_B$$

$$R_A = 2R_B \quad (i)$$

$$T = R \ln 2 (R_A + 2R_B)C$$

$$20\mu = \ln 2 (R_A + 2R_B) (680p)$$

$$R_A + 2R_B = \frac{20\mu}{(\ln 2)(680p)}$$

$$R_A + 2R_B = 42.432k$$

$$\rightarrow (i) \quad R_A + R_A = 42.432k$$

$$2R_A = 42.432k$$

$$\boxed{R_A = 21.216k}$$

$$\rightarrow (ii) \quad \frac{R_A}{2} = R_B \Rightarrow \frac{21.216k}{2} = R_B$$

$$\boxed{R_B = 10.5k\Omega}$$

Q. Use a 555 timer to generate a ^{50%} sq. wave signal of 10kHz using a capacitor of 1μF.

$$f = \frac{1}{T}$$

$$T = \frac{1}{f} = \frac{1}{10k} = 0.1ms = 100\mu s$$



Date: _____

$$0.5 = \frac{R_A}{R_A + R_B}$$

$$R_A + R_B$$

$$0.5 R_A + 0.5 R_B = R_A$$

$$\cancel{0.5 R_A} - \cancel{0.5 R_B} \quad (i) \Rightarrow R_B = +0.5 R_A$$

$$0.5$$

$$\tau = \ln 2 (R_A + R_B) C$$

$$100 \mu s = \tau = R_A + R_B$$

$$(\ln 2)(1 \mu)$$

$$R_A + R_B = 144.26 \text{ k}\Omega \quad (ii)$$

$$R_A = 144.26 \text{ k}\Omega - R_B$$

$$R_A = 144.26 \text{ k}\Omega - (+R_A)$$

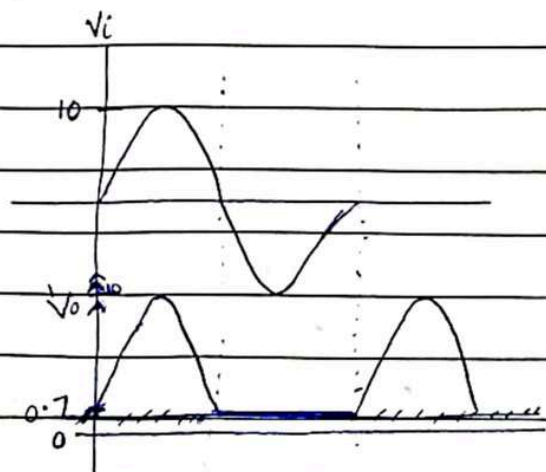
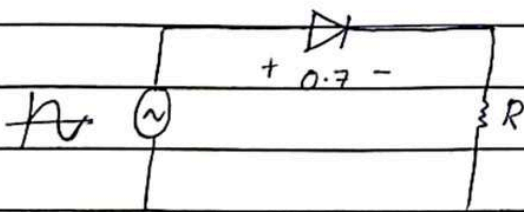
$$2 R_A = 144.26 \text{ k}\Omega \Rightarrow 2 R_A = 144.26 \text{ k}\Omega$$

$$R_A = 72.13 \text{ k}\Omega$$

$$R_B = 144.26 - 72.13$$

$$R_B = 72.13 \text{ k}\Omega$$

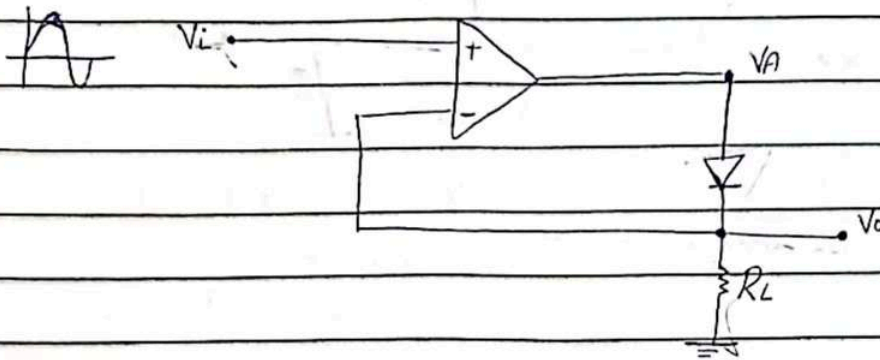
ORDINARY HALF-WAVE RECTIFIER:



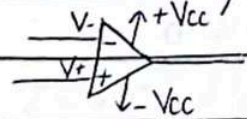
PRECISION HALF-WAVE RECTIFIER:

Date: _____

SUPER-DIODE:



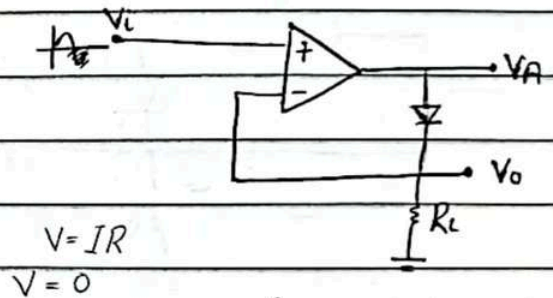
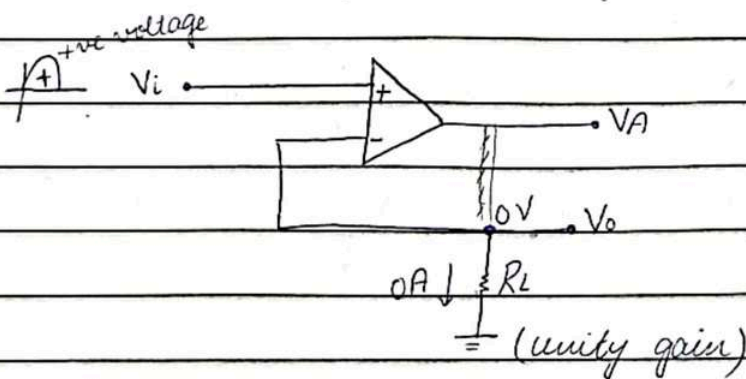
Diode as comparator



$$V_+ > V_- \Rightarrow V_o = L_+$$

$$V_+ < V_- \Rightarrow V_o = L_-$$

- During the +ve half cycle ($V_i > 0$)
- For a moment, ignore that ^{the} diode is existing.



$$V = IR$$

$$V = 0$$

forward biased $\Rightarrow$ short circuit
 $\Rightarrow V_i = V_o$

$$V_+ = +ve$$

$$V_- = 0V$$

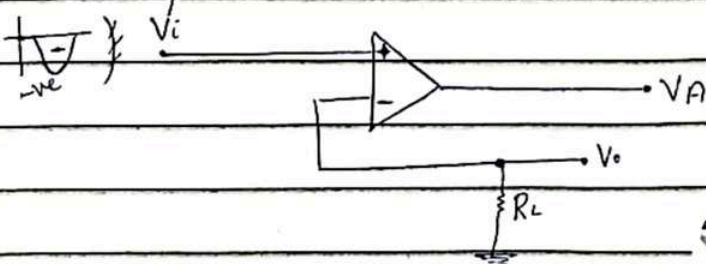
$$+ve - 0 = +ve$$

$$V_A = A(V_+ - V_-)$$

$$V_A = L_+$$

$$\Rightarrow \boxed{V_i = V_o}$$

- DURING THE -ve HALF CYCLE ($V_i < 0$)



Date: _____

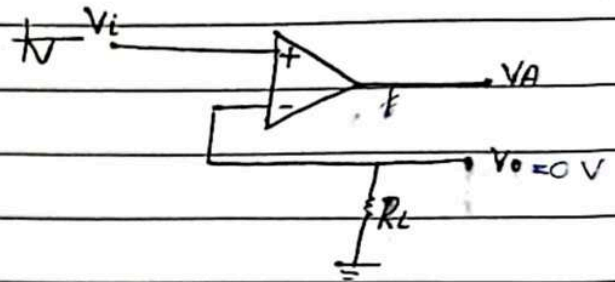
$$V_A = -v_e$$

$$V_- = 0$$

$$V_A = A(V_+ - V_-)$$

$$= A(-v_e - 0) \Rightarrow -v_e$$

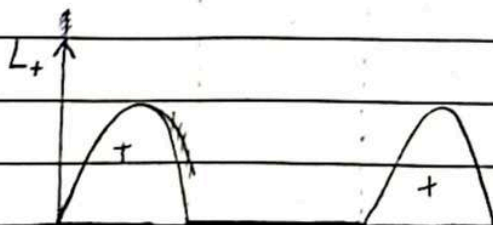
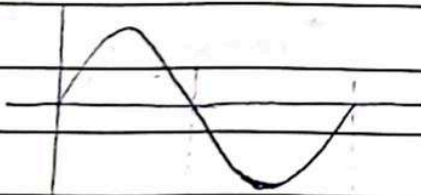
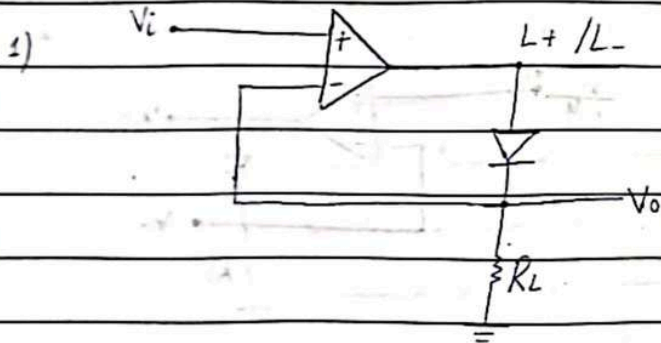
$$\Rightarrow \boxed{V_o = 0V}$$



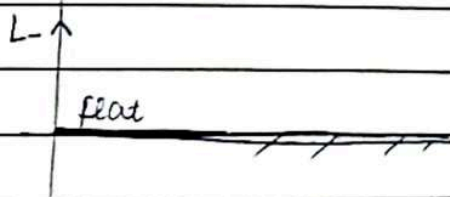
diode reverse biased
→ open circuit

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ACTUAL CIRCUIT

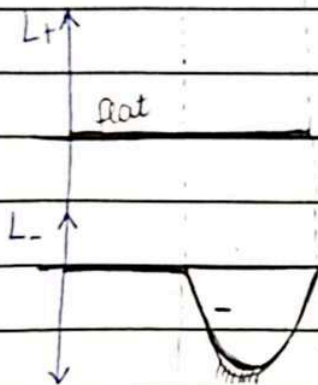
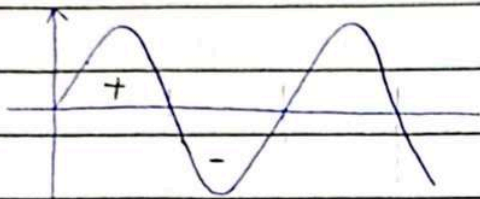
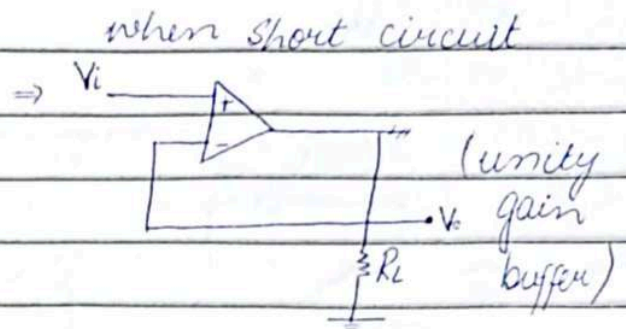
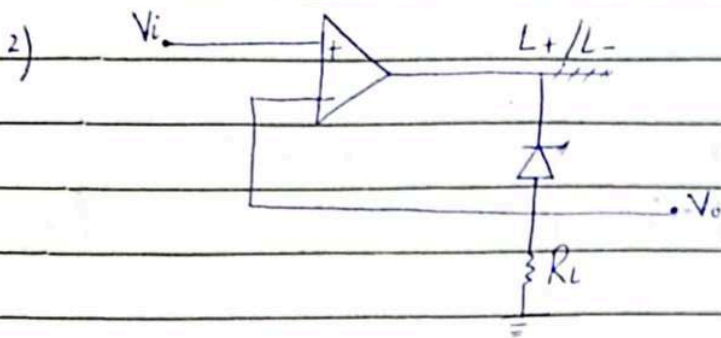


when short circuit



when open circuit

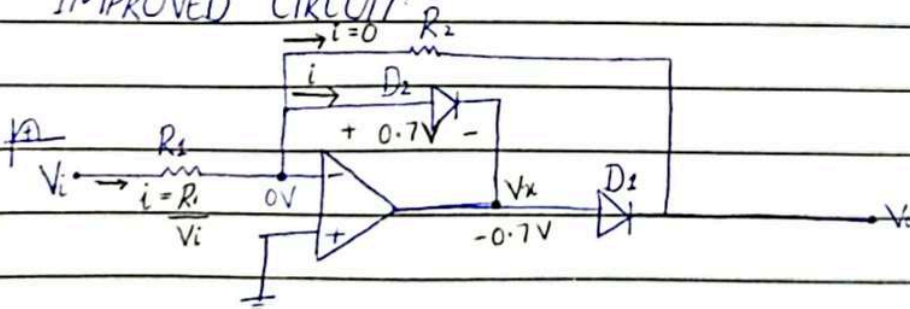




when open circuit

when short circuit
(unity gain buffer)

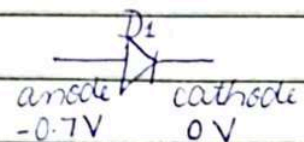
IMPROVED CIRCUIT:



$V_i \rightarrow +ve$

FOR POSITIVE HALF CYCLE:

- D_2 is forward biased (short circuit)
- 'i' passes through D_2 .
- D_1 will be reverse biased (open circuit)

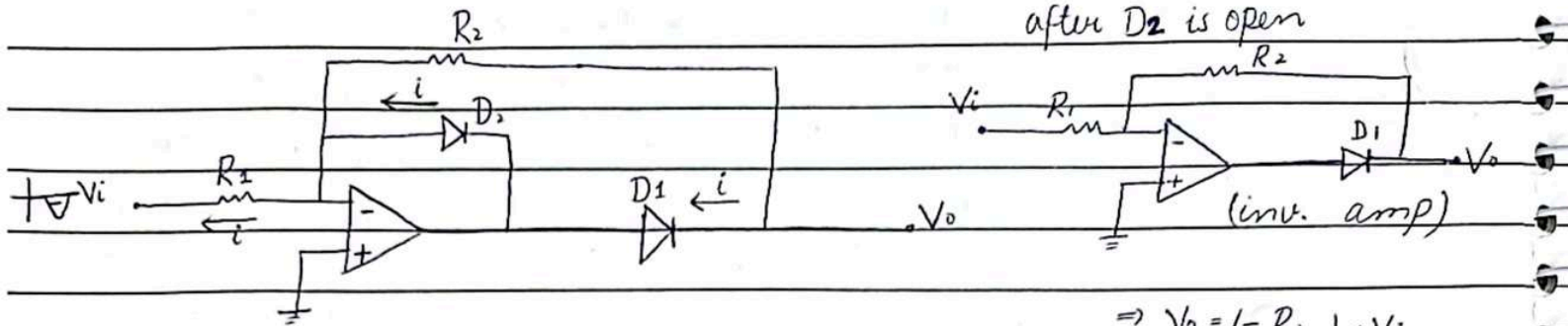


$\Rightarrow$ open circuit ($V_{anode} < V_{cathode}$)



Date: _____

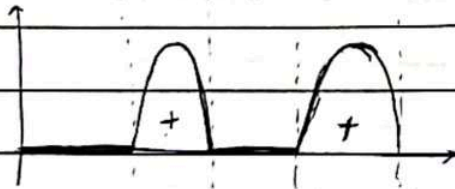
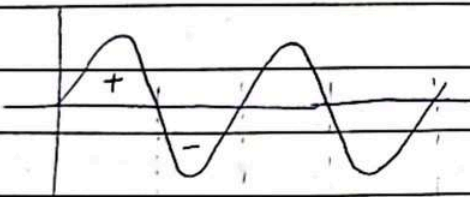
$V_i \rightarrow -ve$



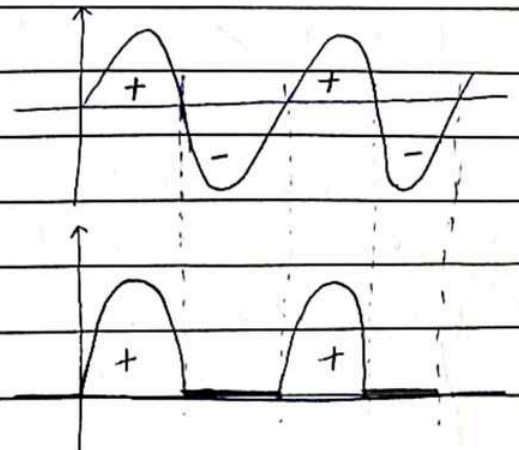
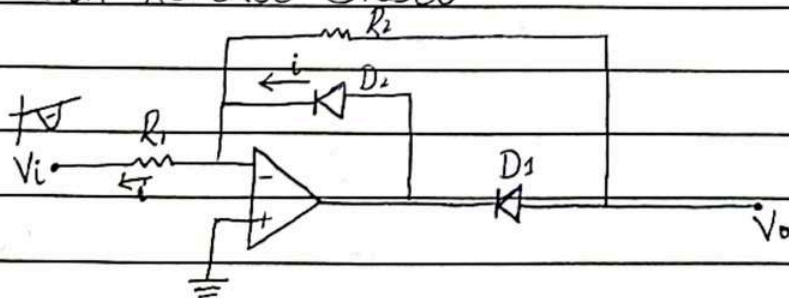
$$\Rightarrow V_o = \left(-\frac{R_2}{R_1} \right) \times V_i$$

FOR NEGATIVE HALF CYCLE

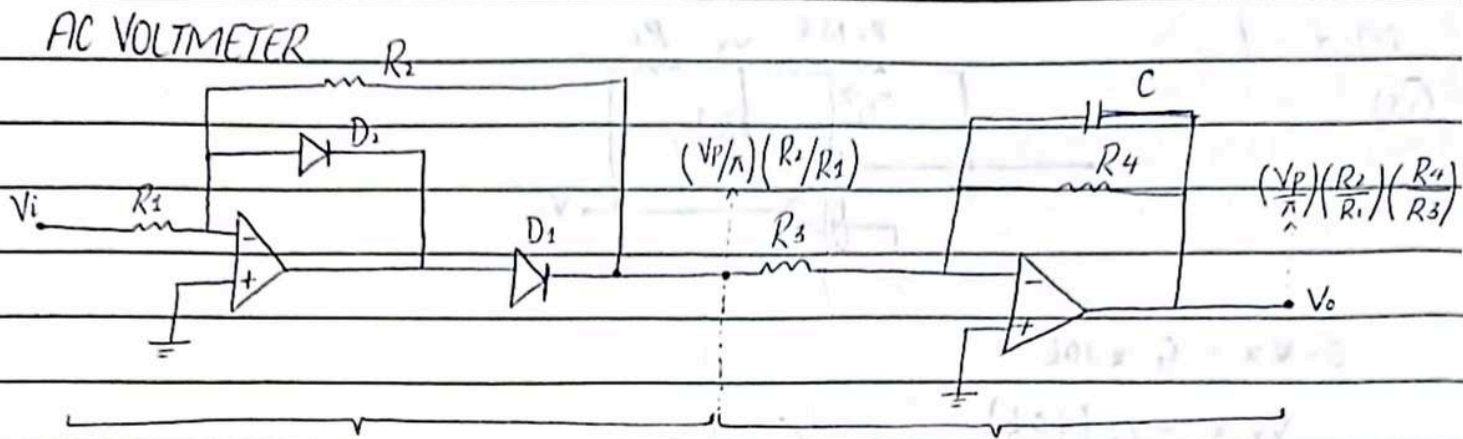
- D_2 is open circuit
- 'i' passes through D_1
- D_1 is forward biased (open circuit)



FOR REVERSE DIODES



Date: _____

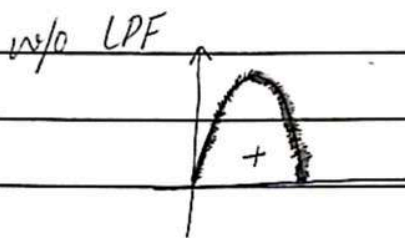


Precision rectifier
DC gain = R_2/R_1

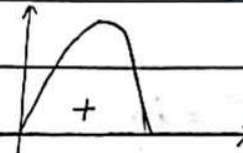
Low pass filter
DC gain = R_4/R_3

Average value = $\frac{V_p}{\pi}$

odd harmonics
→ high freq. noise
↓
LPF is used to remove noise



→ after LPF



* $V_{o, avg} = \frac{V_p}{\pi} \times \frac{R_4}{R_3} \times \frac{R_2}{R_1}$

$V_{o, avg} \propto V_p$

* $\omega_c = \frac{1}{CR_4}$

filter design:
cut off frequency
 $\omega = 1/CR_4$
 $f = 1/2\pi CR_4$

* $V_p = (\pi V_{o, avg}) \left(\frac{R_1}{R_2} \right) \left(\frac{R_3}{R_4} \right)$

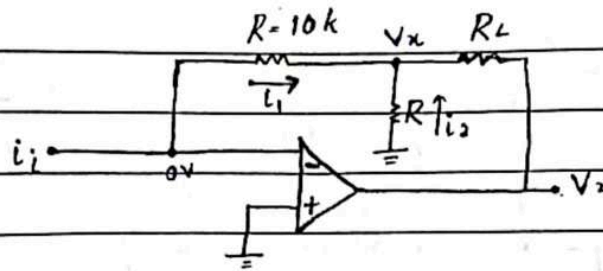
∴ $V_{o, avg}$ - given



Date: _____

Mid - I

Q3)



$$0 - V_x = i_1 \times 10k$$

$$V_x = -i_1 (10k) \quad (i)$$

$$0 - V_x = i_2 \times R$$

$$V_x = -i_2 \times R \quad (ii)$$

$$-i_1 (10k) = -i_2 R$$

$$i_2 = \left(\frac{10k}{R} \right) i_1$$

$$i_1 + i_2 = i_o$$

$$i_1 + \left(\frac{10k}{R} \right) i_1 = i_o$$

$$\left(1 + \frac{10k}{R} \right) i_1 = i_o$$

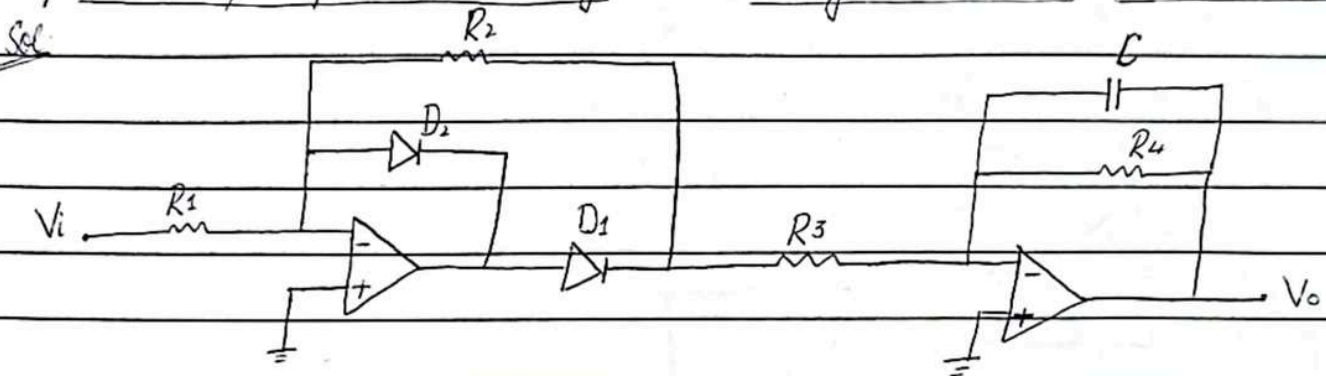
$$1 + \frac{10k}{R} = 21$$

$$R = 500 \Omega$$



AC VOLTMETER

Q. Design an AC voltmeter for a 10Hz input signal. It is calibrated for a sine wave signal to provide a +10V output for input of 1.0 V_{rms}. The R_{in} must be as high as possible, keeping the gain of AC rectifier part to unity. The largest resistor available is 1 MΩ.



AC rectifier gain is unity $\Rightarrow \frac{R_2}{R_1} = 1.0$

input resistor must be as large $\Rightarrow R_1 = 1.0 \text{ M}\Omega$
 $\therefore R_2 = 1.0 \text{ M}\Omega$

$V_{in} = 1.0 \text{ V}_{rms}$

$\therefore V_{rms} = V_p / \sqrt{2}$

$\therefore = (1.0)(\sqrt{2}) = \sqrt{2}$

$\Rightarrow V_p = V_{rms} \times \sqrt{2}$

$V_o = \frac{V_p}{\pi} \times \frac{R_2}{R_1} \times \frac{R_4}{R_3}$

$+10 = \frac{\sqrt{2}}{\pi} (1) \left(\frac{R_4}{R_3} \right)$

* if R_3 is $1 \text{ M}\Omega$ then R_4 would become also larger than $1 \text{ M}\Omega$.

Highest is $1 \text{ M}\Omega \Rightarrow$ let $R_4 = 1.0 \text{ M}\Omega$

So, that R_3 is lesser than R_4

$10 = \frac{\sqrt{2}}{\pi} (1) \left(\frac{1 \times 10^6}{R_3} \right)$

$R_3 = 45 \text{ k}\Omega$

$\omega = 1/CR_4$

$f = 1/2\pi CR_4$

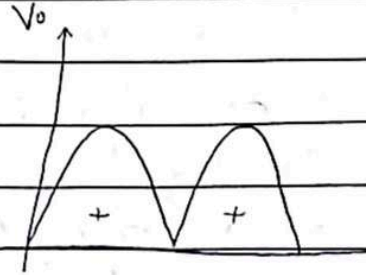
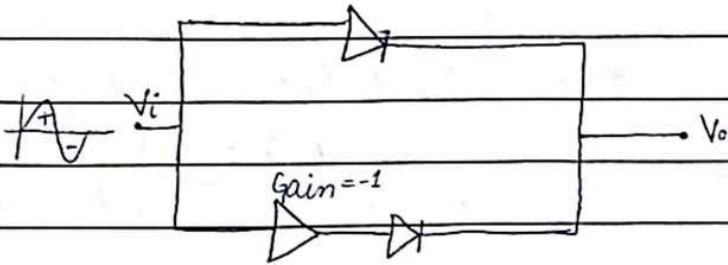
$10 = 1/2\pi C(10^6)$

$C = 15.9 \text{ nF}$

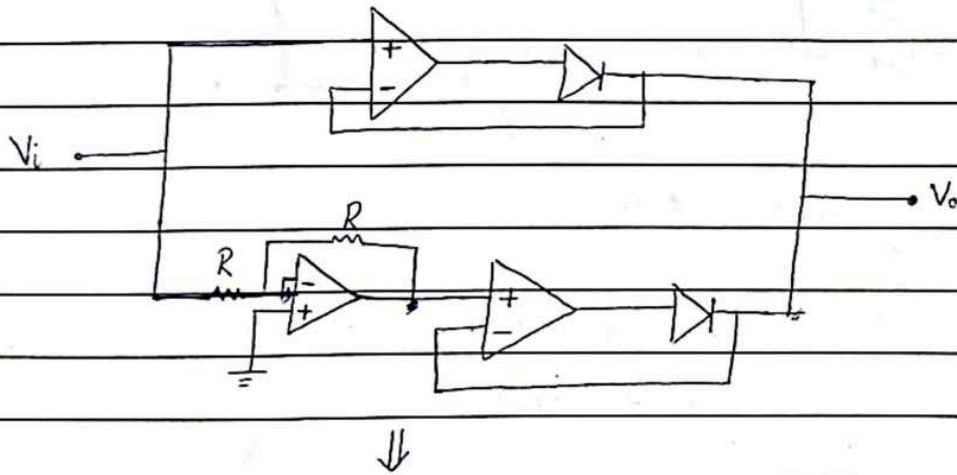


Date: 21/11

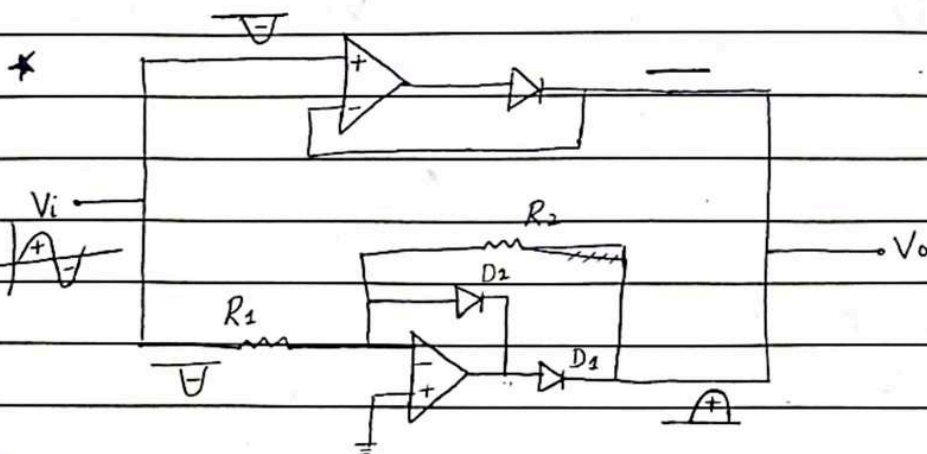
PRECISION FULL WAVE RECTIFIER:



Full-wave rectifier

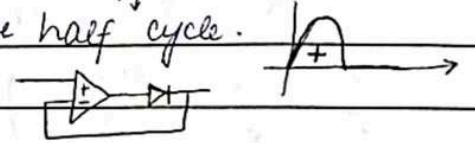


⇓



use improved circuit for -ve half cycle.

and super diode for +ve half cycle.

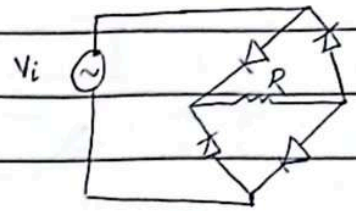
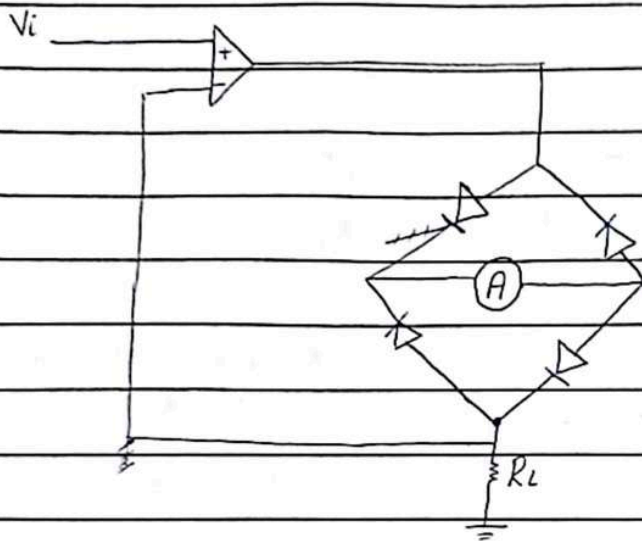


(precision full-wave rectifier)

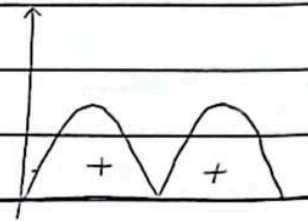


full wave rectifier using BRIDGE (IDEAL)

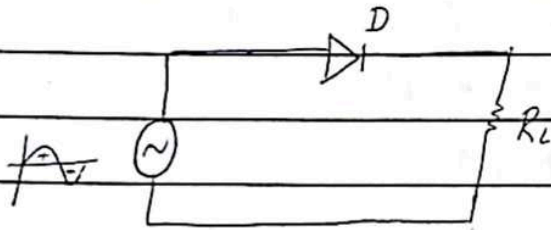
bridge



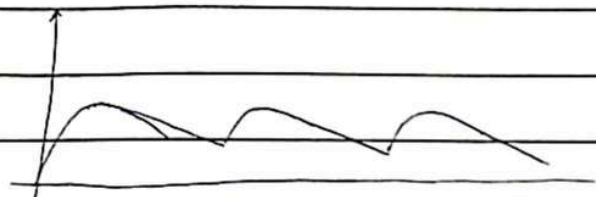
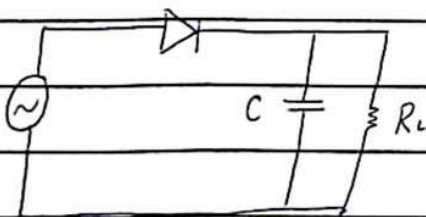
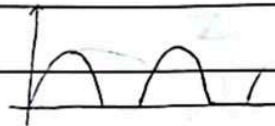
Ammeter 0.1Ω



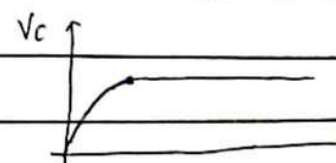
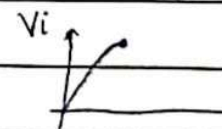
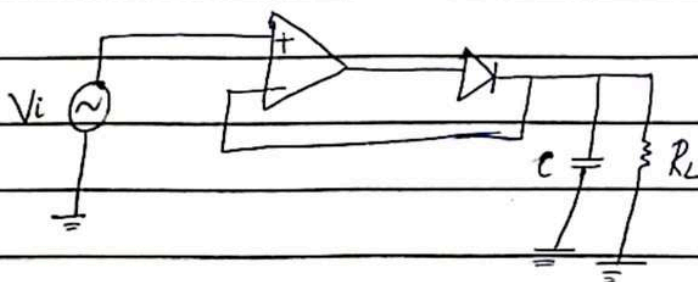
PEAK RECTIFIER



(Half wave)

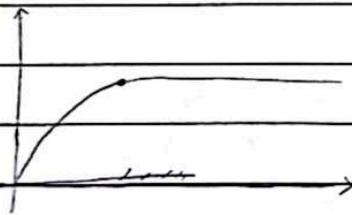
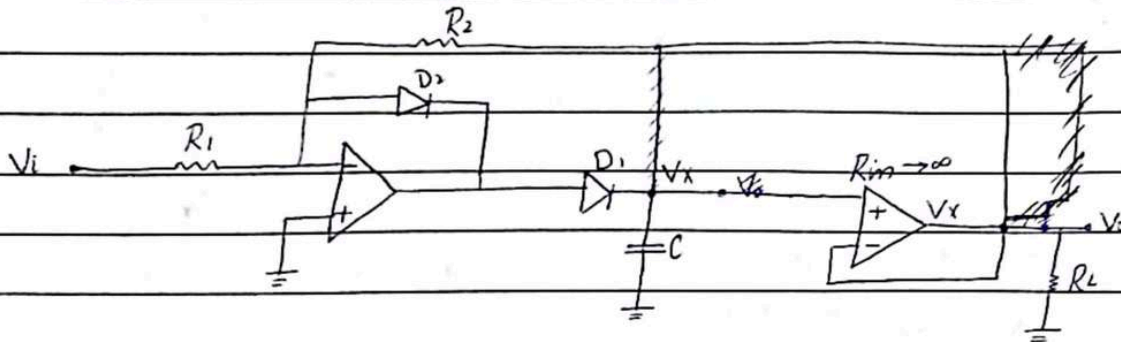


PRECISION IN PEAK RECTIFIER



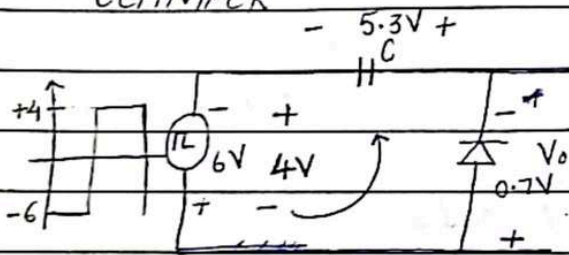
Date: _____

BUFFERED PRECISION RECTIFIER



* feedback loop is closed where R_L is connected.

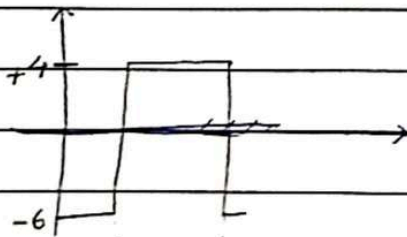
CLAMPER



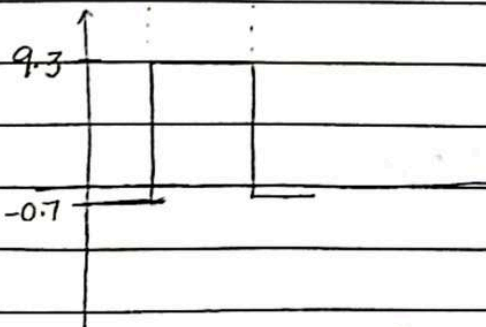
for $-6V$

for $+4V$

$$6 - 0.7 = 5.3$$



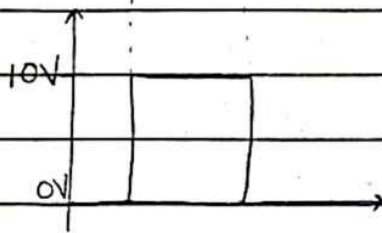
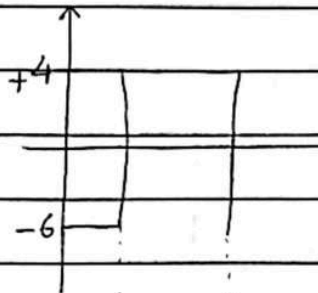
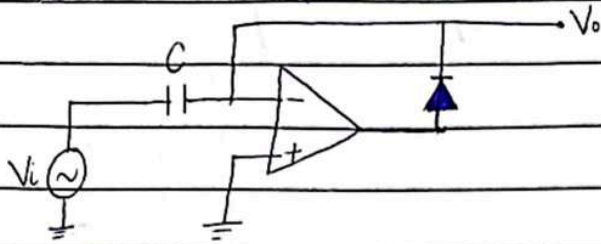
overall



$$10 - 0.7 = 9.3V$$



PRECISION CLAMPER



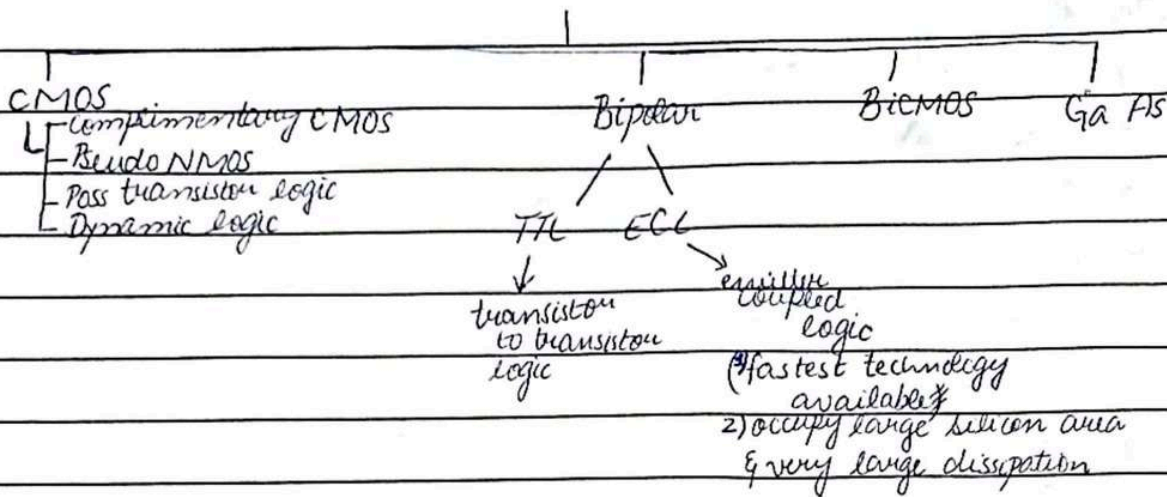
26/11

DIGITAL IC TECHNOLOGIES & LOGIC CIRCUIT FAMILIES: (Qualitative)

Logic circuit families are as follows:

- 1) Logic feasibility
- 2) Speed of operation
- 3) Availability of complex function
- 4) Noise immunity
- 5) Operating temp. range
- 6) Power dissipation
- 7) Cost
- 8) Size



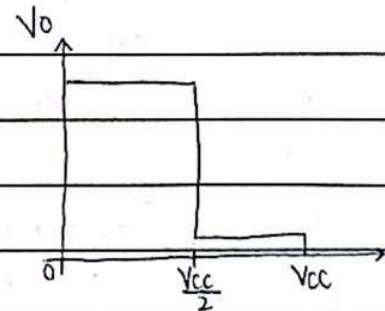
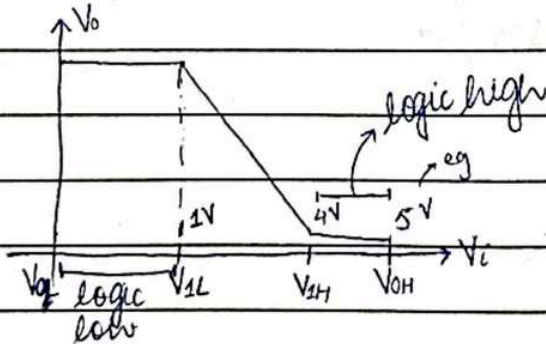


→ Complementary CMOS:

- 1) low power dissipation
- 2) low space
- 3) channel size = $0.06 \mu m$
- 4) high input impedance
- 5) high input optimization

LOGIC CIRCUIT CHARACTERISTICS: (Quantitative)

1) NOISE MARGIN: (NM)



best noise margin

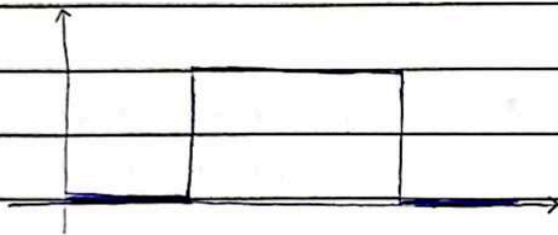
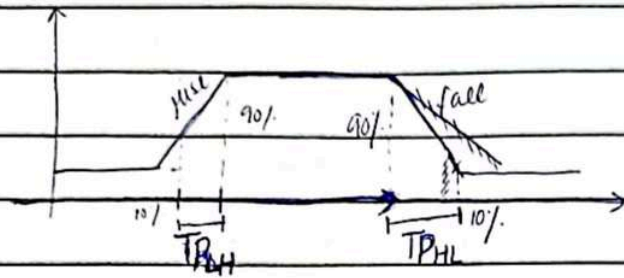
$$NM_H = V_{OH} - V_{IH}$$

$$NM_L = V_{IL} - V_{OL}$$



Date: _____

2) PROPAGATION DELAY - Time taken by the signal going from 10% to 90%.



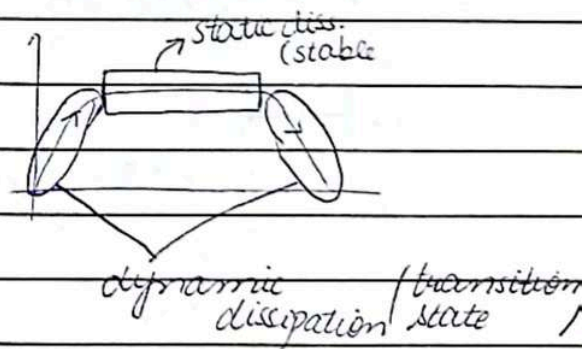
$$T_P = \frac{1}{2} (T_{PLH} + T_{PHL})$$

CMOS
* capacitor at output stores voltage

3) POWER DISSIPATION:

i) Static dissipation $\Rightarrow P_S = \frac{V_{CC}^2}{R}$

ii) Dynamic dissipation $\Rightarrow P_D = f C V_{CC}^2$

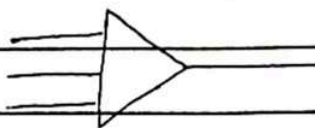


4) DELAY POWER PRODUCT

$$DP = T_P \times P_D$$

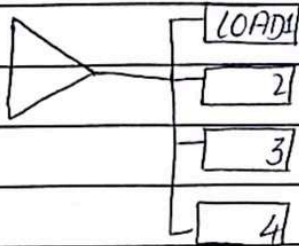
* DP value - less as possible $\rightarrow$ best IC.

5) FAN-IN (max. no. of input)



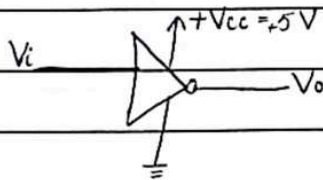
Date: _____

ii) FAN OUT: (max. no. of load / circuits at o/p)

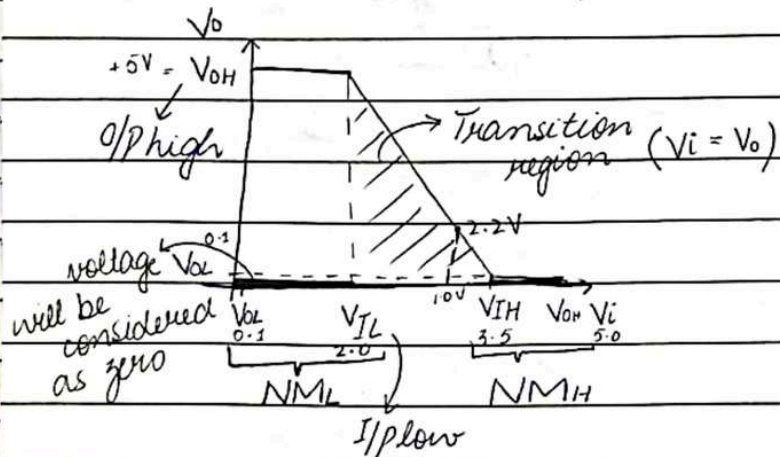


28/11 DIGITAL LOGIC INVERTER.

• FUNCTION OF INVERTER



VOLTAGE TRANSFER CHARACTERISTICS:



$[0.1 - 2.0] \rightarrow \text{logic } 0 \Rightarrow \text{logic } '1' \rightarrow V_{OH} = 5V$

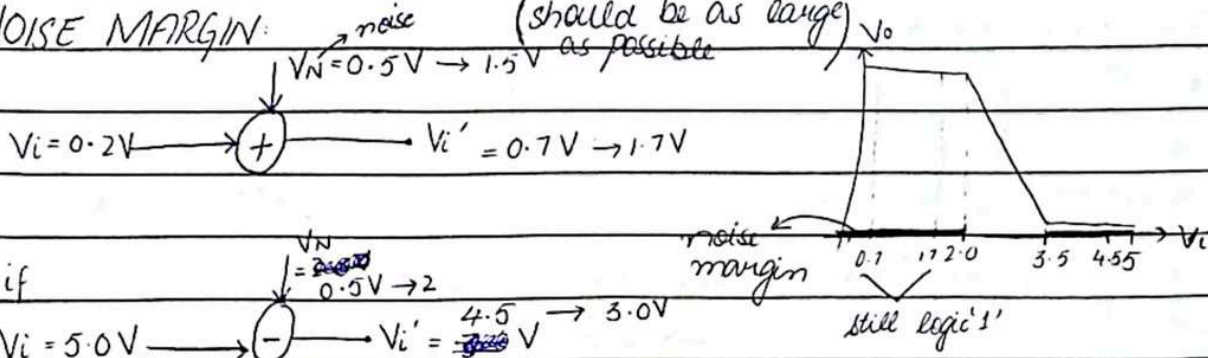
$[3.5V - 5.0V] - \text{logic } 1 \Rightarrow \text{logic '0'} \rightarrow V_{OL} = 0.1V$

$V = 2.2V \rightarrow$ undefined



Date: _____

• NOISE MARGIN: (should be as large as possible)



* when

$$V_{OL} \leq V_i < V_{IL} ; V_o = 0 \text{ } V_{OH}$$

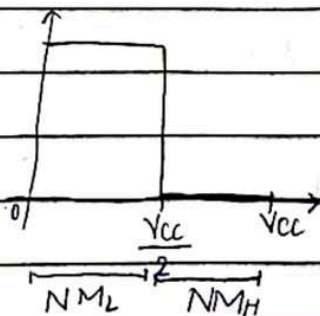
$$V_{IH} < V_i \leq V_{OH} ; V_o = V_{OL}$$

V_{IL} : The max I/P of an inverter that can be interpreted as logic low '0'.

V_{IH} : Min. I/P of an inverter that can be interpreted as logic high '1'.

$$* \begin{cases} NML = V_{IL} - V_{OL} \\ NMH = V_{OH} - V_{IH} \end{cases}$$

• INVERTER IMPLEMENTATION:



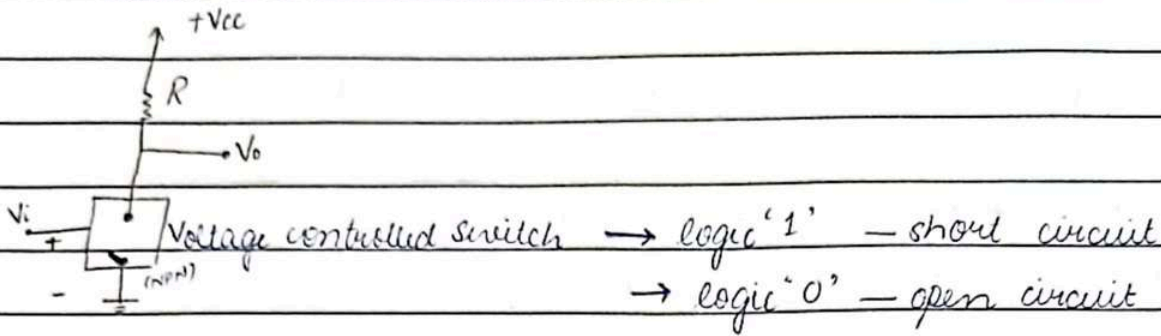
$$NMH = NML = V_{cc}/2$$

$$V_{IL} = V_{IH} = V_{cc}/2$$

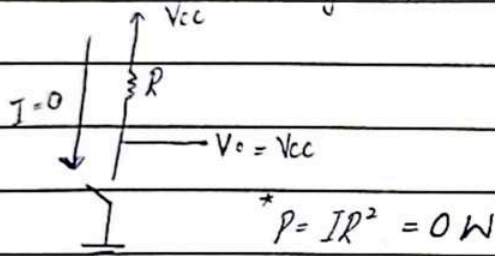


Date: _____

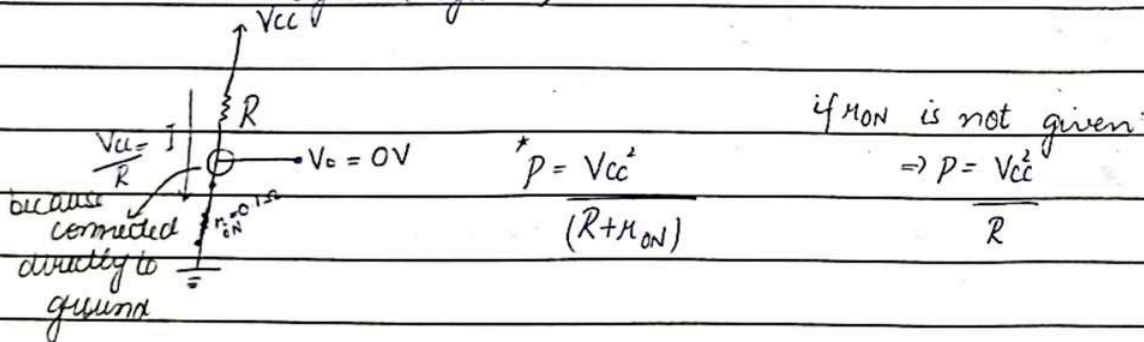
• INVERTER IMPLEMENTATION



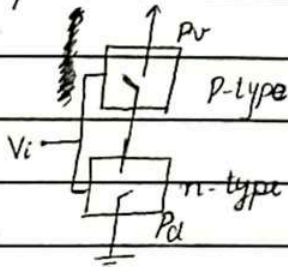
→ when V_i is low (logic '0')



→ when V_i high (logic '1')



USING COMPLEMENTARY SWITCHES



$I = 0$ (one switch open)
(static power = 0 (always))

n-type:

V_i is low logic '1' — switch close

logic '0' — switch open

p-type

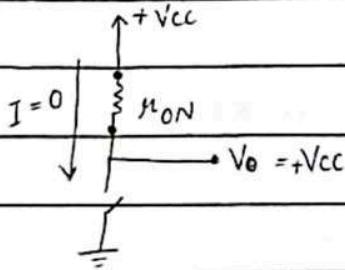
logic '1' — switch open

logic '0' — switch close



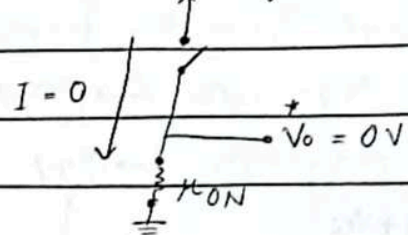
Date: _____

V_i is low



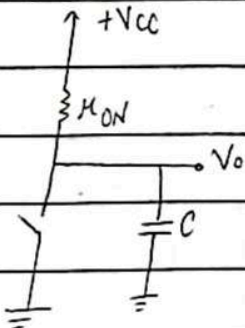
$$P = 0W$$

V_i is high

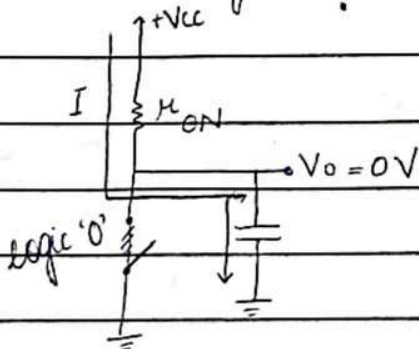


$$P = 0W$$

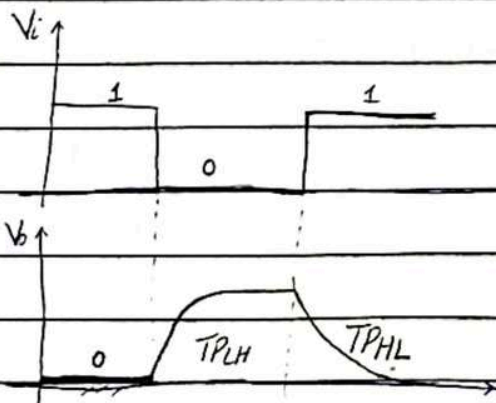
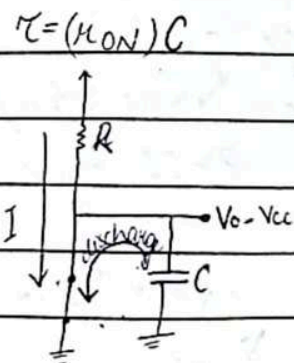
DYNAMIC POWER DISSIPATION:



Switch - $I/p = \text{logic '0'}$



$\Rightarrow$ logic '1'



when capacitor is fully charged
current stops flowing

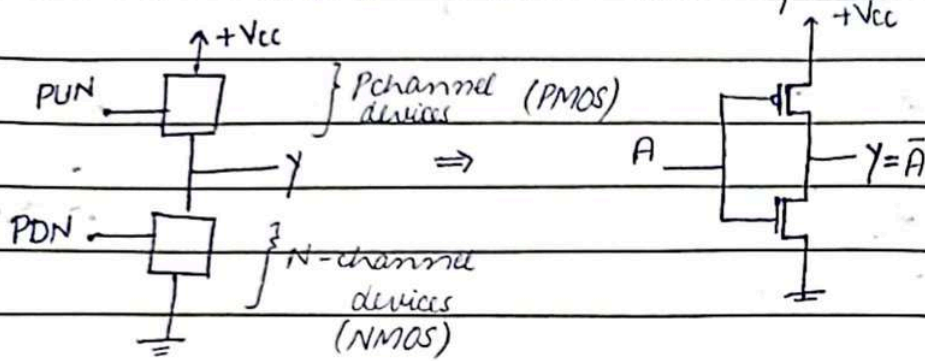
$$P_{dynamic} = fC V_{cc}^2$$



Date: 3/12

CMOS LOGIC GATE CIRCUITS

- 1) Logic level inverter contains a PUN & PDN.
- 2) NMOS transistor $\rightarrow$ PDN (Pull down network)
- PMOS transistor $\rightarrow$ PUN (Pull up network)



Operated in Complementary function

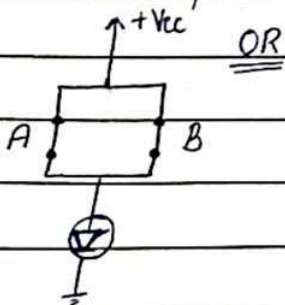
- * $A = '1'$, NMOS is ON, PMOS is OFF, $Y = '0'$
- * $A = '0'$, PMOS is ON, NMOS is OFF, $Y = '1'$

For other logic gates

- * PDN contain NMOS transistor only - conducts when gate I/p is high.
- * PUN contain PMOS transistor only - conducts when gate I/p is low.

TWO NMOS TRANSISTOR IN PDN OR TWO PMOS TRANSISTOR IN PUN:

- * Perform OR function when connected in parallel.
- * Perform AND function when connected in series.

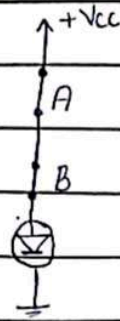


A	B	Y
1	0	1
0	1	1
1	1	1
0	0	0



Date: _____

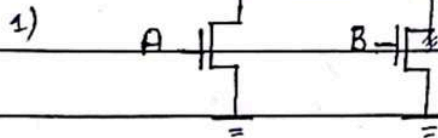
AND



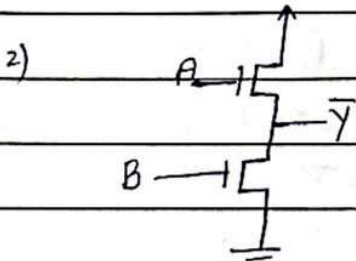
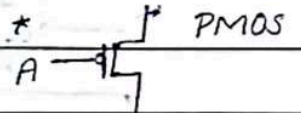
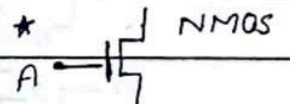
A	B	Y
1	0	0
0	1	0
1	1	1
0	0	0

* PDN is exact offset of PUN ie two series NMOS transistors in PDN are interpreted as two parallel PMOS transistors in PUN.

PDN

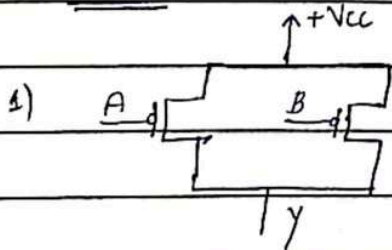


$$\bar{Y} = A + B$$

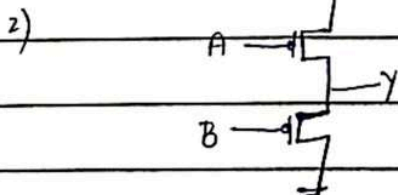


$$\bar{Y} = A \cdot B$$

PUN



$$Y = \bar{A} + \bar{B}$$



$$Y = \bar{A} \cdot \bar{B}$$



st METHOD OF CMOS IMPLEMENTATION

Date: _____

TWO INPUT NOR GATE

Q: $Y = \overline{A+B} = \bar{A} \cdot \bar{B}$

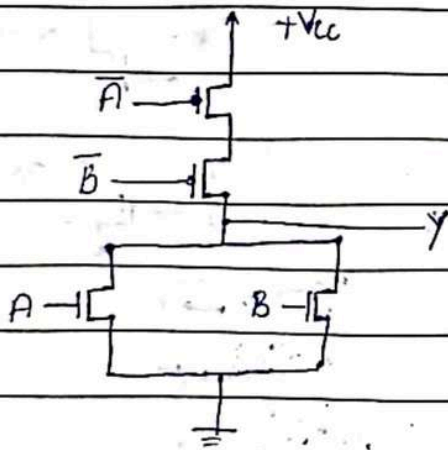
PUN: $Y = \overline{A+B}$

PDN: $\bar{Y} = \overline{\overline{A+B}}$ } Complement of PUN
 $\Rightarrow \bar{Y} = A+B$

implement PDN first!

Complement PDN to get PUN!

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

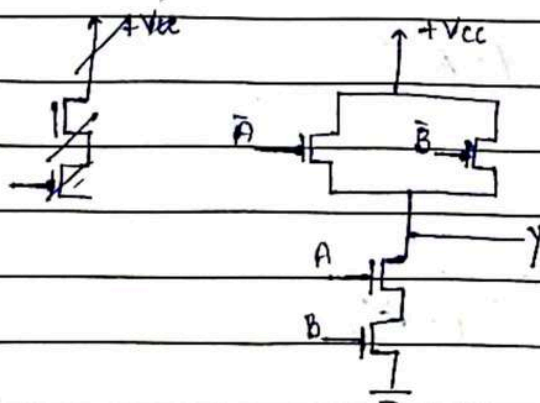


TWO I/P NAND GATE

Q: $Y = \overline{AB}$

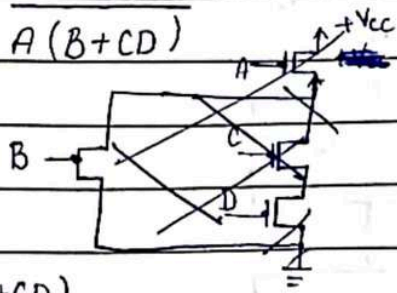
PUN $\Rightarrow Y = \overline{AB}$

PDN $\Rightarrow \bar{Y} = AB$



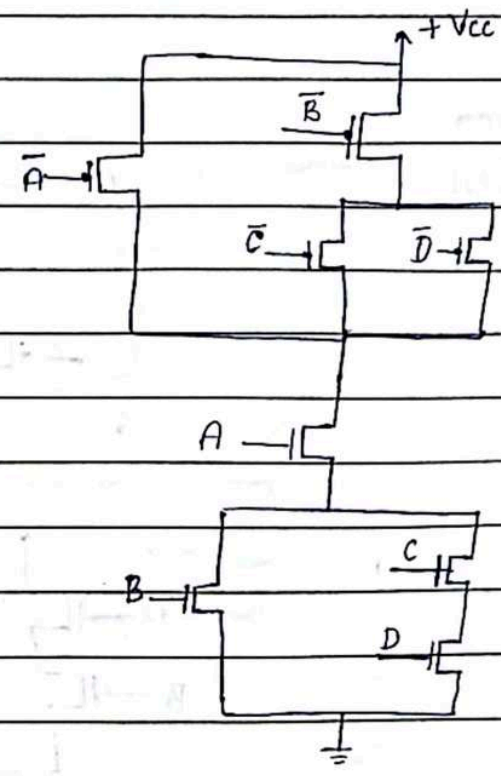
EXAMPLE #1

$$Y = A(B+CD)$$



PUN: $A(B+CD)$

PDN: $\overline{A(B+CD)}$

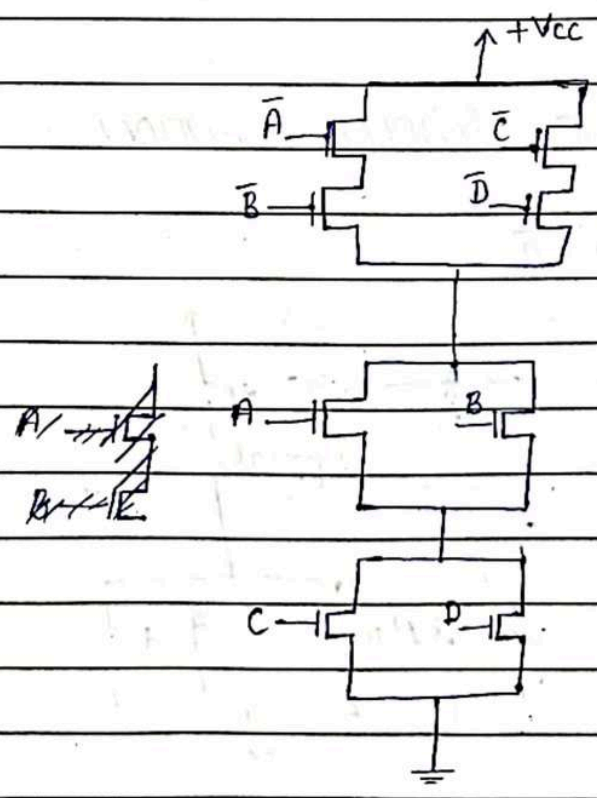


EXAMPLE #2

$$Y = (A+B)(C+D)$$

PUN: $(A+B)(C+D)$

PDN: $\overline{(A+B)(C+D)}$



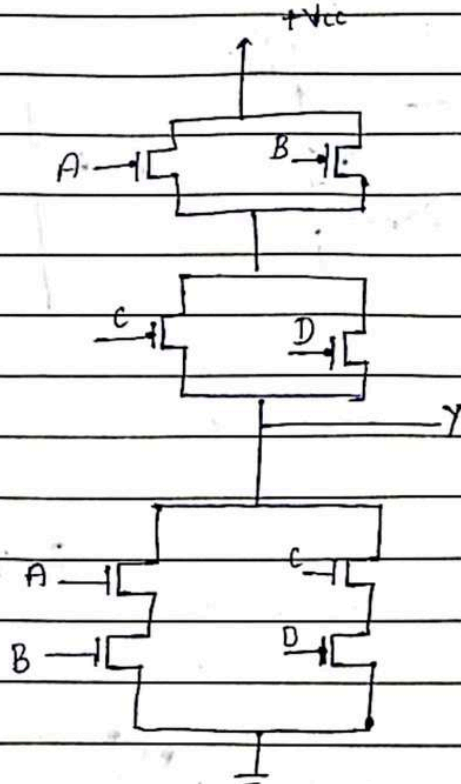
Date: _____

EXAMPLE #3

$$F = \overline{AB + CD}$$

$$\text{PUN: } AB + CD$$

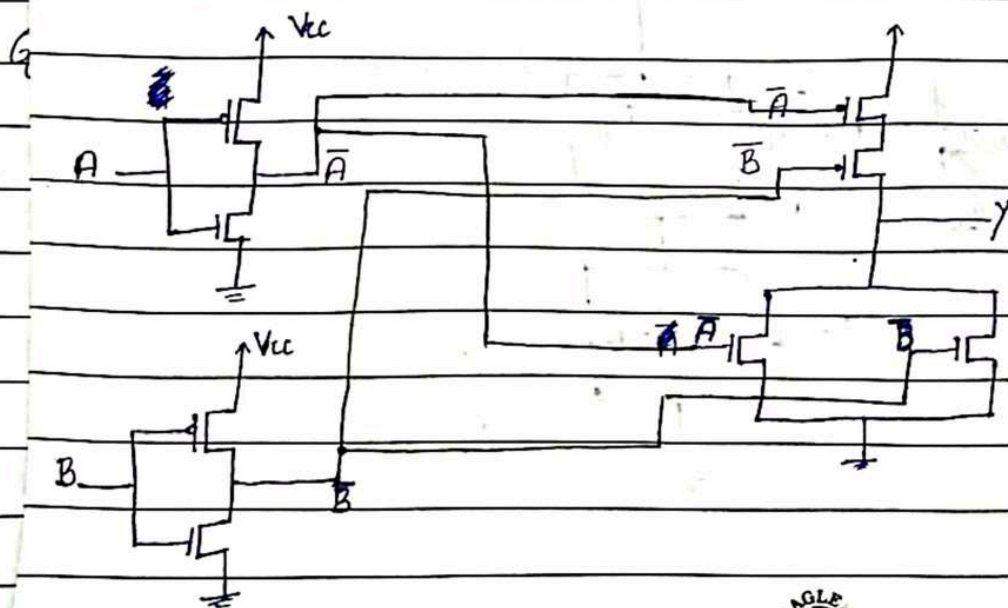
$$\text{PDN: } \overline{AB + CD}$$



2ND METHOD OF CMOS IMPLEMENTATION: (PRE-inversion)

$$Y = AB$$

$$\text{PDN: } \overline{Y} = \overline{AB} = \overline{A} + \overline{B}$$



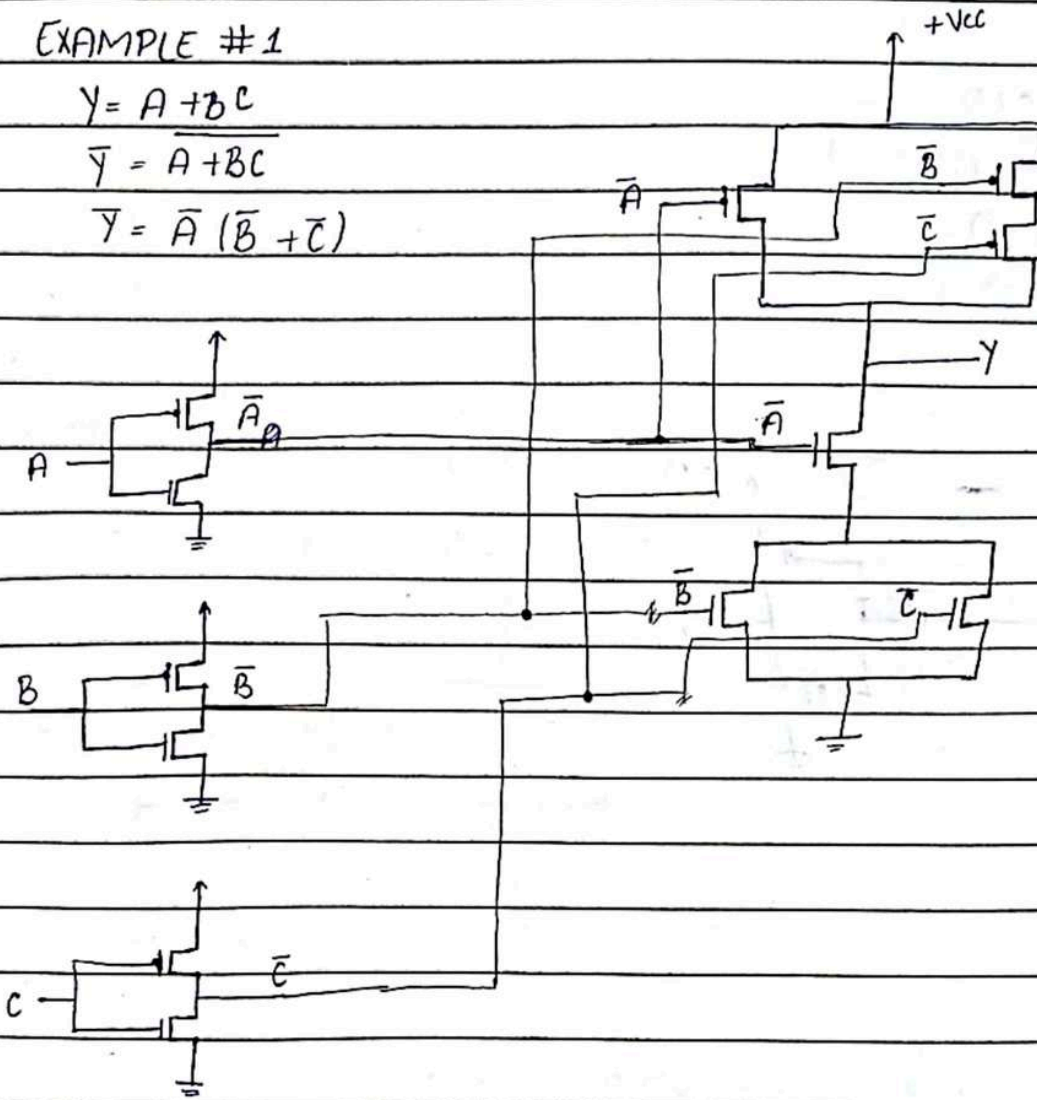
Date: _____

EXAMPLE #1

$$Y = A + BC$$

$$\bar{Y} = \overline{A + BC}$$

$$Y = \bar{A} (\bar{B} + \bar{C})$$

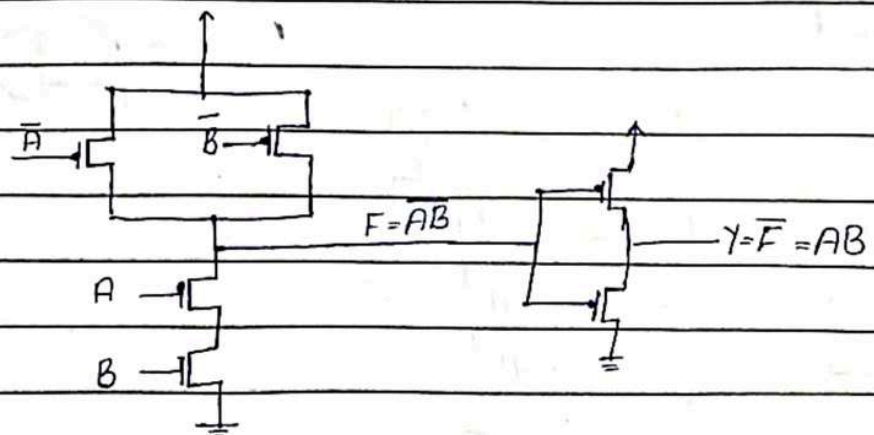


3RD CMOS METHOD OF CMOS IMPLEMENTATION: (Post inversion)

EXAMPLE #1: $Y = AB$

PUN: $F = \bar{A}\bar{B}$

PDN: $\bar{F} = AB$



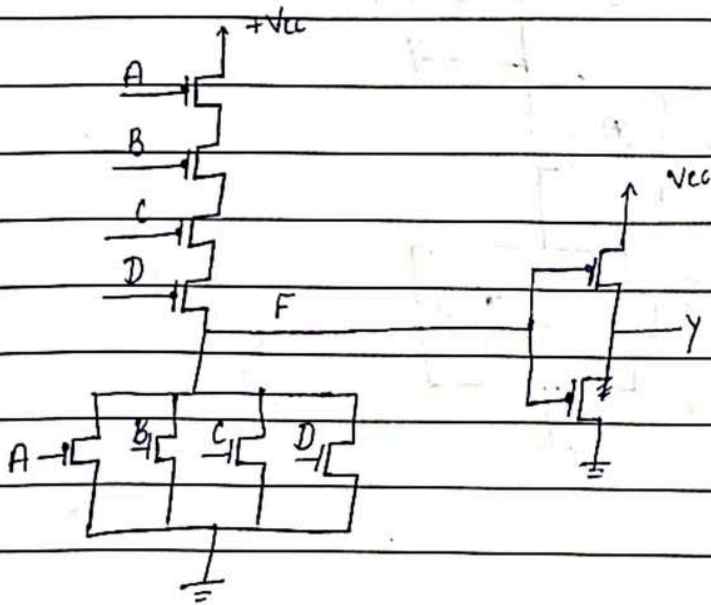
Date: _____

EXAMPLE #2

$$Y = A + B + C + D$$

PUN: $Y = \overline{A + B + C + D}$

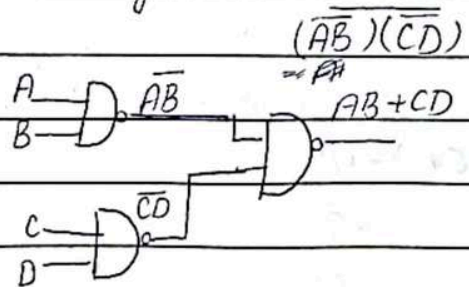
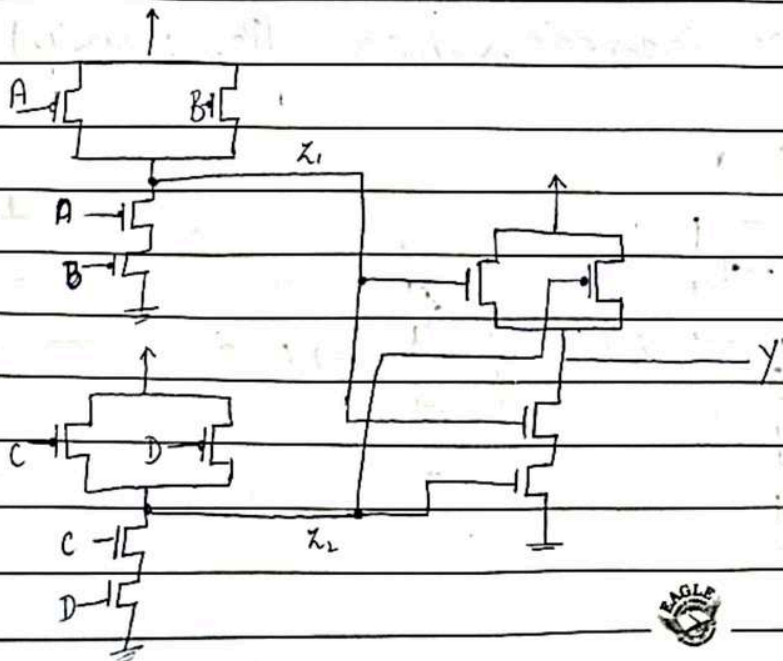
PDN: $\overline{Y} = A + B + C + D$



EXAMPLE:

5/12 $Y = AB + CD$

Make the CMOS circuit using NAND gate only.

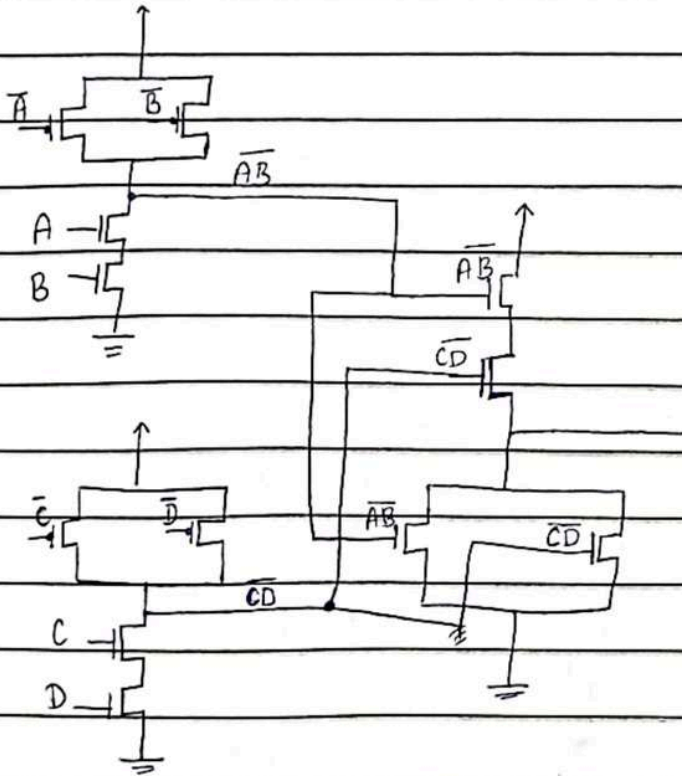
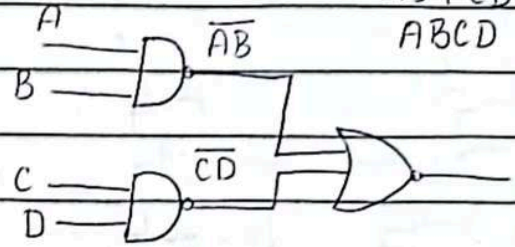


Date: _____

EXAMPLE:

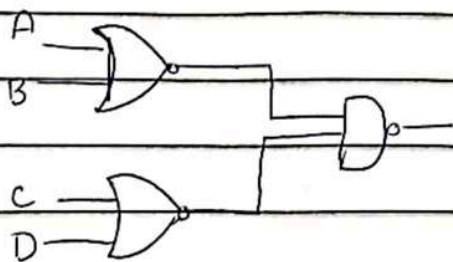
$$Y = ABCD$$

$$\overline{AB} + \overline{CD} = \overline{AB} \cdot \overline{CD}$$



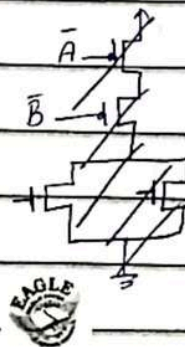
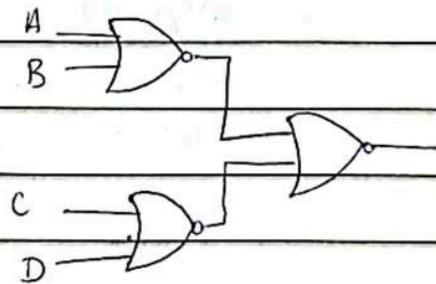
EXAMPLE:

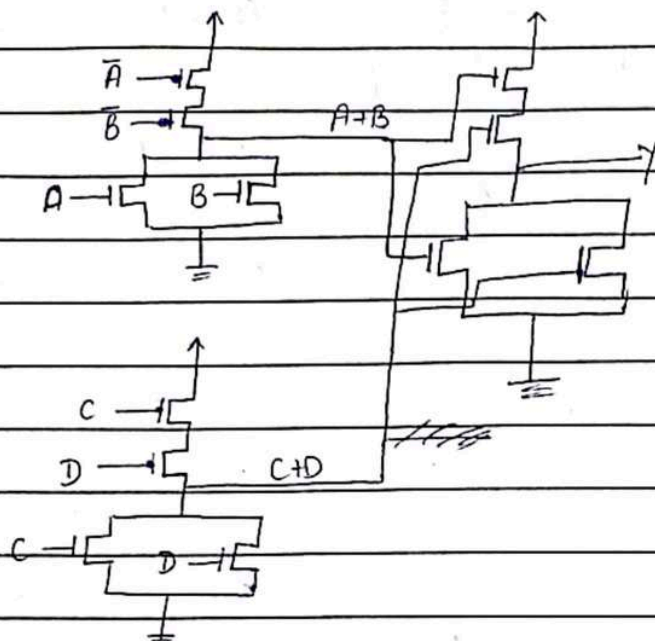
$$Y = (A+B)(C+D)$$



EXAMPLE

$$Y = A + B + C + D$$



for $A+B+C+D$ 

EXAMPLES:

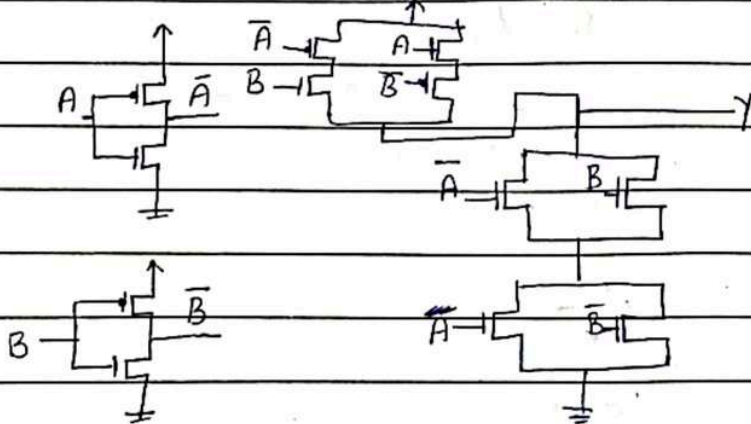
1) $Y = AB + C + D$

2) $Y = AB(C+D)$

XOR GATE

1) PUN: $Y = A\bar{B} + \bar{A}B$

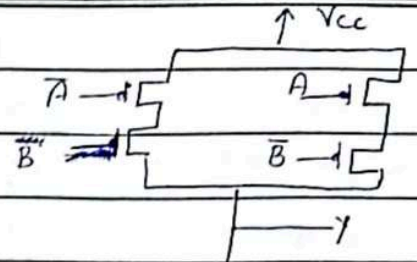
PDN: $A\bar{B} + \bar{A}B = (\bar{A} + B)(A + \bar{B})$



Date: _____

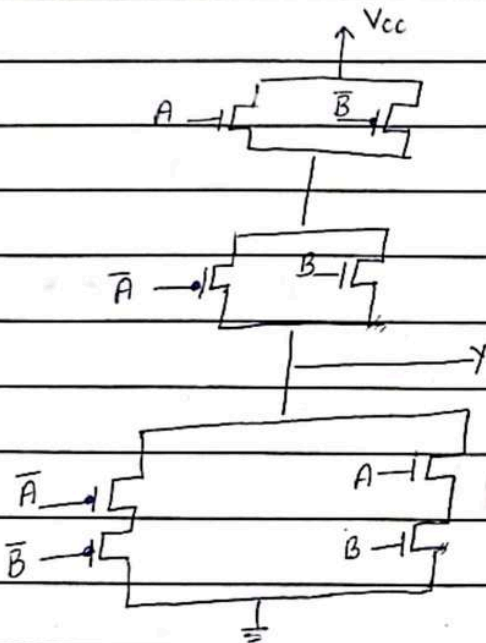
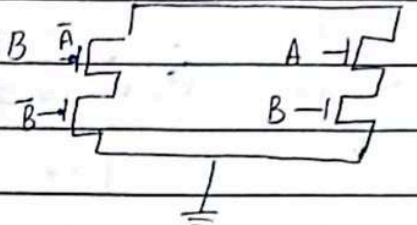
2) PUN: $Y = A\bar{B} + \bar{A}B$

PDN: $\bar{Y} = \bar{A}\bar{B} + AB$
 $= \bar{A}\bar{B} + \bar{A}\bar{B} + AB + AB$
 $= \bar{A}\bar{B} + AB$



3) PUN: $Y = (A+B)(\bar{A}+\bar{B}) \Rightarrow \bar{Y} = (\bar{A}+\bar{B})(A+B) = \bar{A}B + A\bar{B}$

PDN: $\bar{Y} = \bar{A}B + A\bar{B}$



4) PUN: $Y = (A+B)(\bar{A}+\bar{B}) \Rightarrow Y = \bar{A}\bar{B} + AB$

$\Rightarrow Y = \bar{A}\bar{B} + AB$

PDN: $\bar{Y} = (\bar{A}+B)(A+\bar{B}) \Rightarrow \bar{Y} = \bar{A}B + A\bar{B}$

$\Rightarrow \bar{Y} = \bar{A}B + A\bar{B}$

